NRC
Natural Resources College

Mathematics Module

Module Code: BS-MAT

Module Writer:
Symon B. Chibaya
Mathematics and Statistics Lecturer, Natural Resources College

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MODULE OBJECTIVES

By the end of the Module, students should be able to:

a) describe number systems, functions and graphs, equations, inequalities and linear programming, matrix algebra, set theory and series.

b) solve problems involving number systems, functions and graphs, equations, inequalities and linear programming, matrix algebra, set theory and series.
c) apply mathematical knowledge and techniques to solve practical problems in agriculture and other science sectors.

TOPIC 1: NUMBER SYSTEMS

OBJECTIVES

By the end of this topic, students should be able to;

(a) Convert numbers from base ten to any base.
(b) Convert numbers from any base to base ten.
(c) Add numbers other than in base ten.
(d) Subtract numbers other than in base ten.
(e) Simplify complex numbers.

DECIMAL NUMBERS (BASE TEN)

When recording in base ten, only ten symbols are needed. These are 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

Decimal numbers can be expanded using base ten. For example,

$143=1x100+4x10+3x1$

$=1x10^2+4x10^1+3x10^0$

$5894=5x1000+8x100+9x10+4x1$

$=5x10^3+8x10^2+9x10^1+4x10^0$

Decimal numbers can be added or subtracted together.

Example. Simplify the following:

(a) $178_{10}+469_{10}$

(b) $587_{10}-396_{10}$

Solutions

(a) 178_{10}

$+469_{10}$

647_{10}

(b) 587_{10}

-396_{10}

191_{10}

We probably use base ten because we were born with four fingers and a thumb on each hand.

2. BINARY NUMBERS (BASE TWO)

Binary numbers have two digits and these are 0 and 1. Because of their application to computers, number base two are widely used. Such numbers soon become very long. Some examples of binary numbers are 1100011_{two} and 1111100111_2

3. OCTAL NUMBERS (BASE EIGHT)

Octal numbers have eight digits namely: 0, 1, 2, 3, 4, 5, 6 and 7.

Some examples of octal numbers: 1543_8 and 1764_8

CONVERTING OF OTHER NUMBER BASE TO BASE TEN NUMBER

a) You expand the given number using its base
b) In other ways, each digit must be multiplied by the base.
c) The power of the last right is always equal to 0.
d) Write the powers of the base as 0,1, 2, 3,…going to left hand side.

Example.
Convert 1101_2 to the corresponding decimal number.
Solution
$1101_2=1x2^3+1x2^2+0x2^1+1x2^0$
$=1x8+1x4+0x2+1x1$
$=8+4+0+1$
$=13_{10}$
Example
Express the following octal number in base ten 1750
Solution
$1750_8=1x8^3+7x8^2+5x8^1+0x8^0$
$=1x512+7x64+5x8+0x1$
$=512+448+40+0$
$=1000_{10}$

CONVERTING DECIMAL NUMBERS TO ANY OTHER BASE NUMBER
Steps to be followed
a) Divide the decimal number by the new base continuously until you get zero.

b) Remember to write the remainders.

c) The answer comes from the remainders. Starting from the last remainder the go upwards.

Example

Express 47_{10} in base two

Solution

2 47 remainder
2 23 remainder1
2 11 remainder1
2 5 remainder1
2 2 remainder1
2 1remainder 0
2 0 remainder1

$4710 = 101111_2$

Example

Express 176_{10} in base eight

Solution

8 177
8 22 remainder 0
8 2 remainder 6
8 0remainder2

$176_{10} = 260_8$

ADDITION AND SUBTRACTION IN OTHER THAN TEN

The ordinary arithmetic operations, addition and subtraction are carried out in exactly the same manner as in base ten, but we have to take extra care when carrying, since this requires considering the number base.

Example
Add the octal numbers 174 and 233
Solution

$$\begin{array}{r} 174 \\ +233 \\ \hline 427_8 \end{array}$$

Example
Simplify $567_8 - 374_8$
Solution

$$\begin{array}{r} 567_8 \\ -374_8 \\ \hline 173_3 \end{array}$$

4. HEXADECIMAL NUMBERS (BASE SIXTEEN)

If you work with computer programming or computer engineering, you will encounter base sixteen or hexadecimal.
Now as mentioned before, decimal numbers does not have =one single solitary digit that represents the value of ten. Instead, we use two digits a 1 and a 0:10

Now in base ten, we had digits 0 through 9. In base eight, we had digits 0 through 7. In any base system, you will have digits 0 through one-less-than-base. Then in hexadecimal, we would need single solitary digits that stand for the values of ten, eleven, twelve, thirteen, fourteen and fifteen. But we do not. So instead, we use letters. That is counting in hexadecimal, the sixteen numerals are 0,1 ,2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F. In other words, A is ten, B is eleven, C is twelve, D is thirteen, E is fourteen and F is fifteen.

Example
Convert 357_{10} to the corresponding hexadecimal number
Solution
Divide repeatedly by 16, keeping the remainders as you go.

16|357
17|22 remainder 5
16|1 remainder 6
0 remainder 1

So $357_{10}=165_{16}$

Example
Convert 165_{16} to the corresponding decimal number.

Solution

$165_{16}=1x16^2+6x16^1+5x16^0$

$=1x256+6x16+5x1$

$=256+96+5$

$=357_{10}$

Example

Convert 63933_{10} to the corresponding hexadecimal number.

Solution

16|63933
16|3995 remainder 13-D
16|249 remainder 11-B
16|15 remainder 9
0 remainder 15-F

We can write the hexadecimal number as 15 9 11 13 because this would be confusing and imprecise. So we use letters for the digits that are otherwise too large, letting F stand in for fifteen, B stand in for eleven and D stand in for thirteen.

Thus $63933_{10}=F9BD_{16}$

Example

Convert *F9BD* to the decimal number

Solution

$F9BD= Fx16^3+9x16^2+Bx16^1+Dx16^0$

$=15x16^3+9x16^2+11x16+13x16^0$

$=63933_{10}$

5. NATURAL NUMBERS

The set of all counting numbers i.e. 1, 2, 3, 4 …is called the set of natural numbers, denoted $\aleph$ i.e. = 1,2,3,…
Two elements of $\aleph$ may be added or multiplied together to give another member of $\aleph$
Thus $2 \in \aleph$, $7 \in \aleph$ and $2+7=9 \in \aleph$, $2x7=14 \in \aleph$
However, it is not true that the difference between two members of $\in \aleph$ is a member of $\in \aleph$.Thus $5 \in \aleph$, $8 \in \aleph$ but $5-8=-3 \notin \aleph$

6. INTEGERS

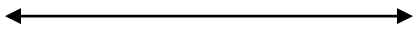
The set of integers is denoted by Z
$Z=\{...,-3,-2,-1,0,1,2,3,...\}$
Now if a, b $\in Z$ then $a+b \in Z$, $ab \in Z$ and –b $\in Z$ .However it is not always true that $\frac{a}{b} \in Z$ .For example $3 \in Z$, $5 \in Z$ but $\frac{3}{5} \notin Z$

7. RATIONAL NUMBERS

Rational numbers are detonated by Q .This is a set of all numbers $\frac{p}{q}$ where p, q $\in Z$ and q$\neq$0. In other ways, rational numbers are set of fractions.
NOTE: The denominator should not be equal to zero.
i.e. $\frac{a}{0}$ is meaningless.
Thou shall not divide by zero.

8. REAL NUMBERS

The set of numbers that corresponds to all points on a number line is called the set of real numbers or $\Re$. A graph of the real numbers is shown on a number line by shading all points.



All rational numbers are real numbers but there are points on a number line that do not correspond to rational numbers.

9. COMPLEX NUMBERS

Suppose, we want to solve the following equation $x^2+1=0$
i.e. $x^2=-1$

$x=\pm\sqrt{-1}$

Note that $\sqrt{-1}$ is not a real number
We denote i, such that $i^2=-1$ or $i=\sqrt{-1}$

DEFINITION

A complex number is a number Z of the form
$Z=x+iy$ where $x,y\in\Re$
The set of all such numbers is denoted by *C,* the set of complex numbers.

DEFINITION

Suppose $Z_1=x_1+iy_1$ and $Z_2=x_2+iy_2$ are complex numbers (where x_1, x_2, y_1,y_2). Then

(a) $Z_1+Z_2= (x_1+x_2)+i(y_1+y_2)$

(b) $Z_1Z_2=(x_1x_2-y_1y_2)+ix_1y_2+x_2y_1)$

Example
Let Z=2 +i and Z=3-4i , find Z_1+Z2 and Z_1Z_2

Solutions
(i) $Z_1+Z_2=(2+3)+i(1-4)$
$=5-3i$
(ii) $Z_1Z_2=(2x3-1x-4)+i(2x-4+3x1)$
$=(6+4)+i(-8+3)$
$=10-5i$

REAL AND IMAGINARY PARTS
Let Z = x+ yi where x, y$\in \Re$ then the real part of Z is x, denoted by R(Z) = x and the imaginary part of Z is y, denoted I(Z)= y.
Let *$Z_1= x_1+y_1i$ and $Z_2 = x_2+ y_2i$ (*where *$x_1, x_2, y_1, y_2\in \Re$)* Then $Z_1= Z_2$ if and only if $x_1= x_2$ and $y_1= y_2$

Example
Find all real x and y such that (3+2i)x+(1-4i)y = 7
Solution
(3+2i)x+(1-4i)y=7
Removing the brackets, we get
3x+2xi+y-4yi=7 or 3x+y+(2x-4y)I =7
Equating the real parts and imaginary parts, we have,
3x+y =7
2x-4y=0
Solving these simultaneously we obtain x= 2 and y=1

TUTORIAL1: NUMBER SYSTEM

1. Convert (a) 416_{10} to base 8
   (b) 100101000_2 to decimal number.
   (c) 225_{10} to base 16
2. Work out the following
   (a) $22_{six} + 45_{six}$
   (b) $45_9 - 28_9$
3. Express the following in the form x+iy
   (a) (2+i)-(3-2i)
   (b) (5+i)+(2-3i)
   (c) (3-2i)(6+i)
4. Evaluate the following
   (a) i^3
   (b) i^4
   (c) i^{-2}
5. Find all real x and y such that:
   (a) 3x+4i=6+iy
   (b) (1+2i)x+(3-i)y=2+i

TOPIC 2: COMPUTATION AND ACURACY

OBJECTIVES

By the end of this topic, students should be able to;

(a) Define accuracy and error.
(b) Round numbers to the given significant figures.
(c) Round off numbers.

ACCURACY

This is the degree to which a given quantity is correct and free from error. For example, a value quantity specified as 100 ± 1 has an accuracy of ± 1 (mean its true values can fall in the range 99-100.

ERROR

This is the difference between the measured value x_0 and its actual value x given by $\Delta x = x_0 - x$
The term is also used to mean a mistake.

SIGNIFICANT FIGURES

In any number, the first significant figure is the first digit which is not a 0.
To round to any number of significant figures:
(i). Look at the first unwanted digit.
(ii) If it is 5, 6, 7, 8 or 9 add one to the last wanted digit. And if it is 0, 1, 2, 3, 4 ignore it.
(iii) Be careful to keep the number about the right size and fill in the spaces with zeros.

Example
Write 34597 (i) correct to 1 s.f
(ii) Correct to 3 s.f
(iii) Correct to 3 s.f
(iv) Correct to 4 s.f
Solutions
34597=30000, correct to 1 s.f
=35000, correct to 2 s.f
=34600, correct to 3s.f
=34600, correct to 4 s.f

DECIMAL NUMBERS

To determine how many decimal places are given in a number, we start counting at the decimal marker (apoint or a comma in some books).

For example 0.01254=0.01 correct to 2 decimal places
=0.013 correct to 3 decimal places
=0.0125 correct to 4 decimal places

Notice that we count zeros after the decimal marker.

ROUNDING NUMBERS

The process of dropping unnecessary extra digits in a number is known as rounding off. The rule is if the digit to be removed is equal or greater than 5, then the immediately preceding digit is increased by 1.

Numbers can also be rounded to the nearest 10, 100 or 1000.

Example
Round the number 63 to the nearest 10.
Solution
63 is near to 60 than 70. Therefore, it is rounded to 60 to the nearest 10.

Example
Round the number 683 to the nearest 100
Solution
683 is between 600 and 700 but it is near to 700. Therefore, it is rounded to 700 to the nearest 100.

Example
Round the number 4500 to the nearest 1000
Solution
4500 is halfway between 4000 and5000. Therefore, it is rounded to5000 to the nearest 1000.

TUTORIAL 2: COMPUTATION

1. Evaluate the following quantities shown, correct to 4 significant figures:
   (a) 3.249^2
   (b) $\sqrt{0.0256}$
   (c) Log 41.62
1. Evaluate correct to 3 decimal places:

(a) $\frac{29.12}{(5.81)^2 - (2.96)^2}$

(b) $\sqrt{53.98} - \sqrt{21.78}$

TOPIC 3: FORMULA

OBJECTIVES

By the end of this topic, students should be able to;

(a) Evaluate a given formula by substituting values for the symbols or letters.

(b) Change subject of formula.

A formula is an equation in which letters represent quantities. When we know the values of all the unknowns except one we can substitute these values in the formulas and find the value of the unknown.

Example

The formula $F=\frac{9c}{5}+32$ shows the relationship between temperature in degrees Fahrenheit (F) and degree centigrade (c) .Find F when c=40.

Solution

$F=\frac{9c}{5}+32$

When c=40, we have

$F=\frac{9x40}{5}+32$

$=72x32$

$=104$

Therefore, the temperature is 104^0F

The area of a triangle of base length b and height h is given by the formula $A=\frac{1}{2}bh$ where h and b are in the same units. In this formula, A is the subject of the formula.

CHANGE OF SUBJECT OF FORMULA

It is often necessary to change the subject of a formula. The process of making anew subject of a formula is known as transposition of a formula. To do this, think of the formula as an equation. Solve the equation for the letter, which is to become the subject.

STEPS TO BE FOLLOWED WHEN CHANGING THE SUBJECT OF THEFORMULAE

(a) Begin by clearing fractions, brackets and roots sign.

(b) Rearrange the formula so that all the terms which contain the new subjects are on one side of the equals sign and the rest on the other.

(c) If more than one term contains the subject take it outside a bracket.

(d) Divide both sides by the bracket, then simplify as far as possible

Example

The period of oscillation, T seconds, of a pendulum is given by the formula.

$$T=2\pi\sqrt{\frac{k^2+h^2}{gh}}$$

Where h is the distance in metres of the centre of gravity of the pendulum from the point of support and k and g are positive constants. Express k in terms of T, h and g.

Solution

$$T=2\pi\sqrt{\frac{k^2+h^2}{gh}}$$

Squaring both sides, we have

$$T^2=4\pi\left(\frac{k^2+h^2}{gh}\right)$$

Multiplying both sides by gh, we get

$$-ghT^2=4\pi^2(k^2+h^2)$$

dividing each term by $4\pi^2$, we obtain

$$\frac{ghT^2}{4\pi^2}=k^2+h^2$$

Subtracting h^2 from each side, we have

$\frac{ghT^2}{4\pi^2} - h^2 = k^2$

Taking positive square root since k is known to be positive

$\sqrt{\frac{ghT^2}{4\pi^2} - h^2} = k$

Example

$T^2 = \frac{ax}{a+x}$

(a) Make x the subject of the formula.

(b) Calculate the value of x if a=6 and t=2

Solution

(a) $T^2 = \frac{ax}{a+x}$

Multiplying both sides by a+x, we have

$T^2(a+x) = ax$

Removing brackets, we get

$aT^2 + xT^2 = ax$

collecting like terms with x

$aT^2 = ax - xT^2$

taking out x as a common factor

$aT^2 = x(a - T^2)$

dividing each term by $(a - T^2)$, we have

$\frac{aT^2}{a - T^2} = x$

(b) $x = \frac{6x2^2}{6 - 2^2}$

$= \frac{24}{2}$

$= 12$

TUTORIAL 3: FORMULAE

1. Make b the subject of the formula

$$a = \frac{x - y}{\sqrt{bd + be}}$$

2. The power P watts dissipated in an electrical circuit may be expressed by the formula $P = {}^{v^2}\!/_{R}$. Evaluate the power, correct to 3 significant figures, given that V= 17.48v and R= 36.12Ω.
3. A formula for kinetic energy is

$$K = \frac{1}{2}mv^2$$

   Transpose the formula to make v the subject.

4. The velocity, v, of water in a pipe appears in the formula $h = \frac{0.03Lv^2}{2dg}$. Express v as the subject of the formula and evaluate v when h= 0.712, L= 150, d= 0.30 and *g*= 9.81.
5. Transpose the formula

$$y = \frac{pq^2}{r + q^2} - t$$

to make q the subject.

TOPIC 4: RATIO, PROPORTION AND PERCENTAGES

OBJECTIVES

By the end of this topic, students should be able to;

(a) Differentiate a ratio from a proportion.
(b) Simplify a given ratio in its simplified form.
(c) Express a given ratio as a percentage.

RATIOS

A ratio is a comparison of two numbers by division.
The ratio that compares 44 to 77 can be written as follows: 44 to 77, 44:77, 44out of 77, $\frac{44}{77}$

Example

Simplify these ratios if possible

(a) 9:6
(b) 20:15
(c) 9:2

Solutions

(a) The common factor of 9 and 6 is 3. So dividing each term by 3 we get

$\frac{3}{2}$

Therefore the ratio is 3:2

(b) The common factor of 20 and 15 is 5. So each term should be divided by 5

i.e. $\frac{20}{15} = \frac{4}{3}$

Therefore the simplified ratio is 4:3.

(c) There is no common factor of 9 and 2. So 9:2 can not be simplified any further.

PROPORTIONS

A proportion is a statement that two ratios are equal. There are two ways to determine if a pair of ratios forms a proportion or not.

1. The simplified ratios of proportion are equal
2. The cross product of proportion are equal

Example

Tell whether each pair of ratios forms a proportion or not.

1. $\frac{8}{12}, \frac{10}{15}$
2. $\frac{3}{2}, \frac{4}{6}$

Solutions

1. $\frac{8}{12} = \frac{2}{3}$ and $\frac{10}{15} = \frac{2}{3}$ since these ratios are equivalent then they form a proportion

2. $\frac{3}{2}$ can not be simplified further and $\frac{4}{6} = \frac{2}{3}$ since the ratios are not equivalent then they do not form a proportion.

PERCENTAGES

A percentage is a ratio that compares a number to 100. The symbol for percent is %. Percentages are another way of writing fractions. For example:Mark has $\frac{68}{100}$ in Mathematics test. This means that he scored 68 marks out of 100. The fraction $\frac{68}{100}$ can be written as 68%. This only works when the denominator is 100.

Example

Express $\frac{32}{50}$ as a percentage.

Solution

The denominator is not equal to 100. So we need to multiply both the numerator and denominator by 2 $\frac{32}{50}$ becomes $\frac{64}{100}$. Therefore $\frac{32}{50}$ is the same as 64%

In general, to change a fraction to a percentage you multiply by 100. Then add % to the answer.

Example

In a town of 4280 inhabitants, there were 56 births during 2003. Find the percentage birth rate.

Solution

The fraction is $\frac{56}{4280}$. Changing it to a percentage, we have

$\frac{56}{4280}$ x100=1.31% (to 3 s.f.)

We calculate the percentage change by using the starting value. We use the following formula in order to compute the percentage change

$$\text{Percentage change} = \frac{\text{actual change}}{\text{Starting value}} \times 100\%$$

Example

Find the percentage decrease when a shirt is reduced from K250 to K170.

Solution

The actual decrease= K250-K170

=K80

The starting price was K250.

Therefore Percentage decrease= $\frac{80}{250}$ x 100%

=32%

Example

Emily buys eggs at K15 each and sells for K18. Find her gain percentage.

Solution

Actual profit=K18-K15

=K3

Starting value (cost price)=K15

Gain Percent= $\frac{k3}{k15}$ x100%

=20%.

TOPIC 5: EXPONENTS AND LOGARITHMS

OBJECTIVES

By the end of this topic, students should be able to;

(a) State laws of exponents and logarithms.

(b) Simplify terms using laws of exponents and logarithms.

(c) Solve indicial and logarithmic equations.

EXPONENTS

Exponent is another word for index or power.

An exponential function is one where the variable is an index.

Some examples of exponential functions are 2^x, 3^{-x}, 10^{x+1}

LAWS OF EXPONENTS

1. For any real numbers pand q, $x^p.x^q=x^{p+q}$

2. $\frac{x^p}{x^q}=x^{p-q}$

3. $(x^p)^q=x^{p.q}$

4. For any real number p, $x^{-p}=\frac{1}{x^p}$

5. For any real number $a\neq 0$,$a^0=1$

6. For any real number p, $a^{\frac{1}{p}} = \sqrt[p]{a}$

7. For any real numbers p and q $a^{\frac{p}{q}} = \left(a^{\frac{1}{p}}\right)^p = \left(\sqrt[q]{a}\right)^p$

Example

Simplify the following

(a) $\frac{3^5 x 3^6}{3^4}$

(b) $\frac{18x^2y^5}{3x^{4y}}$

(c) $\left(3x^5\right)^2$

(d) $(64)^{\frac{-1}{3}}$

Solutions

(a) $\frac{3^5 x 3^6}{3^4} = \frac{3^{5+6}}{3^4}$

$= \frac{3^{11}}{3^4}$

$= 3^{11-4}$

$= 3^7$

(b) $\frac{18x^2y^5}{3x^{4y}} = \frac{18x^{2-4}y^{5-1}}{3}$

$= 6x^{-2}y^4$

(c) $\left(3x^5\right)^2 = 3^2.x^{5x2}$

$= 9x^{10}$

(d) $(64)^{\frac{-1}{3}} = \frac{1}{64^{\frac{1}{3}}}$

$= \frac{1}{\sqrt[3]{64}}$

$= \frac{1}{4}$

LOGARITHMS

For any positive number $b \neq 1$ the logarithm function with base b, written $\log_b x$
Is defined by $y = \log_b x$ if and only if $x=b^y$ or if is the power to which the base must be raised in order to equal the number.

LAWS OF LOGARITHMS
The following are the laws of logarithms
1.$\log_a xy = \log_a x + \log_a y$
2.$\log_a \frac{x}{y} = \log_a x - \log_a y$
3.$\log_a x^b = b\log_a x$

PROPERTIES OF LOGARITHMS
1. $\log_a 1 = 0$
2. $\log_a a = 1$

Example
Simplify the following as far as possible without using the calculator or four figure tables

(a)$\log_a \sqrt{a}$
(b)$\log_2 14 - \log_2 7$
(c)$\log 8 + \log 5$
(d)$\frac{\log 16}{\log 2}$

Solutions

(a)$\log_a \sqrt{a} = \log_a a^{\frac{1}{2}}$

$= \frac{1}{2}\log_a a$

$= \frac{1}{2}x1$

$= \frac{1}{2}$

(b) $\log_2 14 - \log_2 7 = \log_2 \frac{14}{7}$

$= \log_2 2$

$= 1$

(c) $\log 8 + \log 5 = \log 8 \times 5$

$= \log 40$

(d) $\frac{\log 16}{\log 2} = \frac{\log 2^4}{\log 2}$

$= 4 \frac{\log 2}{\log 2}$

$= 4 \times 1$

$= 4$

Example

Given that $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$ and $\log_{10} 7 = 0.8451$. Evaluate the following.

(a) $\log_{10} 5$

(b) $\log_{10} 42$

(c) $\log_{10} 8$

Solutions

(a) $\log_{10} 5 = \log_{10} \frac{10}{2}$

$= \log_{10} 10 - \log_{10} 2$

$= 1.0000 - 0.3010$

$= 0.699$

(b) $\log_{10} 42 = \log_{10} 7 \times 2 \times 3$

$= \log_{10} 7 + \log_{10} 3 + \log_{10} 2$

$=0.8451+0.3010+0.4771$

$=1.6232$

(c) $\log_{10}8=\log_{10}2x2x2$

$=\log_{10}2+\log_{10}2+\log_{10}2$

$=0.3010+0.3010+0.3010$

$=0.9030$

The most common bases used for logarithms are 10 and e. In value $e\approx 2.7182818$ and logarithms taken to this base are called *Natural logarithms or Naperian logarithms*. To avoid confusion with other bases, $\log_e x$ is written as $\ln x$. $\ln x$ means $\log_e x$ so $\ln x=y \Leftrightarrow e^y=x$.

The laws used for working with logarithms to a general base, apply equally well to natural logarithms.

$\ln a+\ln b=\ln ab$

$\ln a-\ln b=\ln \frac{a}{b}$

$\ln a^n=n\ln a$

Example

Solve the following equations

(a) $2^x=64$

(b) $2^{2x+3}=32$

(c) $\log_8 64=m$

(d) $2\log_2 x-2\log_2 2=3$

(e) $3^x=20$

(f) $4^{3x+1}=79$

Solutions

(a) $2^x=64$

$2^x=2^6$ expressing 64 in index form whose base is 2

$x=6$ equating the powers

(b) $2^{2x+3}=32$

$2^{2x+3}=2^5$

$2x+3=5$

$2x=2$

$x=1$

(c) $\log_8 64=m$

$8^m=64$

$8^m=8^2$

$m=2$

(d) $2\log_2 x-2\log_2 2=3$

$\log_2 x^2-\log_2 2^2=3$

$\log_2 \frac{x^2}{4}=3$

$2^3=\frac{x^2}{4}$

$4x8=x^2$

$32=x^2$

$\sqrt{32}=x$

(e) $3^x=20$

Taking ln of both sides, we have

$\ln 3^x=\ln 20$

$x\ln 3=\ln 20$

$x=\frac{\ln 20}{\ln 3}$

$x=\frac{2.995732274}{1.098612289}$

$x=2.73$ (3 s.f)

(f) $4^{3x+1}=79$

Taking ln of both sides, we have

$\ln 4^{3x+1}=79$

$(3x+1)\ln 4=\ln 79$

$3x+1=\frac{\ln 79}{\ln 4}$

$3x=\frac{4.369447852}{1.386294361}-1$

$x=\frac{2.151890374}{3}$

$x=0.717$ (3 s.f)

TUTORIAL5: EXPONENTS AND LOGARITHMS

1. Suppose $\log_3 4=o.6309$ and $\log_3 5=1.46498$. Find
   (a) $\log_3 4$
   (b) $\log_3 6$
   (c) $\log_3 10$
   (d) $\log_3 80$
2. Solve the following equations
   (a) $\log_{10} x=-2$
   (b) $\log x-\log(2x-1)=1$
   (c) $8^{x-1}=16$
   (d) $2^x=0.25$
3. Evaluate the following, without using the calculators or four figure tables
   (a) $3\log_{10}2+\log_{10}20-\log_{10}1.6$
   (b) $\frac{\log\sqrt{5}}{\log 5}$
   (c) $\log_3 21-\log_3 7$
   (d) $2x^{\frac{1}{4}}(4x^{\frac{1}{2}})^{\frac{1}{2}}$
   (e) $(8^{\frac{1}{2}}\times 5^{\frac{3}{2}})^{\frac{2}{3}}$

TOPIC 6: FUNCTIONS AND GRAPHS

OBJECTIVES

By the end of this topic, students should be able to;

(a) Differentiate between Cartesian and polar co-ordinates.

(b) States types of functions.

(c) Identify linear and quadratic, exponential and logarithmic graphs.

(d) Calculate gradient of a straight line.
(e) Find equations of straight line and circle.
(f) State the condition of parallel lines and perpendicular lines.
(g) Calculate distance between two points.

CO-ORDINATE SYSTEMS

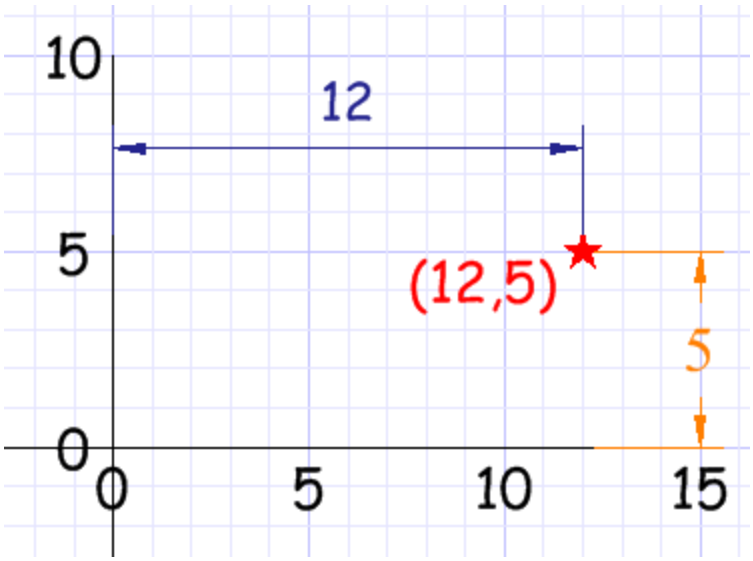
Coordinates are numbers, which describe the location of points in a plane or in space. For example, the height above sea level is a coordinate, which is useful for describing points near the surface of the earth. A coordinate system, in a plane or in space, is a systematic method of assigning a pair or a triple of numbers to each point in the plane or in space (respectively) which describe its position uniquely. For example, the triple consisting of latitude, longitude and altitude (height above sea level) define a coordinate system near to the surface of the earth.

In the two-dimensional Cartesian coordinate system, a point P in the xy-plane is represented by a pair of numbers (x,y).

x is the signed distance from the y-axis to the point P, and

y is the signed distance from the x-axis to the point P

Using Cartesian Coordinates you mark a point by how far along and how far up it is:

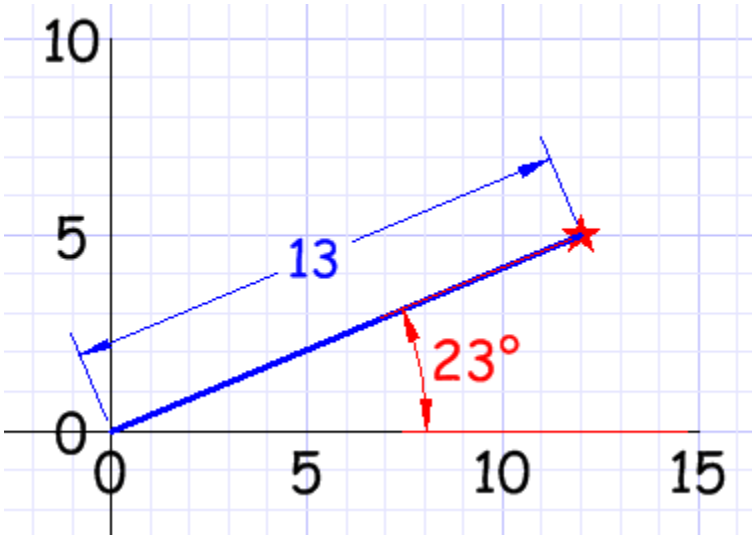


The polar coordinate system is a two-dimensional coordinate system in which each point on a plane is determined by an angle and a distance. The polar coordinate system is especially useful in situations where the relationship between two points is most easily expressed in terms of angles and distance; in the more familiar Cartesian or rectangular coordinate system, such a relationship can only be found through trigonometric formulation.

As the coordinate system is two-dimensional, each point is determined by two polar coordinates: the radial coordinate and the angular coordinate. The radial coordinate (usually denoted as *r*) denotes the point's distance from a central point known as the *pole* (equivalent to the *origin* in the Cartesian system). The angular coordinate (also known as the polar angle or the azimuth angle, and usually denoted by θ or *t*) denotes the positive or anti-clockwise (counterclockwise) angle

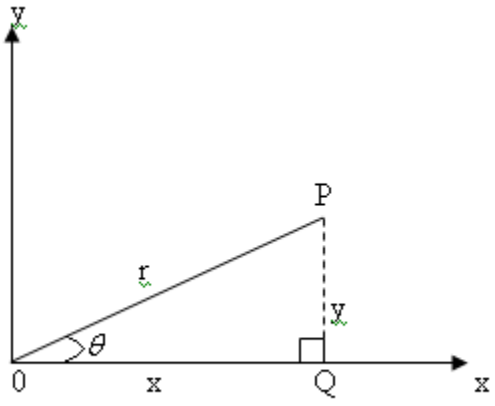
required to reach the point from the 0° ray or *polar axis* (which is equivalent to the positive x-axis in the Cartesian coordinate plane.

Using Polar Coordinates you mark a point by how far away, and what angle it is:



CHANGING FROM CARTESIAN INTO POLAR COORDINATES

Consider the following right-angle triangle;



r can be obtained from Pythagoras theorem

i.e. $r^2 = x^2 + y^2$ *or* $r = \sqrt{x^2 + y^2}$ (1)

Recall that $\tan\theta = \frac{y}{x}$, making θ the subject of the formula, we get

$\theta = \tan^{-} \frac{y}{x}$ (2)

Formulae (1) and (2) are needed to change from Cartesian to polar coordinates.

Example

Change the Cartesian (3, 4) into polar coordinates.

Solution

Here x = 3 and y =4 , but $r = \sqrt{x^2 + y^2}$

$r = \sqrt{3^2 + 4^2}$

$r = \sqrt{25}$

$r = 5$

And $\theta = \tan^{-} \frac{y}{x}$

$\theta = \tan^{-} \frac{4}{3}$

$\theta = 53.13^0$

Therefore, (3, 4) in polar is (5, 53.13^0)

Example

Express in polar coordinates the position (-4, 3)

Solution

Here x = -4 and y = 3

But $r = \sqrt{x^2 + y^2}$

$r = \sqrt{-4^2 + 3^2}$

$r = \sqrt{25}$

$r = 5$

And $\theta = \tan^{-} \frac{y}{x}$

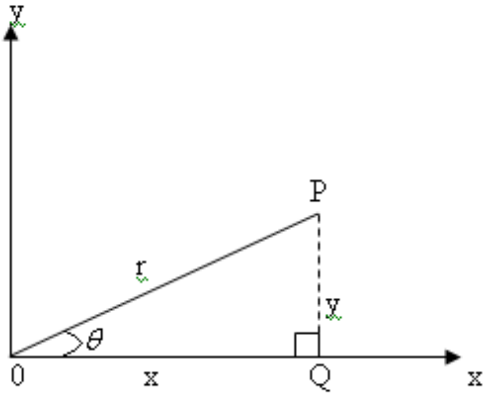
$\theta = \tan^{-} \frac{3}{-4}$

$\theta = 143.13^{0}$

Therefore, (-4, 3) in polar is (5, 143.13^{0})

CHANGING FROM POLAR INTO CARTESIAN COORDINATES

Consider the following triangle;



$$Cos\theta = \frac{x}{r} \Rightarrow x = r\cos\theta \qquad (1)$$

$$Sin\theta = \frac{y}{r} \Rightarrow y = r\sin\theta \qquad (2)$$

These above formulae are used to change from polar to Cartesian coordinates.

Example

Change (4, 32^{0}) into Cartesian coordinates.

Solution

$r = 4 \quad and \quad \theta = 32^{0}$

Here $x = 4\cos 32^{0} = 3.39$

$y = 4\sin 32^{0} = 2.12$

Therefore, (4, 32^0) in Cartesian is (3.39, 2.12)

RELATIONS AND FUNCTIONS

Let A and B be sets, a function is a rule which assigns to each element of A a unique element of B.

Such a "rule" can be specified in a variety of ways:

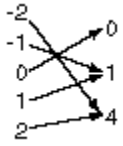
As a set of ordered pairs: for example,

$\{(-2,4),(-1,1),(0,0),(1,1),(2,4),...\}$

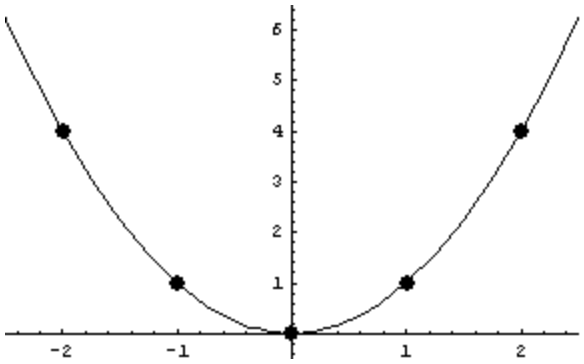
As a table of values:

x	y
-2	4
-1	1
0	0
1	1
2	4
.	.
.	.

As an arrow diagram:



As a 2-dimensional plot:



As an equation: $y=x^2$.

If *f* is a function from A to B, we write;

$f : A \to B \quad or \quad A \xrightarrow{f} B$

If f assigns $a \in A$ the element $b \in B$ then b is called the image of a under *f.*

Example

If $f(x) = x^3 + 4x - 3$. find f(1), f(-1) and f(0).

Solutions

(d) If $f(x) = x^3 + 4x - 3$, then

$f(1) = 1^3 + 4 \times 1 - 3$

$f(1) = 2$

(e) If $f(x) = x^3 + 4x - 3$, then

$f(-1) = (-1)^3 + 4 \times (-1) - 3$

$f(-1) = -8$

(f) If $f(x) = x^3 + 4x - 3$, then

$f(0) = 0 + 0 - 3$

$f(0) = -3$

TYPES OF FUNCTIONS

There are different types of functions. Here, we are going to briefly describe the following types;

(a).ALGEBRAIC FUNCTIONS

These are names for functions of first, second and third order polynomial functions, respectively. What this means is that the highest order of x (the variable) in the function is 1, 2 or 3.
The generalized form for a linear function (1 is highest power):
$f(x) = ax+b$, where a and b are constants, and a is not equal to 0
The generalized form for a quadratic function (2 is highest power):
$f(x) = ax^2+bx+c$, where a, b and c are constants, and a is not equal to 0
The generalized form for a cubic function (3 is highest power):
$f(x) = ax^3+bx^2+cx+d$, where a, b, c and d are constants, and a is not equal to 0

(b). TRANSCENDENTIAL FUNCTIONS

A transcendental function is a function that does not satisfy a polynomial equation whose coefficients are themselves polynomials, in contrast to an algebraic function, which does satisfy such an equation. In other words, a transcendental function is a function that "transcends" algebra in the sense that it cannot be expressed in terms of the algebraic operations of addition, multiplication, and root extraction. Examples of transcendental functions include the exponential function, the logarithmic, and the trigonometric functions.

ONE-ONE FUNCTIONS

A function *f* from X to Y is one-one function if, whenever $a \neq b$ in X then $f(a) \neq f(b)$ in Y.

Example

(a) If *f(x)* = 3x +2, where x is real, prove that f is one-one.

(b) If *g(x)* = x^2 +5, where x is real, prove that *g* is not one-one.

Solutions

(a) If $a \neq b$, then $3a \neq 3b$ and hence $3a+2 \neq 3b+2$, or $f(a) \neq f(b)$.

Hence *f* is One-one by defition.

(b) $-1 \neq 1$, but $g(-1) = g(1) = 6$. Therefore *g* is not one-one

(c)INVERSE FUNCTIONS

Suppose $f : A \to B$. Then, provided it exists, the inverse of *f* is the function

$f^{-1} : B \to A$ such that if $f : x \to y, \quad then \quad f^{-1} : y \to x$.

Example

Suppose f(x)= 2x-3. Find $f^{-1}(x)$.

Solution

Let y= $f^{-1}(x)$, then x= f(y)=2y-3

Now 2y=x+3

$y = \frac{x+3}{2}$ = $f^{-1}(x)$

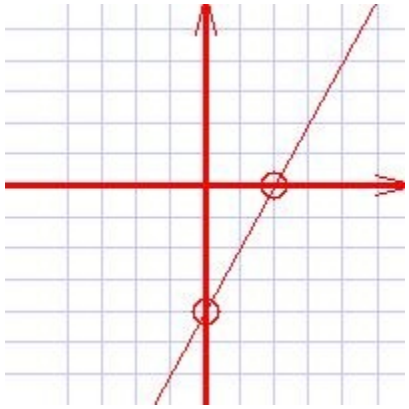
GRAPHS OF FUNCTIONS

(a) *LINEAR GRAPHS*

They are represented by straight lines. They are given by equations such as

y=3x-6

Their graphs are always represented by a straight line. You have to use a ruler to plot them. They look like this

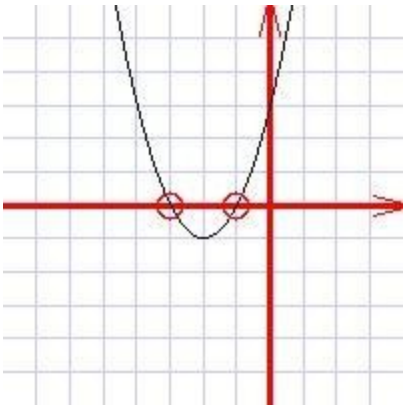


(b) *QUADRATIC GRAPHS*

They are represented by parabolas, which are curvy (not straight) lines sort of like cow horns. By the way, if you throw a rock, its line of fall is represented by a parabola (with horns obviously pointing down). Try it where windows and people are not in danger. Parabolas represent quadratic equations such as

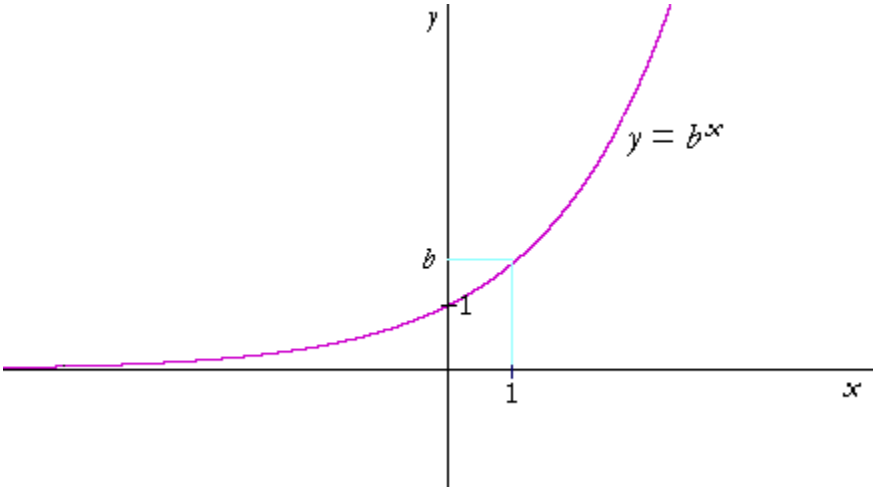
$y=x^2+4x+3$

Because parabolas are curvy, you cannot use a ruler to plot them. They look like this:



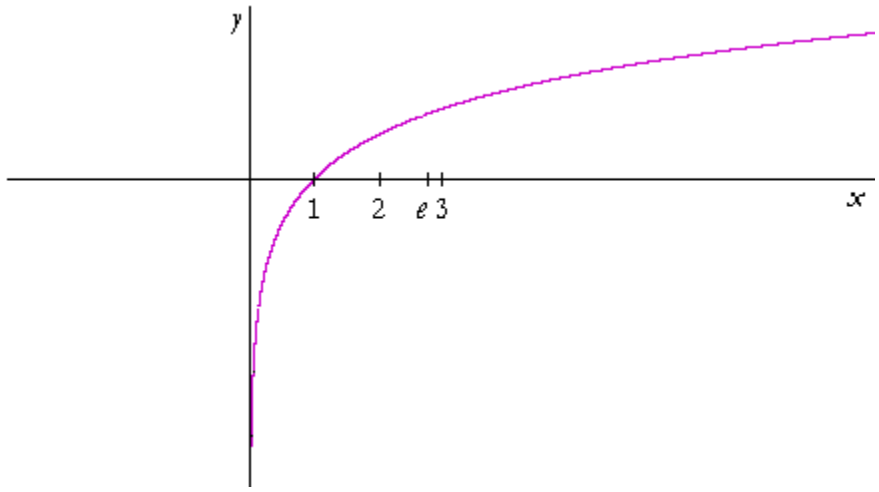
(c) *EXPONENTIAL GRAPHS*

As a function of the real variable x, the graph of $y=e^x$ is always positive (above the x axis) and increasing (viewed left-to-right). It never touches the x axis, although it gets arbitrarily close to it (thus, the x axis is a horizontal asymptote to the graph). Exponential graphs look like this;



(d) *LOGARITHMIC GRAPHS*

Since logarithm is the inverse of exponential then its graph is also the inverse of exponential graph. In general the graph of $f(x) = \log_e x$ looks like the following graph;



STRAIGHT LINES

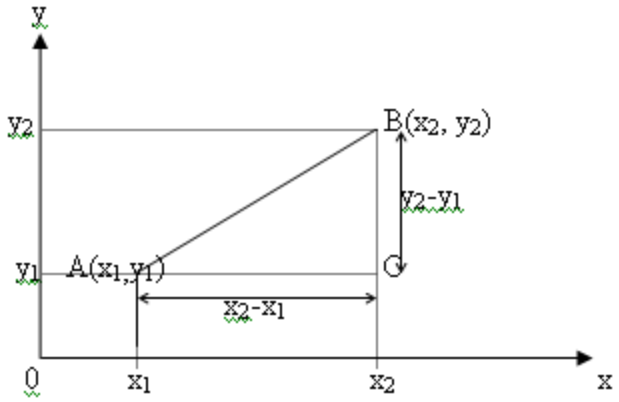
SLOPE/GRADIENT

The gradient is the ratio of the vertical displacement to the horizontal displacement.

The gradient can be calculated algebraically using the rule

$$\text{Gradient} = \frac{\text{change in y-coordinate}}{\text{Change in x-coordinate}}$$

Consider the following graph



The gradient of the straight line joining $A(x_1,y_1)$ and $B(x_2,y_2)$ is given by

$$\text{Gradient, } m=\frac{y_2-y_1}{x_2-x}$$

Example

Find the gradient of the straight line joining M(2, 3) and N(5, 7).

Solution

$$M=\frac{y_2-y_1}{x_2-x_1}$$

$$M=\frac{7-3}{5-2}$$

$$M=\frac{4}{3}$$

Example

If the gradient of the straight line joining P(a, 3) and Q(2, 8), find the value of a.

Solution

$m=\frac{y_2-y_1}{x_2-x_1}$

$5=\frac{8-3}{2-a}$

$5=\frac{5}{2-a}$

$10-5a=5$

$5a=5$

$a=1$

EQUATION OF STRAIGHT LINE

Recall that $m=\frac{y_2-y_1}{x_2-x_1}$ and this equation can be rearranged as $y_2-y_1=m(x_2-x_1)$. This equation can also be written as $(y-b)=m(x-a)$ where (a, b) is the given point and m is the gradient.

Example

Find the equation of straight line through (2, 3) with slope $\frac{1}{2}$.

Solution

Given: $m=\frac{1}{2}$ and (a,b)=(2, 3)

But $y-b=m(x-a)$

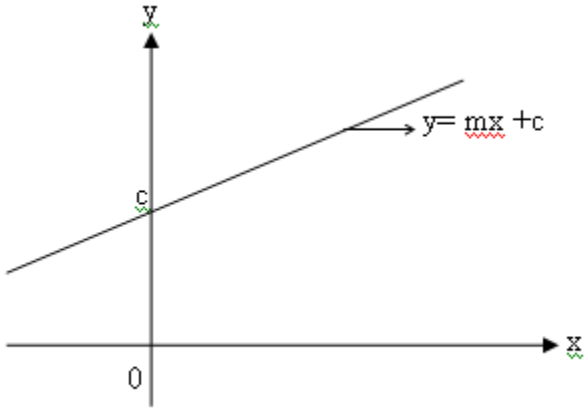
$y-3=\frac{1}{2}(x-2)$

$y=\frac{1}{2}x-1+3$

$y=\frac{1}{2}x+2$

Example

Find the equation of a line with gradient m, passing through the point (0, c)



Solution

Equation of a straight line is y-b=m(x-a)

y-c=m(x-0)

$$y=mx+c$$

In general, if y is the subject of the equation of the straight line then the coefficient of x is the gradient and the constant tem is the y-intercept.

Example

For each of the following equations, state the gradient of the line.

(a) y=6x+3

(b) y-5x=2.5

Solutions

(a) y=6x+3

6 is the gradient

(b) y-5x=2.5

y=5x+2.5

So 5 is the gradient.

Example
Find the equation of the line through (2, 6) and (-5, 3)
Solution
There is no gradient, so the first step is to find it.

Recall that $m=\frac{y_2-y_1}{x_2-x_1}$

$=\frac{3-6}{-5-2}$

$=\frac{-3}{-7}$

$=\frac{3}{7}$

Then choose one point and use the formula of the equation of straight line

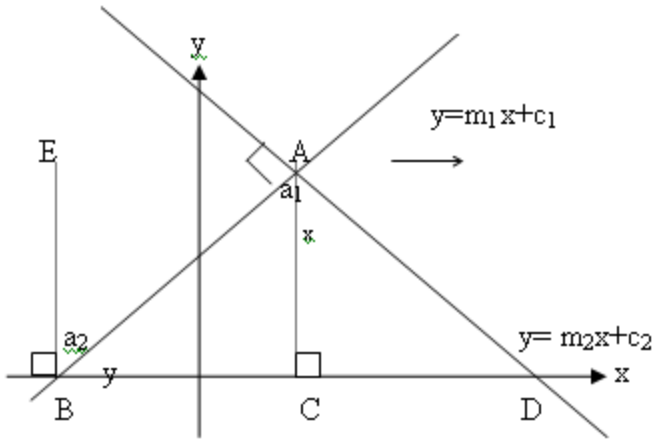
$y-6=\frac{3}{7}(x-2)$

$y-6=\frac{3}{7}x-\frac{6}{7}$

$y=\frac{3}{7}x-\frac{6}{7}+6$

$y=\frac{3}{7}x+\frac{36}{7}$

CONDITION OF PERPENDICULARITY
Consider the following graph;



Let us suppose $m_1>0$ and $m_2<0$

Triangle ABC is similar to triangle DAC,

i.e. $a_1=a_2$(alternate angles AC//BE)

$a_1+x=90$ (alternate angles on a straight line)

$a_2+y=90$ (given)

thus $a_1+x=a_2+y$

$x=y$ since $a_1=a_2$(above)

so in triangle ABC and triangle DAC

angle ACB=angle DCA=90(given)

$y=x$ (proved above)

therefore triangle ABC is congruent to triangle DAC (A.A)

hence $\frac{AC}{BC} = \frac{CD}{AC}$

but $\frac{AC}{BC}=m_1$ and $\frac{AC}{CD}= -m_2$

but $\frac{CD}{AC}=\frac{AC}{BC}=m_1$

then $\frac{1}{-m_2}=m_1$, so $-1=m_1m_2$

We have shown that provided $m_1, m_2 \neq 0$ then $y_1=m_1x_1+c_1$ is perpendicular to $y_2=m_2x_2+c_2$ if and only if $m_1m_2=-1$

i.e. the product of the gradient of any pair of perpendicular lines is always -1.

Example
Find the equation of the through (2, -3) perpendicular to y=2x-5.
Solution
The gradient of y=2x-5 is 2, so the gradient of the perpendicular is $-\frac{1}{2}$ (m_1m_2)

But the equation of the straight line is y-b=m(x-a)

$$y- -3=-\frac{1}{2}(x-2)$$

$$y+3=\frac{-1}{2}x+1$$

$$y=\frac{-1}{2}x-2$$

PARALLEL LINES

In general, if $y_1=m_1x+c_1$ and $y_2=m_2x+c_2$ are equations of the lines then their graphs are parallel if and only if $m_1=m_2$

Example
Show that the following pairs of lines are parallel:
y=2(3x+1), 2y-12x+6=0
Solution
Equation 1 is y=2(3x+1) becomes y=6x+2 after removing brackets
Equation 2 is 2y-12x+6=0

$$2y=12x-6$$

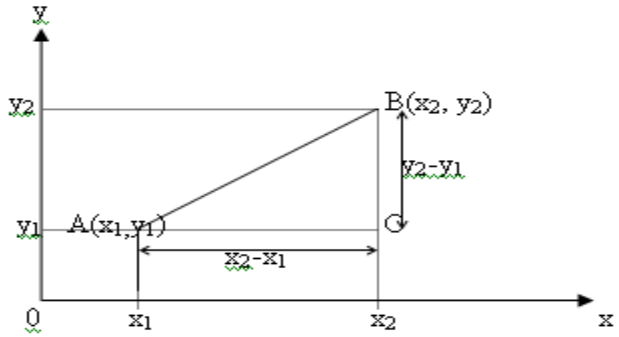
Y=6x-3

So its gradient is 6

Since both equations have the same gradients, then the lines are parallel.

DISTANCE BETWEEN TWO POINTS

Consider the following graph



Step 1: Use the coordinates of A and B to find the lengths of the shorter side

i.e. $BC=y_2-y_1$

$AC=x_2-x_1$

Step 2: State Pythagoras theorem and use it to find the length of the line joining A and B

i.e. $AB^2= BC^2+AC^2$

$AB^2=(x_2-x_1)^2 + (y_2-y_1)^2$

$AB=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Example

Find the distance between A=(2, 3) and B=(4, -1)

Solution

$AB=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$AB=\sqrt{(4-2)^2+(-1-3)^2}$

$AB=\sqrt{(2)^2+(-4)^2}$

$AB=\sqrt{4+16}$

$AB=\sqrt{20}$ or $2\sqrt{5}$

Example

Let A= (1, 4), B=(2, 2) and C=(-2, 0). Show that triangle ABC is a right-angled .

Solution

$AB^2=(x_2-x_1)^2+(y_2+y_1)^2$

$AB^2=(2-1)^2+(2-4)^2$

$AB^2=1^2+(-2)^2$

$AB^2=1+4$

$AB^2=5$

$AC^2=(-2-1)^2+(0-4)^2$

$AC^2=9+16$

$AC^2=25$

$BC^2=(-2-2)^2+(0-2)^2$

$BC^2=16+4$

$BC^2=20$

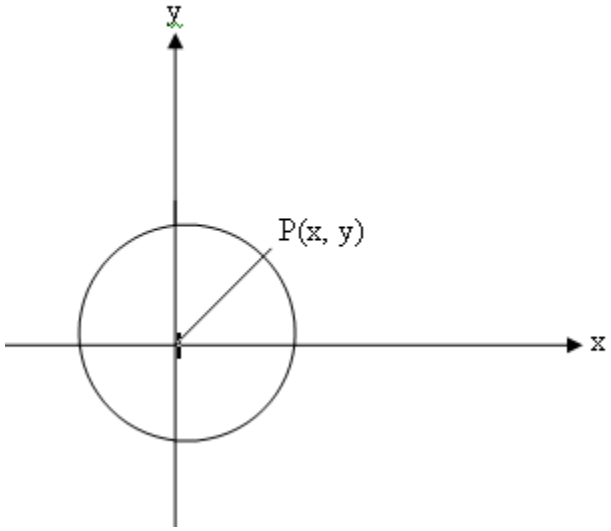
So $AB^2+BC^2=5+20$

$=25=AC^2$

Therefore, triangle ABC is a right angle (converse of Pythagoras theorem).

EQUATION OF A CIRCLE

Consider a circle of radius r whose centre is at the origin and let P(x, y) be any point on the circle.



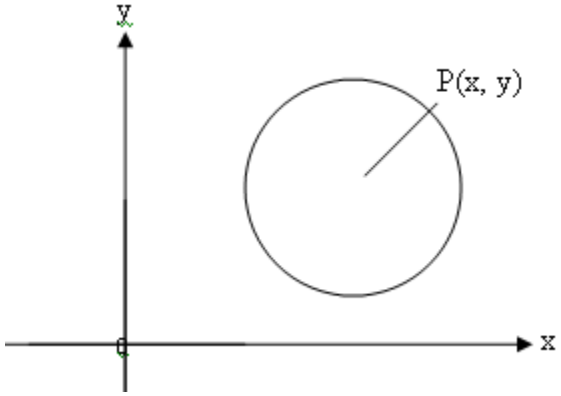
Now using the distance formula

$OP=\sqrt{(x-0)^2+(y-0)^2}$

$OP^2=x^2+y^2$

Or $x^2+y^2=r^2$

The above equation is an equation of a circle of radius r whose centre is at the origin (0, 0). Suppose we wish instead to find the circle, centre Q=(a, b) with radius r.



Let P=(x, y) be any point on this circle. Then PQ=r. By the distance formula,

$PQ^2 = (x-a)^2 + (y-b)^2$

Or $(x-a)^2 + (y-b)^2 = r^2$

This is the equation of the a circle of radius r whose centre is the point (a, b)

It can also be written in another form by multiplying out the bracketed terms and then collecting like terms.

i.e. $(x-a)^2 + (y-b)^2 = r^2$

$x^2 - 2ax + a^2 + y^2 - 2b + b^2 = r^2$

$x^2 + y^2 - 2ax - 2by + a^2 + b^2 - r^2 = 0$

since a, b and r are all constants this can be simplified to the general form of the equation of a circle

$x^2 + y^2 - 2ax - 2by + c = 0$

NOTE: The centre (a, b) can be identified from the coefficient of the x and y terms.

Example

State the radius and centre of the following circle: $(x-3)^2 +(y+4)^2 =81$

Solution

$(x-3)^2 +(y- 4)^2 =9^2$

Therefore the centre of this circle is (3,-4) and the radius is 9.

Example

Find the equation of the circle of radius $\sqrt{7}$ and centre (3, -2)

Solution

The equation of the circle is $(x-a)^2 +(y-b)^2=r^2$

So the equation of this circle is

$$(x-3)^2+(y+2)^2=(\sqrt{7})^2$$

$$\text{Or } (x-3)^2+(y+2)^2= 7$$

Example

The equation of a circle is $x^2+ 6x+y^2=4y+4=0$. Find the centre and the radius of the circle.

Solution

Rewriting the equation of the circle, we have

$x^2 +y^2 +6x+4y+4=0$

Equating the coefficients of x and y we get

$-2a= 6$ and $-2b=4$

$a=-3$ and $b=-2$

Thus the centre is (-3, -2)

The constant term, 4 should be equated to $a^2+b^2-r^2$ in order to find r

i.e. $a^2+b^2-r^2 =4$

$(-3)^2+(-2)^2-r^2=4$

$9+4-r^2=4$

$9=r^2$

$3=r$

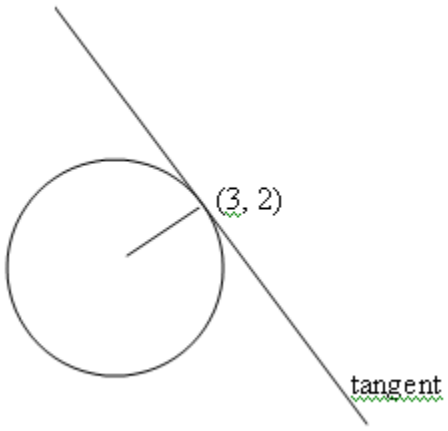
Therefore the radius of this circle is 3

Example

Find the equation of the tangent to the circle $x^2+y^2-3x-y-2=0$ at the point (3, 2)

Solution

A tangent is a straight line that touches the curve at one distinct point



Step 1: find the values of a and b

i.e. equating coefficients of x and y, we get -2a=-3 and -2b=-1

It follows that a= $\frac{3}{2}$ and b= $\frac{1}{2}$

So the centre of the circle is ($\frac{3}{2}$, $\frac{1}{2}$)

Step 2: Find the gradient of the radius through $(\frac{3}{2}, \frac{1}{2})$ and (3, 2)

i.e $m=\frac{y_2-y_1}{x_2-x_1}$

$$m=\frac{2-\frac{1}{2}}{3-\frac{3}{2}}$$

$$m=\frac{\frac{3}{2}}{\frac{3}{2}}$$

m=1

step 3 :Find the gradient of the tangent

i.e $m_1.m_2=-1$

so the gradient of the tangent is -1 (tangent is perpendicular to radius)

Step 4; Find the equation of the tangent

i.e y-a=m(x-b)

y-2=-1(x-3)

y-2=-x+3

y=-x+5

TUTORIAL 6: FUNCTIONS AND GRAPHS

1. If $f(x) = x^2 - 5x + 4$, find f(1), f(-2) and f(3)
2. Find the inverses of the following functions, if they exists.

   (a) f(x)= 3x+5 (b) f(x)= 3-x (c) $f(x)=\frac{1}{x}$

3. Find the gradient, m of the following lines

   a)y=2x+4

   b) 3x+8y=24

c)The line through (-2, -1) and)6,-1)

d) The perpendicular to 5x+3y=1

4. Find the equations of the following lines

a) The line through (4, 5) with slope 2

b) The line through (5, -3) and (2, 0)

c) The line through (1, -2) parallel to 2x-4y=3

d) The line through (-1, -2) perpendicular to 3x-7y+6=0

5. A, B, C are the points (3, -2), (-7, 4), (-7, -8) respectively. Find the lengths of the lines AB, BC and CA. What can you say about the triangle ABC?

6. Find the centre and radius of the following circles

(a) $(x-5)^2 + (y-3)^2 = 7^2$

(b) $x^2 + y^2 - 6x - 2y + 6 = 0$

7. Write the equations of the circles given centre and radii

(a) Centre (-1, 3), radius 2

(b) Centre (3, -3) radius $3\sqrt{2}$

8. Verify that the point P=(0, 1) lies on the circle $x^2 + y^2 - 4x = 1$, and find the equation of the tangent to the circle at P.

TOPIC 7: EQUATION, INEQUALITIES AND LINEAR PROGRAMMING

OBJECTIVES

By the end of this topic, students should be able to;

(a) Solve simple linear equations.

(b) State three methods of solving simultaneous equations.

(c) Solve simultaneous equations by elimination, substitution or graphing method.

(d) Solve quadratic equations by ether factor, completing the square, quadratic formula or graphing methods.

(e) Graph linear inequalities.

(f) Solve inequalities.

(g) Find maximum and minimum values of linear programming problems by using graphing or algebraic methods.

LINEAR EQUATIONS

An equation like 2x-1=3 is called a linear

Linear equations may be put in the form y=ax+b where a and b are constants (real numbers)

When graphed, these equations produce straight lines, which is why they are called linear.

Example

Solve the following equation

4x+ 3 = 6x -4

Solution

Collect like terms, we get

3+4 =6x -4x

7=2x

$\frac{7}{2}$=4 or x=$\frac{7}{2}$

SIMULTANEOUS EQUATIONS

A system of two linear equations can sometimes be solved simultaneously, using:

1. Substitution
2. Elimination
3. Graphical methods

Example

Solve the following equations x +2y= 7 and 2x +3y =10 simultaneously by substitution, elimination and graphical methods

Solutions

(i) substitution method

label the equations as 1 and 2

i.e. x +2y =7 …1

2x +3y=10 …2

Make one letter the subject of one of the equations

i.e. x=7- 2y making x the subject of (1)

Substitute 7-2y for x in the either remaining equation (20

i.e. 2(7-2y) +3y =10

14-4y +3y =10

14-y=10

14-10=y

4=y

Substitute the value of y in one of the original equations

i.e. substitute 4 for y in (1), we get

x+ 2×4 =7

x+8=7

x=-1

Therefore x=-1 and y=4

(ii) elimination method

labeling the equations

i.e. x +2y =7 …1

$2x + 3y = 10$ …2

Multiply each equation by a value that will make the coefficient of one of the variables the same in both equations

i.e. multiply equation (1) by 2 and equation (2) by 1 to get the following

$$2x + 4y = 14$$
$$\underline{2x + 3y = 10}$$
$$y = 4$$

NOTE: IF the sign of the equal coefficients is the same, we subtract and if the signs are different, we add

Substitute 4 for yin one of the original equations

i.e. substitute 4 for y in (1), we obtain

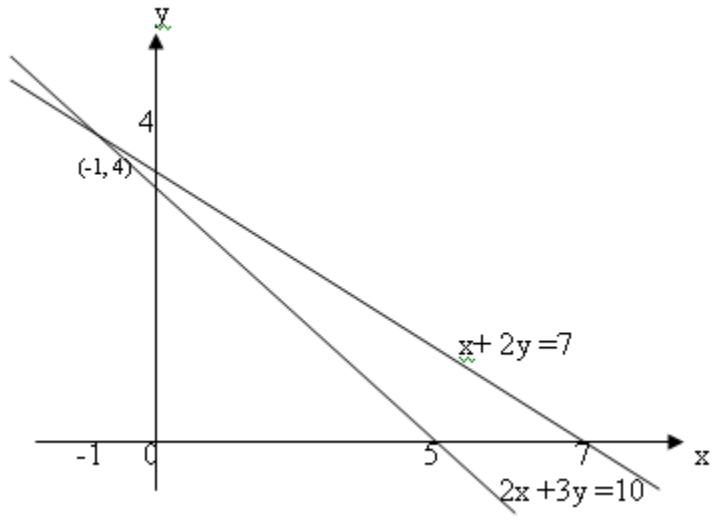
$$x + 2 \times 4 = 7$$
$$x + 8 = 7$$
$$x = -1$$

Therefore, x=-1 and y-4

(iii)GRAPHICAL METHOD

- On the same axes, draw the two graphs.
- Look at the point of intersection. This point gives the root of the simultaneous equations.

i.e



QUARDRATICS EQUATIONS

A quadratic equation is one in which the highest power of x (whatever letter is used for the unknown) is 2, and there are no fractions or negatives powers

The main techniques used to solve the quadratic equations are :

1. factorizing
2. completing the square
3. using the formula and
4. graphical methods

1. *FACTORISING METHOD*

Example

Solve the equation x^2 +5x-14=0 by factor method

Solution

x^2 +5x-14=0

(x+7)(x-2)=0 (factorizing the quadratic expression)

Either x+7=0 or x-2=0

X=-7 or x=2

Therefore x=-7 or 2

2. *COMPLETING THE SQUARE METHOD*

STEPS TO BE FOLLOWED

1. Mark the coefficient of x^2 to be unity (1)
2. Separate the constant term from the variable terms
   i.e. x^2+5x-14=0 would be x^2+5x=14
3. Add (half the coefficient of x) ^{2} to both sides of the equation

Example

Solve the equation x^2+5x-14=0 by completing the square method.

Solution

x^2 +5x-14=0

x^2 +5x=14

$x^2+5x+(\frac{5}{2})^2=14+(\frac{5}{2})^2$

$x^2+5x+\frac{25}{4}=14+\frac{25}{4}$

$(x+\frac{5}{2})^2=\frac{56+25}{4}$

$(x+\frac{5}{2})^2=\frac{81}{4}$

$x+\frac{5}{2}=\sqrt{\frac{81}{4}}$

$X=-\frac{5}{2}\pm\frac{9}{2}$

Either $x=-\frac{5}{2}-\frac{9}{2}$ or $x=-\frac{5}{2}+\frac{9}{2}$

$x=\frac{9-5}{2}$ or $x=\frac{9+5}{2}$

$x=\frac{4}{2}$ or $x=\frac{-14}{2}$

Therefore x=2 or -7

Example

Solve the quadratic equation $3x^2-4x-8=0$ by completing the square method.

Solution

$3x^2-4x-8=0$

$X^2-\frac{4}{3}x-\frac{8}{3}=0$ making the coefficient of x^2 to be 1

$X^2-\frac{4}{3}x=\frac{8}{3}$

$X^2-\frac{4}{3}x+\frac{4}{9}=\frac{8}{3}+\frac{4}{9}$

$(x-\frac{2}{3})^2=\frac{24+4}{9}$

$(x-\frac{2}{3})^2=\frac{28}{9}$

$x-\frac{2}{3}=\pm\sqrt{\frac{28}{9}}$

$x-\frac{2}{3}=\mp\frac{\sqrt{28}}{3}$

$x=\frac{2}{3}\mp\frac{\sqrt{28}}{3}$

either $x=\frac{\sqrt{28}}{3}+\frac{2}{3}$ or $-\frac{\sqrt{28}}{3}-\frac{2}{3}$

$x=\frac{2\pm\sqrt{28}}{3}$ or $-\frac{2\pm\sqrt{28}}{3}$

3. *QUADRATIC FORMULA*

Suppose we wish to solve for x the general equation quadratic equation $ax^2+bx+c=0$.

We are going to solve this equation by completing the square method.

i.e. $x^2+\frac{b}{a}x+\frac{c}{a}=0$ making the coefficient of x^2 to be equal to 1

$x^2+\frac{b}{a}x+\frac{b^2}{4a^2}=\frac{b^2}{4a^2}-\frac{c}{a}$

$(x+\frac{b}{2a})^2=\frac{b^2}{4a^2}-\frac{c}{a}$

$(x+\frac{b}{2a})^2=\frac{b^2-4ac}{4a^2}$

$x+\frac{b}{2a}=\pm\sqrt{\frac{b^2-4ac}{4a^2}}$

$x=-\frac{b}{2a}\pm\frac{\sqrt{b^2-4ac}}{2a}$

$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

The above equation is the quadratic formula where a is the coefficient of x^2, b is the coefficient of x and c is the constant

b^2-4ac is called the discriminant. It can be used to give an indication of how many times the graph will cross the x-axis

(a) If b^2-4ac>0 then the equation ax^2+bx+c=0 has two distinct roots (solutions)

(b) If b^2-4ac=0, then the equation ax^2+bx+c=0 has one repeated root (solution)

(c) If b^2-4ac<0, then the equation ax^2+bx+c=0 has no real; roots (solutions)

Example

Solve the quadratic equation $3x^2$-4x-8=0 by quadratic formula

Solution

Given: a=3, b=-4 and c=-8

But $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$$x=\frac{--4\pm\sqrt{(-4)^2-4\times3\times(-8)}}{2\times3}$$

$$x=\frac{4\pm\sqrt{16+96}}{6}$$

$$x=\frac{4\pm\sqrt{112}}{6}$$

$x\frac{4\pm\sqrt{4\times28}}{6}$

$$x=\frac{4\pm2\sqrt{28}}{6}$$

$$x=\frac{2(2\pm\sqrt{28}}{6}$$

$$x=\frac{2\pm\sqrt{28}}{3}$$

(d) GRAPHICAL METHOD

Draw the quadratic graph on the graph paper

Look at the point where the graph intersect the x axis

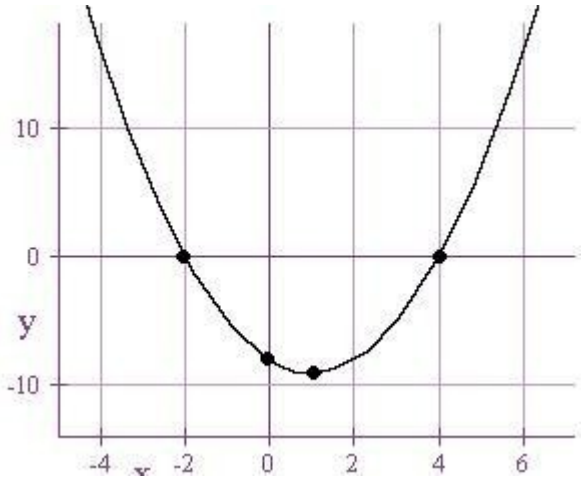
The value of x at these points of intersections is the roots of the quadratic equation.

Example

Solve the equation $x^2-2x-8=0$ by graphing method.

Solution

x	-2	-1	0	1	2	3	4
y	0	-5	-8	-9	-8	-5	0



Either x = -2 or 4

LINEAR INEQUALITIES

There are four inequalities symbols:

$>$means is 'greater than.'
$\geq$means is 'greater than or equal to'
$<$means is 'less than'
$\leq$means is 'less than or equal to'

ORDER AXIOMS

1. For every $x\in$ exactly one of the following is true: $x>0$, $x<0$, $x=0$
2. If x, y $x>0$ and $y>0$ the $x+y>0$ and $xy>0$
3. If $x>0$ then $-x<0$ and $x^{-1}>0$
4. If $x>y$ and $y>z$ then $x>z$
5. If $x>y$ then $x+c>y+c$ for any $c\in$
6. If $x>y$ and $c>0$ then $cx>cy$
7. If $x>y$ and $c<0$ then $cx<cy$.

Property (axiom 7) says if one multiplies an inequality by a negative number, the sign of the inequality is reversed. This means that one can not multiply an inequality by an unknown if the sign of the unknown may be positive or negative.

Example
Solve the inequality $3x-5>7$
Solution
Adding 5 to both sides (i.e. property 5), we have

$$3x>12$$

Multiplying both sides by $\frac{1}{3}$ and using property 6 we get

$x>4$

Example

Solve $7-2x \leq 4$

Solution

$7-2x \leq 4$

$-2x \leq -3$ subtracting 7

$x \geq \frac{3}{2}$ property 7

Example

Solve $x^2+x-6<0$

Solution

Factorize the quadratic expression

i.e. $x^2+x-6=(x+3)(x-2)$

$(x+3)(x-2) <0$

Equate each factor to zero (0) and solve for x

i.e. $x+3=0 \Rightarrow x=-3$

$x-2=0 \Rightarrow x=2$

Then produce the following table

		-3	2
x+3	-	+	+
x-2	-	-	+
(x+3)(x-2)	+	-	+

We are interested in the case where the product is negative. So the solution set is

(-3, 2). That is x is a solution if and only if $-3<x<2$.

Example

Solve $x^2-3x-4 \geq 0$

Solution

$x^2-3x-4 \geq 0$

$(x-4)(x+1) \geq 0$

-1 4

x+1	-	+	+
x-4	-	-	+
(x+1)(x-4)	+	-	+

Therefore, the solution is $x \leq -1$ or $x \geq 4$.

LINEAR PROGRAMMING

SHOWING REGIONS REPRESENTED BY INEQUALITIES

Example

Show the region represented by

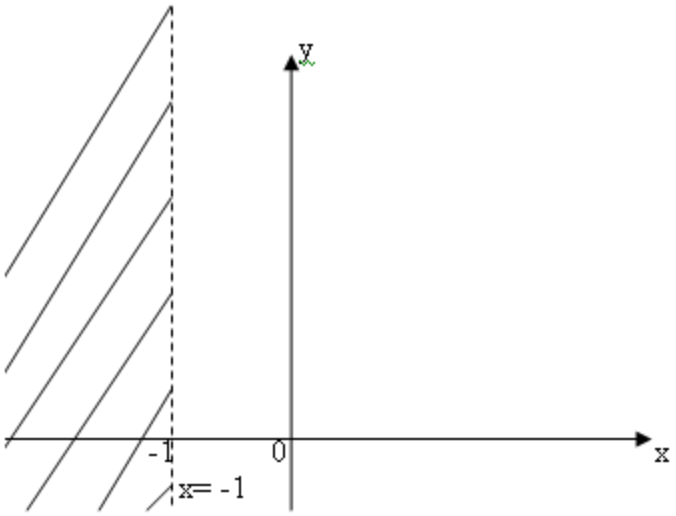
(i) $x>-1$

(ii) $y<5$

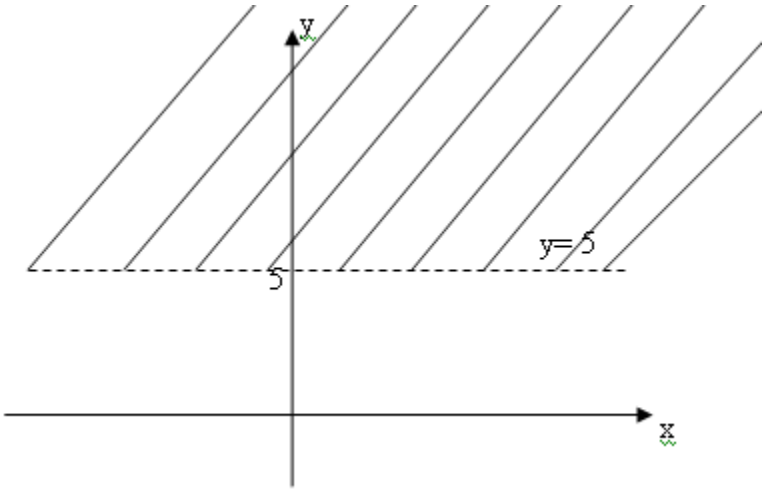
(iii) $y \geq x+2$

Solutions

(i) $x>-1$



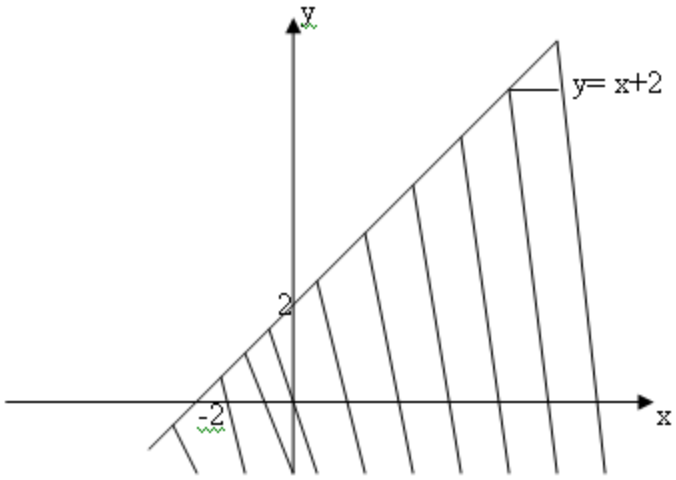
(ii) y<5



(iii)

$y \geq x+2$

y-intercept is 2 and x-intercept is -2



NB (1) Draw continuous line/ solid i.e. _____ when there is an equal Sign

(2) Draw dotted line (broken) ---------- when there is no equal sign.

(3) Shade unwanted regions.

Example

Sketch $2x-y<3$

Working

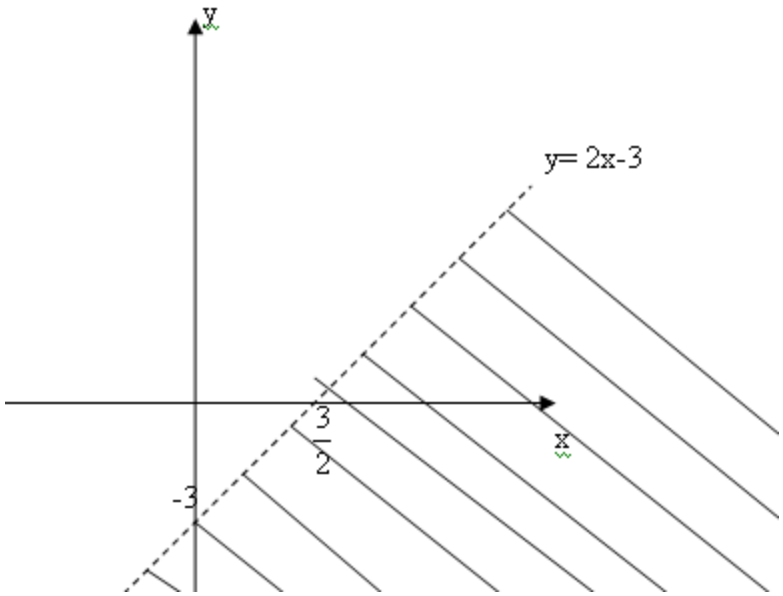
Make y the subject of the inequality

$y>2x-3$

draw a line $y=2x-3$

y-intercept is -3

x-intercept is $\frac{3}{2}$



Example

Sketch the region represented by $4x+y<8$, $2x+3y\leq 6$, $x\geq 0$ and $y\geq 0$

Solution

Making y the subject of the inequality $4x+3y\leq 6$, we have

$y\leq \frac{6}{3}-\frac{2}{3}x$

or $y\leq 2-\frac{2}{3}x$

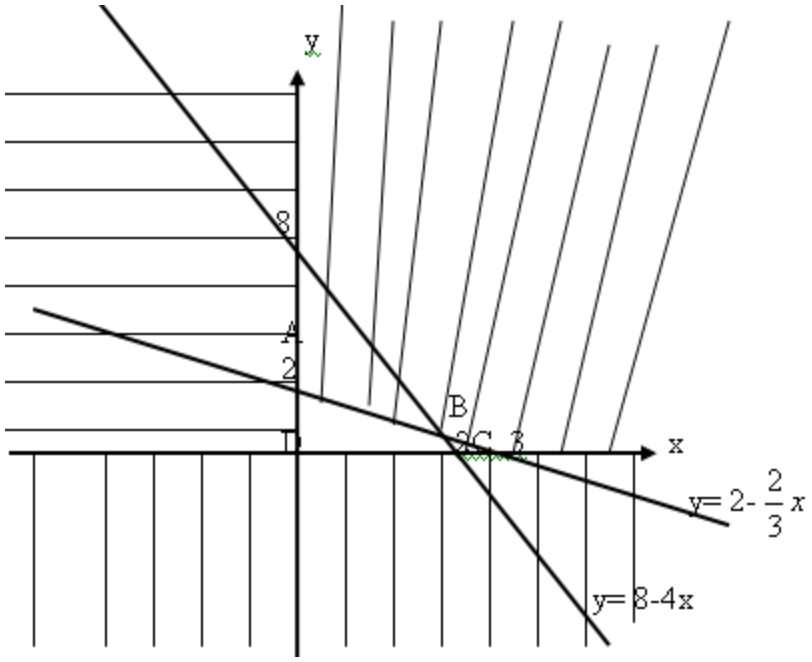
drawing the line $y=2-\frac{2}{3}x$

y-intercept is 2 and x-intercept is 3

making y the $4x+y\leq 8$, we get $y\leq 8-4x$

drawing the line $y=8-4x$

y-intercept is 8 and x-intercept is 2



FEASIBLE REGION

It is the set of all points satisfying the constraints. From the above example, ABCD is a feasible region. Coordinates A=(0, 2), B=?, C=(2,0) and D=(0,5). The coordinate of B is found by solving

$6-2x=24-12x$

$10x=18$

$x=1.8$

Substituting 1.8 for x in * we get

$y=8-4\times1.8$

$y=8-7.2$

$y=0.8$

Therefore, B=(1.8,0.8)

Example

Plot the feasible region defined by $x-3y\leq3$, $2x+5y>10$ $x, y\geq0$

Solution

Making y the subject of $x-3y \leq 3$, we get

$x-3 \leq 3y$

$\frac{x}{3}-1 \leq y$ or $y \geq \frac{x}{3}-1$

Draw the line $y=\frac{x}{3}-1$

y-intercept is -1 and x-intercept is 3

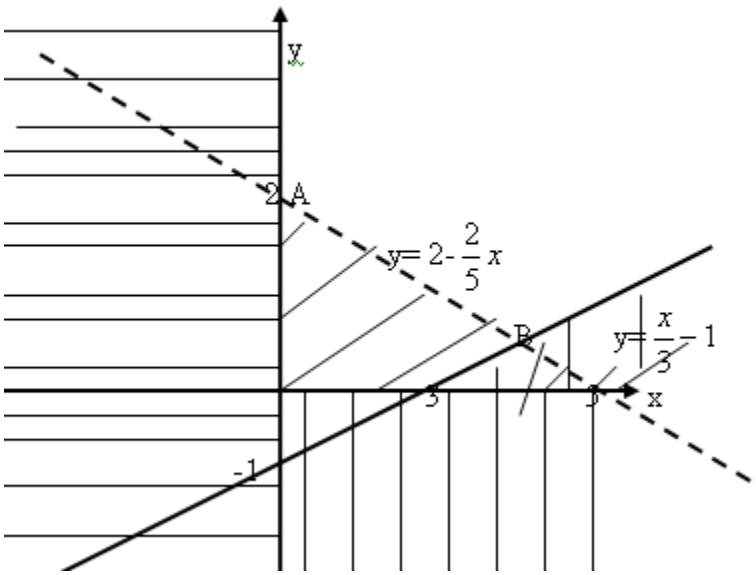
making y the subject of $2x+5y>10$, we have

$5y>10-2x$

$y>2-\frac{2}{5}x$

draw the line $y=2-\frac{2}{5}x$

y-intercept is 2 and x-intercept is 5



The coordinates of A is (0, 2)

In order to find the coordinates of B, we need to solve the following equations: $y=\frac{x}{3}-1$

and y=2-$\frac{2}{5}x$

i.e. 2-$\frac{2}{5}x$=$\frac{x}{3}$-1 *

5x-15=30-6x

11x=45

x=4

Substituting 4 for y in * we have y=$\frac{4}{3}$-1

y=$\frac{1}{3}$

MAXIMISING AND MINIMIZING

STEPS TO BE FOLLWED IN FINDING THE MAXIMUM (MINIMUM) VALUE

1. Sketch the feasible region defined by the constraints
2. Determine whether a maximum (minimum) is attainable
3. If yes, find the coordinates of all corner points of the feasible region
4. Compute the value of ax+by (objective function) at each corner point of the feasible region. The largest (smallest) value obtained is the maximum (minimum) of the objective function on the feasible region.

Example

Maximize x+y subject to the constraints 2x+y$\leq$8, x+3y$\geq$9 x, y$\geq$0

Working

x +y is called the objective function

First, draw the feasible region represented by the inequalities

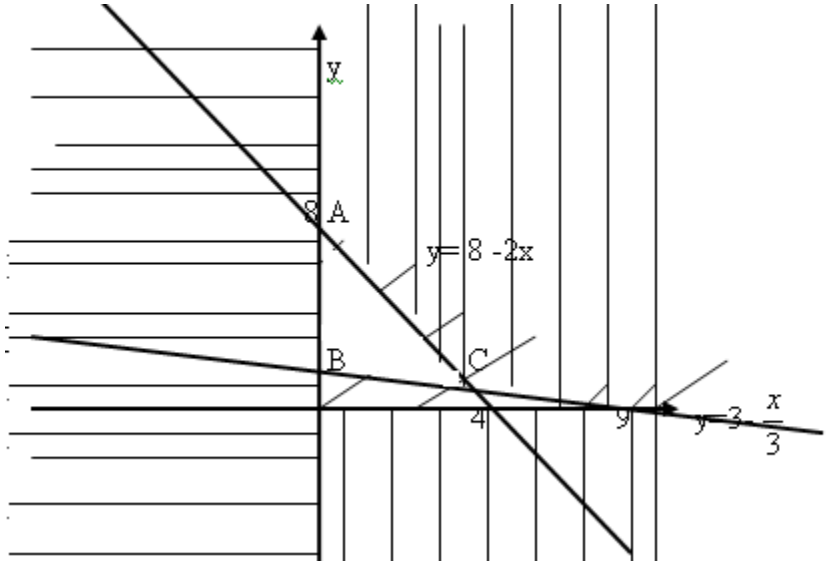
Making y the subject of $2x+y \leq 8$, we get $y \leq 8-2x$

Drawing the line $y=8-2x$

y-intercept is 8 and x-intercept is 4

making y the subject of $x+3y \geq 9$ we get $y \geq 3-\frac{x}{3}$

drawing the line $y=3-\frac{x}{3}$, y-intercept is 3 and x-intercept is 9



The coordinates of A=(0, 8), B=(0, 3) and C=?

To find the coordinate of C equations $y=8-2x$ and $y=3-\frac{x}{3}$ need to be solved simultaneously

i.e. $8-2x= 3-\frac{x}{3}$,

$24-6x=9-x$

$15=5x$

$3=x$

Substitute 3 for x in *, we get

y=2

Therefore C=(3, 2)

Finding maximum or minimum of x+y

The maximum value is 8 at point A(0, 8) and the minimum value is at B(0, 3)

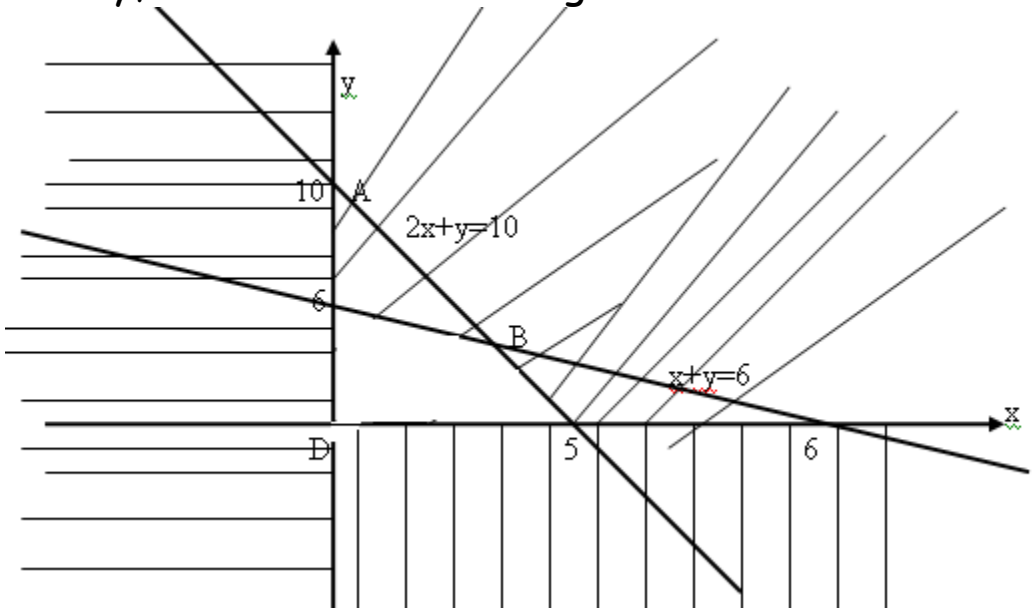
In general, maximum or minimum of the objective function occur at one of the vertices of the feasible region. This means that to find the maximum or minimum value of the feasible region, we need only to consider the values of the vertices

Example

Find the maximum and minimum of 2x-y subject to $2x+y \leq 10$, $x+y \leq 6$, $x,y \geq 0$

Solution

Firstly, draw the feasible region



A(0, 6)=2x-y
=0-6
=-6
B(4, 2)=2×4-2
=8-2
=6
C(5, 0)=2×5-0
=10
D(0, 0)=2×0-0
=0
Therefore the maximum is 10 at C(5, 0) and the minimum is -6 at A(0, 6).

LINEAR PROGRAMMING

Linear programming is a mathematical technique which sort solutions to problems in which a number of variables are considered at the same time.
The basic descriptions involved take the form of linear algebraic inequalities; hence the technique is described as linear programming.
The process involves maximizing or minimizing functions that may have a large number of variables subject to linear objective function.

SOLVING LINEAR PROGRAMMING BY GRAPHING METHOD

Example

A factory produces two types of transformers A and B . Each transformer of type A needs 4 workers while type B needs 3. Furthermore, each type A needs 10m of copper wire, type B needs 20m of copper wire. The manager has 60 men and 200m of copper wire available. If the profit for type A is K20 per transformer and type B is K30 per transformer. What should the manager do to produce the greatest profit?

Solution

Let x be the number of type A and y be the number type B

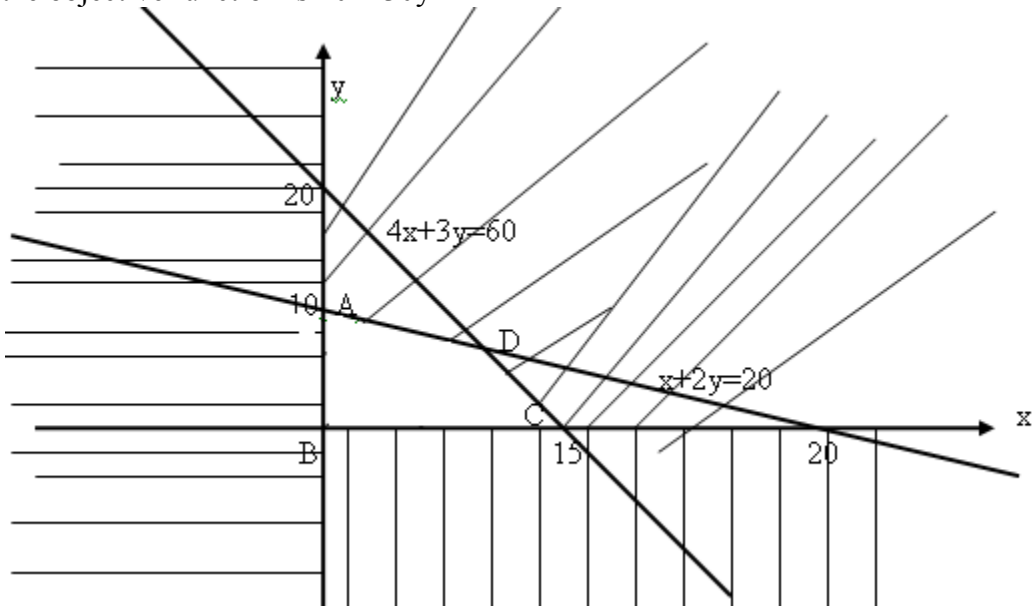
Lower bound, $x \geq 0 \; y \geq 0$

Number of workers; $4x+3y \leq 60$

Length of copper wire; $10x + 20y \leq 200$

i.e. $x+2y \leq 20$

the objective function is $20x+30y$



At point A(0, 10); $20x+30y=0+30\times 10$

$=300$

At point B(0, 0); $20x+30y=0+0$

$=0$

At point C(15, 0); $20x+30y=20\times 15+0$

$=300$

At point D(12, 4); $20x+30y=20\times 12+30\times 4$

$=360$

Therefore, the maximum is 360 at D(12, 4). So the manager should have 12 types of transformer A and 4 types of transformer B.

SOLVING LINEAR PROGRAMMING PROBLEM BY USING ALGEBRAIC METHOD

In practice, linear programming problems that arise deal with maximizing or minimizing the variables can assume only nonnegative values.

STANDARD MAXIMUM PROBLEM

Maximize the linear function $P= cx+cx+\ldots+cx$

Subject to the constraints

$A_{11}x_1+a_{12}x_2+a_{13}x_3+\ldots+a_{1n}x_n\leq b_1$

$A_{21}x_1+a_{22}x_2+a_{23}x_3+\ldots+a_{2n}x_n\leq b_2$

.

.

.

$A_{m1}x_1+a_{m2}x_2+a_{m3}x_3+\ldots+a_{mn}x_n\leq b_m$

Where all b_i are nonnegative and where $x_1\geq 0, x_2\geq 0, \ x_3\geq 0,\ldots \ x_n\geq 0.$

INTRODUCING THE SLACK VARIABLE

The first step in solving a linear programming by algebraic method is to convert the inequalities constraints to linear systems of equations.

Slack variable represents the nonnegative difference the slack, hence it should always be greater or equal to zero but not negative.

In order to solve the system of equations, some variables are set equal to zero. These variables are called*non-basic variables*.

Example

Transform the following inequalities into linear equations

$x_1+x_2+x_3\leq 25$

$2x_1+x_2+3x_3\leq 51$

Solution

$x_1+x_2+x3+s_1=25$

$x_1+x_2+x_3+s_2=51$ where s_1 and $s_2\geq 0$

Algebraic method involves the distribution of zeros and stars in order to come up with exact values of the stars and finding the value of P the objective function

The solution is in the feasible region if each variable is greater than or equal to zero.

Example

Maximize$2x-y$ subject to $2x+y\leq 10$ and $x+y\leq 6$.

Solution

Introducing the slacks, we have

$2x+y+s_1=10$

$x+y+s_2=6$

$P=2x-y$

Distribution of basic and non-basic variables

$$\frac{4}{2(4-2)}=\frac{4\times3\times2\times1}{2\times1\times2\times1}$$

=6 ways

x	y	S_1	S_2	P=2x-y
0	0	*	*	
0	*	0	*	
0	*	*	0	
*	0	0	*	
*	0	*	0	
*	*	0	0	

We need to find the values of the stars.

If x=0 and y=0 from the first row then $s_1=10$ and $s_2=6$ (substitute the values of x and y in (1) and (2)

Substitute x=0 and $s_1=0$ in (1), we get y=10

Substitute x=0 and y=10 in (2), we $s_2=-4$

Substitute x=0 and $s_2=0$ in (2),we get y=6

Substitute y=o and $s_1=0$ in (1), we get x=4

Substitute y=0 and s_1=o in (1), we get x=5

Substitute =5 and y=0 in (2), we get $s_2=1$

Substitute y=0 and $s_2=0$ in (2), we get x=6

Substitute x=6 and y=0 in (1), we get $s_1=-2$

Substitute $s_{1=0}$ and $s_2=0$ in (1) and (2) respectively, we get 2x+y=10 and x+y=6 and solving them simultaneously we find x to be equal to 4 and y to be equal to 2

Replace the stars in the table above with the values we have found above

x	y	S_1	S_2	P=2x-y
0	0	10	5	0
0	10	0	-4	Non-feasible
0	6	4	0	-6
5	0	0	1	10
6	0	-2	0	Non-feasible
4	2	0	0	6

Therefore, the maximum value is 10 occurring when (x, y) =(5, 0).

Example

Find the maximum value of –x+3y subject to the constraints $-2x+y\leq1$, $-x+y\leq4$, $x\geq0$, $y\geq0$

Solution

Introducing the slacks, we have

$-2x+y+s_1=1$

$-x+y+s_2=4$

Distribution of basic and non-basic variables

$$\frac{4}{2(4-2)}=\frac{4\times3\times2\times1}{2\times1\times2\times1}$$

=6 ways

x	y	S_1	S_2	P=-x+3y
0	0			
0		0		
0			0	
	0	0		
	0		0	
		0	0	

Therefore the maximum value is

TUTORIAL: EQUATIONS, INEQUALITIES AND LINEAR PROGRAMMING

1. Solve the following pairs of linear simultaneous equations by substitution and elimination methods
   (a) 2x+y=8 and 3x+3y=14
   (b) X+3y=11 and 5x+y=13
   (c) 3x+y=11 and 2x-5y=11
2. Solve the following quadratic equations by factorizing, completing the square and using the quadratic formula.
   (a) $x^2+5x+4=0$
   (b) $4x^2-19x+12=0$
   (c) $9x^2+24x+16=0$
3. Solve the following inequalities
   (a) x-4<3-x
   (b) 7-3x<13
   (c) 3(3-2x)>2(3+x)
4. Solve the following quadratic inequalities
   (a) $x^2-3>2x$
   (b) (x-2)(x+4)<0
5. Find the maximum and minimum of 2x-y subject to $2x+y\leq10$, $x+y\leq6$, $x,y\geq0$
6. Show that $x^2-4x+5y^2\geq0$ for all real values of x and y

TOPIC 8: MATRIX ALGEBRA

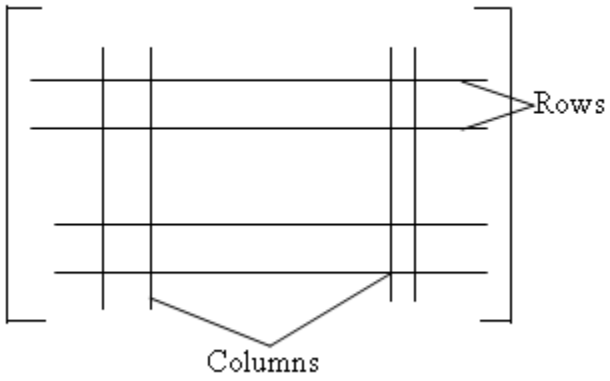
OBJECTIVES

By the end of this topic, students should be able to;

(a) Define the term matrix.
(b) Give five types of matrices.
(c) Find the sum, difference and products of matrices.
(d) Find the transpose, determinants and inverse of matrices.
(e) Solve simultaneous linear equations by using determinants.
(f) Solve simultaneous linear equations by row/column reduction.

WHAT IS A MATRIX?

It is a collection of numbers (items) arranged in *m* rows and *n*columns.



Usually such a collection is called an $m \times n$ matrix

e.g. (1) A matrix with 2 rows and 3 columns

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$$

(2) A matrix with 3 rows and 2 columns

$$\begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix}$$

(3) A matrix with 5 rows and 6 columns

$$\begin{pmatrix} . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \end{pmatrix}$$

We denote the item in the i^{th}row and j^{th} column as a_{ij}
e.g. a_{11} is the item in the first row and first column
a_{12} is the item in the first row and second column
a_{63} is the item in the sixth row and third column
With this notationwe can rewrite the matrix with 4 rows and 6 columns as

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & a_{26} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} & a_{36} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} & a_{46} \end{pmatrix}$$

THE ORDER OF A MATRIX
This is the number of rows and columns.

Example
Find the order of the following matrices

(a) $\begin{pmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{pmatrix}$

(b) $\begin{pmatrix} a & b & c & d \\ e & f & g & i \\ j & h & k & l \end{pmatrix}$

Solutions

(a) $\begin{pmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{pmatrix}$

Its order is 2×3

(b) $\begin{pmatrix} a & b & c & d \\ e & f & g & i \\ j & h & k & l \end{pmatrix}$

Its order is 3×4

TYPES OF MATRICES
1.COLUMN MATRIX
This is a matrix with *m* rows and *1* column .Its order is $m \times 1$
Some examples of column matrices are

(a) $\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$ and (b) $\begin{pmatrix} a \\ b \\ c \\ d \\ e \end{pmatrix}$

3×1 5×1

2. *ROW MATRIX*
This is a matrix with 1 row and m columns
Some examples of row matrices are

(a) $\underset{1\times 3}{\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}}$ and (b) $\underset{1\times 6}{\begin{pmatrix} a & b & c & d & e & f \end{pmatrix}}$

3.NULL or ZERO MATRIX

This is a matrix with every element zero. For example $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

4.SQUARE MATRIX
This is a matrix with the same number of rows and columns.
Some examples of square matrices are

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\begin{pmatrix} a & b & c \\ d & e & f \\ g & i & j \end{pmatrix}$

5.IDENTITY MATRIX
This is a matrix which has a 1 in every element of the leading diagonal, and 0 elsewhere.
Some examples are

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

6.DIAGONAL MATRIX
It is a square matrix whose elements are all zeros except on the leading diagonal.

$\begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$ and $\begin{bmatrix} 12 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

7.TRIANGLE MATRIX
It is a square matrix whose elements either above or below the leading diagonal are all zeros.

$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, upper triangle matrix and $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$, lower triangle matrix.

NOTE: All identity matrices are square matrices.

ADDITION AND SUBTRACTION OF MATRICES

Matrices A and B are conformable to addition if and only if they have the same number of rows and same number of columns. The sum of A and B is a matrix

$A+B=(a_{ij}+b_{ij})$

NOTE: corresponding elements are added together.

Example

If $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B=\begin{pmatrix}5 & -2\\ 1 & 2\end{pmatrix}$ and $C=\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{pmatrix}$

Find (i) A+B

(ii)A+C

(iii)A-B

Solutions

(i) $A+B=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}+\begin{pmatrix}5 & -2\\ 1 & 2\end{pmatrix}$

$=\begin{pmatrix}1+5 & 2+-2\\ 3+1 & 4+2\end{pmatrix}$

$=\begin{pmatrix}6 & 0\\ 4 & 6\end{pmatrix}$

(ii) A+C does not exist since A is 2×2 and C is 2×3

(iii) $A-B=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}-\begin{pmatrix}5 & -2\\ 1 & 2\end{pmatrix}$

$=\begin{pmatrix}1-5 & 2--2\\ 3-1 & 4-2\end{pmatrix}$

$=\begin{pmatrix}-4 & 4\\ 2 & 2\end{pmatrix}$

NOTE: You subtract correspond elements if the two matrices have the same order.

MULTIPLICATION BY A SCALAR

Let the matrix $A=(a_{ij})$ be $m\times n$ and λ be a scalar (i.e λ).We define the matrix λA as (λa_{ij}).

NOTE: Multiply each element of a matrix by a scalar.

Example

If $A=\begin{pmatrix}2 & 3\\ 1 & 2\end{pmatrix}$, find

(1) 2A

(2) 5A
(3) –A

Solutions

(1) $2A=2\begin{pmatrix}2 & 3\\ 1 & 2\end{pmatrix}$

$=\begin{pmatrix}2\times 2 & 2\times 3\\ 2\times 1 & 2\times 2\end{pmatrix}$

$=\begin{pmatrix}4 & 6\\ 2 & 4\end{pmatrix}$

(2) $5A=5\begin{pmatrix}2 & 3\\ 1 & 2\end{pmatrix}$

$=\begin{pmatrix}10 & 15\\ 5 & 10\end{pmatrix}$

(3) $-A=-1\begin{pmatrix}2 & 3\\ 1 & 2\end{pmatrix}$

$=\begin{pmatrix}-2 & -3\\ -1 & -2\end{pmatrix}$

MATRIX MULTIPLICATION

Let A be an $m\times n$ matrix and B be a$p\times q$ matrix. Then the matrix AB is only defined (or A and B are conformable to multiplication) if the number of columns of A is equal to the number of rows of B. i.e. AB is only defined if $n=p$

When finding the element of the product matrix, multiply the element in the row of the first matrix by element in the column of the second matrix then find their sum.

Example

If $A=\begin{pmatrix}2 & 3\\ 1 & 4\end{pmatrix}$, $B=\begin{pmatrix}4 & 1\\ 3 & -2\end{pmatrix}$, $C=\begin{pmatrix}4 & 1 & 0\\ 5 & 2 & 7\end{pmatrix}$ and $D=\begin{pmatrix}-1 & 6\\ 2 & 4\\ 5 & 1\end{pmatrix}$, find

(i) AB
(ii) AC
(iii) CA
(iv) CD

Solutions

(v) $AB=\begin{pmatrix}2 & 3\\ 1 & 4\end{pmatrix}\begin{pmatrix}4 & 1\\ 3 & -2\end{pmatrix}$

$=\begin{matrix}2\times 4+3\times 3 & 2\times 1+3\times -2\\ 1\times 4+4\times 3 & 1\times 1+4\times -2\end{matrix}$

$$=\begin{pmatrix} 8+9 & 2-6 \\ 4+12 & 12-8 \end{pmatrix}$$

$$=\begin{pmatrix} 17 & -4 \\ 16 & -7 \end{pmatrix}$$

(ii) $$AC=\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} 4 & 1 & 0 \\ 5 & 2 & 7 \end{pmatrix}$$

$$=\begin{pmatrix} 2\times 4+3\times 5 & 2\times 1+3\times 2 & 2\times 0+3\times 7 \\ 1\times 4+4\times 5 & 1\times 1+4\times 2 & 1\times 0+4\times 7 \end{pmatrix}$$

$$=\begin{bmatrix} 8+15 & 2+6 & 0+21 \\ 4+20 & 1+8 & 0+28 \end{bmatrix}$$

$$=\begin{pmatrix} 23 & 8 & 21 \\ 24 & 9 & 28 \end{pmatrix}$$

(iii) $CA=\begin{pmatrix} 4 & 1 & 0 \\ 5 & 2 & 7 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$, does not exist because matrix C has 3 columns and matrix A has 2 rows. i.e. $3 \neq 2$

(iv) $$CD=\begin{pmatrix} 4 & 1 & 0 \\ 5 & 2 & 7 \end{pmatrix}\begin{pmatrix} -1 & 6 \\ 2 & 4 \\ 5 & 1 \end{pmatrix}$$

$$=\begin{pmatrix} 4\times -1+1\times 2+0\times 5 & 4\times 6+1\times 4+0\times 1 \\ 5\times -1+2\times 2+7\times 5 & 5\times 6+2\times 4+7\times 1 \end{pmatrix}$$

$$=\begin{pmatrix} -4+2+0 & 24+4+0 \\ -5+4+35 & 30+8+7 \end{pmatrix}$$

$$=\begin{pmatrix} -2 & 28 \\ 34 & 45 \end{pmatrix}$$

TRANSPOSE OF A MATRIX

Given an $m \times n$ matrix $A=(a_{ij})$ then its transpose is $A^T=(a_{ji})$

NOTE: This is an $n \times m$ matrix

In other ways, if the rows and columns of matrix A are interchanged, the resulting matrix $A^/$ (sometimes A^T) is called the *transpose of A*.

Example

If $A=\begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$ and $B=\begin{pmatrix} 1 & 5 \\ 2 & 6 \\ -1 & 7 \end{pmatrix}$, then find A^T and B^T

Solutions

(i) $A^T=\begin{pmatrix}1 & 4\\ 3 & 2\end{pmatrix}$

(ii) $B^T=\begin{pmatrix}1 & 2 & -1\\ 5 & 6 & 7\end{pmatrix}$

DETERMINANTS OF MATRICES

Given a 2×2 matrix $A=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$, we define its determinant as

$\det A=\begin{vmatrix}a & b\\ c & d\end{vmatrix}$

=ad-bc

Instead of detA, we can write the determinant of A as $|A|$.

Only square matrices have determinants.

Example

If $A=\begin{pmatrix}5 & 10\\ -1 & 3\end{pmatrix}$, find $|A|$

Solution

$|A|$=5x3-10x(-1)

=15+10

=25

PROPERTIES OF DETERMINANTS

1. The determinant of a matrix and its transpose are equal

i.e. $A=\begin{pmatrix}2 & 5\\ 3 & 1\end{pmatrix}$

$|A|$=2-15

=-13

$A^T=\begin{pmatrix}2 & 3\\ 5 & 1\end{pmatrix}$

$|A^T|$=2-15

=-13

2. If two rows or columns of A are equal then detA is equal to zero (0)

i.e. $A=\begin{pmatrix}2 & 5\\ 2 & 5\end{pmatrix}$, then $|A|$=2x5-5x2

=10-10

=0

3. If a row or column consists entirely of zeros then detA=0

e.g. A=$\begin{pmatrix} 2 & 5 \\ 0 & 0 \end{pmatrix}$ then $|A|$=2x0-5x0

=0-0

=0

4. If a matrix A is upper or lower triangular then $|A|=a_{11}a_{22}a_{33}a_{44}\ldots a_{nn}$

5. The determinant of matrix AB is the determinant of their product

i.e. $|AB|=|A||B|$

DETERMINANTS OF 3×3 *MATRICES*

Let A=$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

We define the determinant of A as

$\det A=a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}-a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}+a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{33} \end{vmatrix}$

$=a_{11}(a_{22}a_{33}-a_{23}a_{32})-a_{12}(a_{21}a_{33}-a_{23}a_{31})+a_{13}(a_{21}a_{33}-a_{22}a_{31})$

$=a_{11}a_{22}a_{33}-a_{11}a_{23}a_{32}-a_{12}a_{21}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{33}-a_{13}a_{22}a_{31}$

$=a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{33}-a_{11}a_{23}a_{32}-a_{12}a_{21}a_{33}-a_{13}a_{22}a_{31}$

$=a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{33}-(a_{11}a_{23}a_{32}+a_{12}a_{21}a_{33}+a_{13}a_{22}a_{31})$

This way of finding determinant is called*sarrus rule*

A quick way to remember sarrus rule is to use basket weave method or Sarrus rule.

i.e. $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}\begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{matrix}$

$(a_{13}a_{22}a_{31}+a_{11}a_{23}a_{32}+a_{12}a_{21}a_{33})$ $(a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32})$

Then $\det A=(a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32})-(a_{13}a_{22}a_{31}+a_{11}a_{23}a_{32}+a_{12}a_{21}a_{33})$

Example

If A=$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 2 & 2 & 1 \end{pmatrix}$, then find $|A|_1$

Solution

$|A|$=3+16+6-(18+8+2)

=25-28

=-3

Example

If B=$\begin{pmatrix} 2 & 0 & 4 \\ 1 & 3 & 4 \\ -2 & 1 & 1 \end{pmatrix}$, then find $|B|$

Solution

$|B|$=6+0+4-(-24+8+0)

=10+18

=28

MINORS

The determinant of the matrix formedby deleting the ith row and j^{th}column is called the minor of a_{ij} and we denote it as M_{ij}

For example if A=$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, then the minor of a_{11}

$M_{11}=\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{3} \end{vmatrix}$

$M_{12}=\begin{vmatrix} a_{12} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

So we re-write the determinant of A (3x3) as $detA=a_{11}M_{11}-a_{12}M_{12}+a_{13}M_{13}$

Example

If A=$\begin{pmatrix} 1 & 2 & -1 \\ 5 & 6 & 7 \\ 3 & -2 & -5 \end{pmatrix}$, then find M_{11} and M_{32}

Solution

$M_{11}=\begin{vmatrix} 6 & 7 \\ -2 & -5 \end{vmatrix}$

=-30-(-14)

=-30+14

=-16

$M_{32}=\begin{vmatrix} 1 & -1 \\ 5 & 7 \end{vmatrix}$

=7- -5

=12

COFACTORS

The cofactor of a_{ij} is defined as $C_{ij}=(-1)^{i+j}M_{ij}$

Example

If A=$\begin{pmatrix} 3 & -1 & 2 \\ 4 & 5 & 6 \\ 7 & 1 & 2 \end{pmatrix}$, then find C_{11} and C_{12}

Solutions

$M_{11}=\begin{vmatrix}5 & 6\\ 1 & 2\end{vmatrix}$

$=4$

Then $C_{11}=(-1)^{1+1}4$

$=4$

(iii) $M_{12}=\begin{vmatrix}4 & 6\\ 7 & 2\end{vmatrix}$

$=8-42$

$=-34$

Then $C_{12}=(-1)^{1+2}-34$

$=34$

0ur definition of $\det A$ for a 2×2 matrix allows us to calculate det $A_{n\times n}$ y reducing the problem to several determinants of 2×2 matrices. In the definition, we show an example alongside.

Definition 1.6.2 For an $n\times n$ matrix *A,detA* number which can be calculated in the following way:

1.

Choose any row of A, say, a_{i1}, a_{i2}, a_{i3},...,a_{in} (row i, where i is arbitrarily chosen).

2.

Let M_{ij} be the $(n-1)\times(n-1)$ matrix obtained from A by crossing out the i th row and j th column of A

3.

$$\begin{aligned}\det A &= (-1)^{i+1}a_{i1}\det M_{i1}\\ &+(-1)^{i+2}a_{i2}\det M_{i2}\\ &+\cdots\\ &+(-1)^{i+n}a_{in}\det M_{in}\end{aligned}$$

$$A=\begin{pmatrix}1 & -2 & 2\\ 4 & 1 & -3\\ 2 & 1 & 1\end{pmatrix}$$

Consider

1.

$$\begin{pmatrix}1 & -2 & 2\\ \mathbf{4} & \mathbf{1} & \mathbf{-3}\\ 2 & 1 & 1\end{pmatrix}$$

Choose row 2:

2.

$$M_{21} = \begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$$
$$M_{22} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$
$$M_{23} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

3.

$$\begin{aligned} \det A &= (-1)^{3}\mathbf{4}(-2-2) \\ &\quad + (-1)^{4}\mathbf{1}(1-4) \\ &\quad + (-1)^{5}(-\mathbf{3})[1-(-4)] \\ &= 16 - 3 + 15 = 28 \end{aligned}$$

or the number can be calculated in an alternative way:

1.

Choose any column of A, say,

$$\begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{pmatrix}$$

(column *j*, *j* arbitrary).

2.

Let M_{ij} be as in 2 above.

3.

$$\begin{aligned} \det A &= (-1)^{1+j} a_{1j} \det M_{1j} \\ &\quad + (-1)^{2+j} a_{2j} \det M_{2j} \\ &\quad + \cdots \\ &\quad + (-1)^{n+j} a_{nj} \det M_{nj} \end{aligned}$$

1.

$$\begin{pmatrix} 1 & -2 & \mathbf{2} \\ 4 & 1 & -\mathbf{3} \\ 2 & 1 & \mathbf{1} \end{pmatrix}$$

Choose column 3:

2.

$$M_{13} = \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix}$$
$$M_{23} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$
$$M_{33} = \begin{pmatrix} 1 & -2 \\ 4 & 1 \end{pmatrix}$$

3.

$$\begin{aligned} \det A &= (-1)^4 \mathbf{2}(4-2) \\ &\quad + (-1)^5(-\mathbf{3})[1-(-4)] \\ &\quad + (-1)^6 \mathbf{1}[1-(-8)] \\ &= 4 + 15 + 9 = 28 \end{aligned}$$

Note that the determinant of an $n \times n$ matrix is defined as a sum of ± 1 times determinants of $(n-1)\times(n-1)$ matrices. Each of those determinants is calculated as a sum of ± 1 times determinants of $(n-2)\times(n-2)$ matrices. Continuing this process, we work down to determinants of 2×2 matrices, which we know how to compute. It is not at all obvious (but true) that $\det A$ is independent of the choice of row or column for calculation. The determinant is not defined for nonsquare matrices.

Example 2 Calculate

$$\det \begin{pmatrix} 1 & 2 & -1 \\ 4 & 1 & 0 \\ 3 & -1 & 0 \end{pmatrix}$$

(a) by choosing a row and (b) by choosing a column.

Solution

First we show a simple way to remember whether to write $+1$ or -1 in using the definition. For a_{11}, $(-1)^{1+1} = +1$; when we move to the right or down, we have a sign of -1 since $(-1)^{2+1} = -1$ and $(-1)^{1+2} = -1$. That is, the sign changes as we move one row up or down, or one column left or right. So for a 3×3 matrix the signs are

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

(a)

Use the second row. This is preferable to using row 1 because the zero entry will be multiplied by another determinant, which will give a zero. Using the second row (see Fig. 1.6.1) gives

$$\det\begin{pmatrix} 1 & 2 & -1 \\ \mathbf{4} & \mathbf{1} & \mathbf{0} \\ 3 & -1 & 0 \end{pmatrix} = -4\det\begin{pmatrix} 2 & -1 \\ -1 & 0 \end{pmatrix} + \mathbf{1}\det\begin{pmatrix} 1 & -1 \\ 3 & 0 \end{pmatrix} - \mathbf{0}\det\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$$

$$= -4(-1) + 1(3) - 0$$

$$= 7$$

(b)
Use column 3 since it has the most zeros.

$$\det\begin{pmatrix} 1 & 2 & \mathbf{-1} \\ 4 & 0 & \mathbf{1} \\ 3 & -1 & \mathbf{0} \end{pmatrix} = +(-\mathbf{1})\det\begin{pmatrix} 4 & 1 \\ 3 & -1 \end{pmatrix} - \mathbf{0}\det\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} + \mathbf{0}\det\begin{pmatrix} 1 & 2 \\ 4 & 1 \end{pmatrix}$$

$$= -1(-7) - 0 + 0 = 7$$

Example 3 Calculate

$$\det\begin{pmatrix} 1 & -2 & 4 & 0 \\ 7 & 3 & 0 & 3 \\ -1 & 1 & -4 & 0 \\ 0 & 3 & 2 & 1 \end{pmatrix}$$

Solution
Choose column 4 since it has the most zeros. The signs are

$$\begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix}$$

So

$$\det\begin{pmatrix} 1 & -2 & 4 & \mathbf{0} \\ 7 & 3 & 0 & \mathbf{3} \\ -1 & 1 & -4 & \mathbf{0} \\ 0 & 3 & 2 & \mathbf{1} \end{pmatrix} = +\mathbf{3}\det\underbrace{\begin{pmatrix} 1 & -2 & 4 \\ -1 & 1 & -4 \\ 0 & 3 & 2 \end{pmatrix}}_{\text{Use column 1}}$$

$$+\mathbf{1}\det\underbrace{\begin{pmatrix} 1 & -2 & 4 \\ 7 & 3 & 0 \\ -1 & 1 & -4 \end{pmatrix}}_{\text{Use column 3}}$$

$$= \mathbf{3}\left[\mathbf{1}\det\begin{pmatrix} 1 & -4 \\ 3 & 2 \end{pmatrix} - (-1)\det\begin{pmatrix} -2 & 4 \\ 3 & 2 \end{pmatrix}\right]$$

$$+\mathbf{1}\left[\mathbf{4}\det\begin{pmatrix} 7 & 3 \\ -1 & 1 \end{pmatrix} + (-4)\det\begin{pmatrix} 1 & -2 \\ 7 & 3 \end{pmatrix}\right]$$

$$= 3[14 + (-16)] + [4(10) - 4(17)] = -34$$

Example 4 Calculate

$$\det\begin{pmatrix} 2 & -1 & 4 & 6 \\ 3 & -2 & 7 & 17 \\ 0 & 0 & 0 & 0 \\ -4 & 3 & -6 & 12 \end{pmatrix}$$

Solution
Choose row 3, since it has the most zeros.

$$\det\begin{pmatrix} 2 & -1 & 4 & 6 \\ 3 & -2 & 7 & 17 \\ 0 & 0 & 0 & 0 \\ -4 & 3 & -6 & 12 \end{pmatrix} = 0 \cdot \det(\ ) + 0 \cdot \det(\ ) + 0 \cdot \det(\ ) + 0 \cdot \det(\ )$$

$$= 0$$

This example illustrates the fact that if a matrix has a row (or column) containing all zeros, the determinant is zero.

Example 5 Calculate

$$\det\begin{pmatrix} 2 & 16 & 17 & 4 \\ 0 & -3 & 22 & -3 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Solution

Choose column 1 since it has the most zeros.

$$\det\begin{pmatrix} 2 & 16 & 17 & 4 \\ 0 & -3 & 22 & -3 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 2\det\underbrace{\begin{pmatrix} -3 & 22 & -3 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{Use column 1}} = +2(-3)\det\begin{pmatrix} & 0 \\ 0 & 1 \end{pmatrix}$$

$$= +2(-3)(6) = -36$$

INVERSE MATRICES

Given a square matrix A then its inverse (if it exists) is defined as A^{-1} and is such that $A.A^{-1}=A^{-1}.A=I_n$ where I_n is the identity matrix

A square matrix has an inverse if and only if its determinant is nonzero. Given a square matrix A the A^{-1} exists if and only if $|A| \neq 0$

Suppose $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then $A^{-1}=\frac{1}{|A|}\begin{pmatrix} d & -d \\ -c & a \end{pmatrix}$

Example

Find the inverse of each of the following matrices

(a) $\begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$

(b) $\begin{pmatrix} 4 & 2 \\ 3 & 2 \end{pmatrix}$

Solutions

Let $A=\begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$ then $A^{-1}=\frac{1}{|A|}\begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix}$

$$=\frac{1}{|12-10|}\begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix}$$

$$=\frac{1}{2}\begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix}$$

$$=\begin{pmatrix} \frac{3}{2} & \frac{-5}{2} \\ -1 & 2 \end{pmatrix}$$

(b) Let B=$\begin{pmatrix} 4 & 2 \\ 3 & 2 \end{pmatrix}$ then $B^{-1}=\frac{1}{|B|}\begin{pmatrix} 2 & -2 \\ -3 & 4 \end{pmatrix}$

$$=\frac{1}{|8-6|}\begin{pmatrix} 2 & -2 \\ -3 & 4 \end{pmatrix}$$

$$=\frac{1}{2}\begin{pmatrix} 2 & -2 \\ -3 & 4 \end{pmatrix}$$

$$=\begin{pmatrix} 1 & -1 \\ \frac{-3}{2} & 2 \end{pmatrix}$$

If a matrix has an inverse, then the inverse is unique
A matrix is called nonsingular (invertible) if its inverse exists and a matrix is called singular, (vertible) if its inverse does not exists.

MATRIX METHOD FOR SOLVING SIMULTANEOUS EQUATIONS
a.BY USING INVERSES
Consider the following equations
$ax+by=m_1$
$cx+dy=m_2$
In matrix form

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is called coefficient matrix

Let A=$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and suppose det$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ exists. Then we can hit both sides of * with A^{-1} to get $A^{-1}A\begin{pmatrix} x \\ y \end{pmatrix}=A^{-1}\begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$

$$\frac{1}{6-1}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}= A^{-1}\begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$$

Example
Solve the following simultaneous equations
4x+5y=2
2x+3y=0
Solution
In matrix form, we can rewrite as

$$\begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}^{-1} \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{12-10} \begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & \frac{-5}{2} \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3+0 \\ -2+0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

Therefore x=3 and y=-2

Example

Solve the following simultaneous equations

3x+y=1

x+2y=2

Solution

In matrix form, we write as

$$\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Let A= $\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ then $A^{-1} = \frac{1}{6-1} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$

$$= \frac{1}{5} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2}{5} & \frac{-1}{5} \\ \frac{-1}{5} & \frac{3}{5} \end{pmatrix}$$

But $\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{2}{5} & \frac{-1}{5} \\ \frac{-1}{5} & \frac{3}{5} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{2}{5} + \frac{-2}{5} \\ \frac{-1}{5} + \frac{6}{5} \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Therefore x=0 and y=1

b. BY USING ROW/COLUMN REDUCTION

Consider the following equations

$ax+by=m_1$

$cx+dy=m_2$

All the information contained in these equations may be summarized in the augmented matrix

$$\left(\begin{array}{cc|c} a & b & m \\ c & d & m \end{array}\right)$$

Where the vertical bar is used to separate the coefficients of the original unknowns from the right hand side.

The idea of Gaussian elimination is to reduce the augmented matrix to a matrix of the form $\left(\begin{array}{cc|c} 1 & 0 & c_1 \\ 0 & 1 & c_2 \end{array}\right)$

How do we reduce the augmented matrix?

Via row or column operations

(i). Interchange a row by a non-zero number.

(ii).Multiply a row by a non-zero number.

(iii). Add a multiple of one row to a different row.

Example

Solve the following pairs of simultaneous equations by Gaussian elimination method

(a) 4x+5y=2 and 2x+3y=0

(b) 3x+y=1 and x+2y=2

Solutions

(a) 4x+5y=2

2x+3y=0

Its augmented matrix is

$$\left(\begin{array}{cc|c} 4 & 5 & 2 \\ 2 & 3 & 0 \end{array}\right)$$

$\frac{r_1}{4} = r_1$

$$\left(\begin{array}{cc|c} 1 & \frac{5}{4} & \frac{1}{2} \\ 2 & 3 & 0 \end{array}\right)$$

$-2r_1+r_2 \rightarrow r_2$

$$\left(\begin{array}{cc|c} 1 & \frac{5}{4} & \frac{1}{2} \\ 0 & \frac{1}{2} & -1 \end{array}\right)$$

The matrix is now said to be in echelon form. Its equivalent to the system is

$$x+\frac{5}{4}y=\frac{1}{2}$$

$$\frac{1}{2}y=-1$$

From (2) y=-2

Substituting -2 for y in (1) we get

$$x+\frac{5}{4}\times -2=\frac{1}{2}$$

$$x-\frac{5}{2}=\frac{1}{2}$$

$$x=\frac{1}{2}+\frac{5}{2}$$

$$x=3$$

Therefore x=3 and y=-2

(ii) 3x+y=1

X+2y=2

In the augmented matrix

$$\left(\begin{array}{cc|c} 3 & 1 & 2 \\ 1 & 2 & 1 \end{array}\right)$$

$-3r_1+r_2 \rightarrow r_2$

$$\left(\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & -5 & -5 \end{array}\right)$$

This is equivalent to the system of equations

x+2y=2 …(1)

-5y=-5 …(2)

From (2) $-5y=-5 \Rightarrow y=1$

Substituting 1 for y in (1), we get

$$x+2\times 1=2$$

$$x=2-2$$

$$x=0$$

Therefore x=0 and y=1

c.BY USING DETERMINANTS

Cramer's rule is an interesting application of determinants, which we state now. This rule can be used to calculate solutions of simultaneous equations.

For calculating solutions it is primarily useful only when A is of small order and it is a square matrix. We do not prove Cramer's rule.

Let A be an $n \times n$ matrix. The system of equations

$$AX = B$$

has a unique solution if and only if $\det A \neq 0$. The solution can be calculated in that case as follows. Let A_k be the matrix obtained by replacing column k of A by the column matrix B. Then

$$x_1 = \frac{\det A_1}{\det A}$$

$$x_2 = \frac{\det A_2}{\det A}$$

$$\vdots$$

$$x_n = \frac{\det A_n}{\det A}$$

Example

Solve the following equations simultaneously by Cramer's rule;

3x + y = 1

x +2y= 2

Solution

In matrix form, we have

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

So $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$, so $|A| = 6 - 1 = 5$, $B \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$A_1 = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ detA$_1$ = 2-2 =0

$A_2 = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ detA$_2$ = 6-1= 5

$$x = \frac{|A_1|}{|A|} = \frac{0}{5} = 0$$

$$y = \frac{|A_2|}{|A|} = \frac{5}{5} = 1$$

Therefore x= 0 and y= 1

PRACTICAL METHOD OF FINDING A^{-1}

Find an $n \times n$ matrix $\left[A|I_n\right]$ and perform elementary row operations that transform it to $\left[I_{n|} \mid A^{-1}\right]$

Example

Find A^{-1} if (a) $A=\begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$

(b) $A=\begin{pmatrix} 4 & 2 \\ 3 & 2 \end{pmatrix}$

Solutions

(a) $\left(\begin{array}{cc|cc} 4 & 5 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array}\right)$

$\frac{r_1}{4} \rightarrow r_2$

$\left(\begin{array}{cc|cc} 1 & \frac{5}{4} & \frac{1}{4} & 0 \\ 0 & \frac{-5}{2} & \frac{-1}{2} & 1 \end{array}\right)$

$\frac{-2}{5} r_{2 \rightarrow r_2}$

$\left(\begin{array}{cc|cc} 1 & \frac{5}{4} & \frac{1}{4} & 0 \\ 0 & 1 & -1 & \frac{-2}{5} \end{array}\right)$

$\frac{-5}{4} r_2 + r_1 \rightarrow r_1$

$\left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{-1}{5} & \frac{-2}{5} \end{array}\right)$

Therefore $A^{-1}\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{-1}{5} & \frac{-2}{5} \end{pmatrix}$

(b) $\left(\begin{array}{cc|cc} 4 & 2 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array}\right)$

$\frac{r_1}{4} \to r_1$

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{4} & 0 \\ 3 & 2 & 0 & 1 \end{array}\right)$$

$-3r_1+r_2 \to r_2$

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & \frac{1}{2} & \frac{-3}{4} & 1 \end{array}\right)$$

$2r_2 \to r_2$

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{7}{4} & -1 \\ 0 & 1 & -3 & 2 \end{array}\right)$$

Therefore A^{-1} is $\begin{pmatrix} \frac{7}{4} & -1 \\ -3 & 2 \end{pmatrix}$

TUTORIAL8: MATRIX ALGEBRA

1. A matrix and an entry of the matrix are given. State the entry's row and column.

(a) $\begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix}; -3$ (b) $\begin{pmatrix} 1-i & -1 & 2 \\ 0 & 1 & 1 \\ 3i & -6+i & 7 \end{pmatrix}; -6+i$

(c) $\begin{pmatrix} -3 & 2 & 1 \\ 4 & 6 & 7 \end{pmatrix}$; 4 (d) $\begin{pmatrix} 3 & 2 \\ 4 & 1 \\ 6 & 7 \\ 5 & 9 \end{pmatrix}$; 5

2. Let

$$A = \begin{pmatrix} -1 & 2 \\ 2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \end{pmatrix} \quad D = \begin{pmatrix} 2 & 0 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$$

$$E = \begin{pmatrix} i & 1 \\ -1 & i \end{pmatrix}$$

For each part of the problem, perform the matrix calculation shown. If a calculation cannot be performed, say so.

(a) $3A$ (b) $A + C$ (c) CD
(d) $A - B$ (e) $3A + AB$ (f) $AB + DC$
(g) AB + CD (h) $(2 + i)A$ (i) iE
(j) $iA + E$ (k) AE (l) $A(B + C)$
(m) $AB + AC$ (n) $A + B$ (o) $B + A$
(p) AA (q) EE (r) $(AB)C$

4. Use Cramer's rule to solve the following systems.

(a) $\begin{aligned} 3x - 2y &= 7i \\ 4x + y &= 8 \end{aligned}$ (b) $\begin{aligned} x - 2y + z &= 3 \\ -x - y - z &= 5 \\ 3x + y - 7z &= -3 \end{aligned}$

TOPIC 9: TRIGONOMETRY

OBJECTIVES

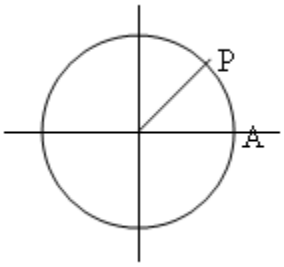
By the end of this topic,students should be able to;

(a) State two units of angles.
(b) State six trigonometric ratios.
(c) Simplify trigonometric functions using addition, subtraction, double angle formulae and many more.

(d) Sketch trigonometric functions.
(e) Solve trigonometric equations.

Trigonometry is the study of angle measurement, and in particular the study of triangle measurement and calculation.

DEGREE MEASURE
Consider the following unit circle (a circle whose radius is 1)



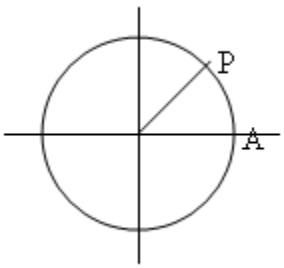
Then we shall say that angle AOP has degree measure if

$$\frac{\text{Length of an arc AP}}{\text{Circumference}} = \frac{\theta}{360^0}$$

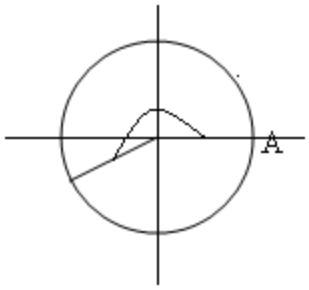
$$\theta = \frac{360^0 \text{x length of arc AP}}{2\pi}$$

In this definition, please note that the length of arc AP is measured in an anticlockwise direction from A.

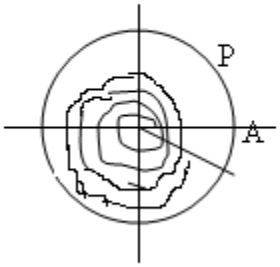
Example
Sketch the points obtained on the unit circle by the following angles
(a) 45^0 (b) 210^0 (c) 1437^0 (d) -80^0
Solutions
(a)



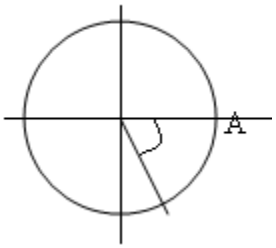
(b)



(c)



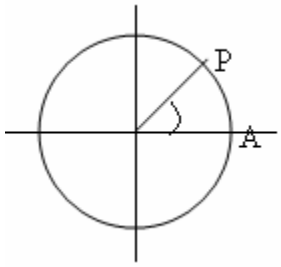
(d)



RADIAN MEASURE

Another way of measuring angles is to compare the length of an arc formed by the angles with the radius of the circle. The unit used in this method is called the radian

We shall say that angle AOP has radian measure if = length of arc AP



In the figure above the arc length AP is θ, so angle AOP has radian measure θ.Notice that if angle AOP is a right angle then

$\theta = \frac{\text{Circumference}}{4} = \frac{2\pi}{4} = \frac{\pi}{2}$

Thus $90 = \frac{\pi}{2}$ radians, $180 = \pi$ radians and 1 radian $= \frac{180^0}{\pi} \cong 57.296$

The following are useful to remember:

(1) $30^0 = \frac{\pi}{6} radians$

(2) $90^0 = \frac{\pi}{2} radians$

(3) $45^0 = \frac{\pi}{4} radians$

(4) $60^0 = \frac{\pi}{3} radians$

(5) $180^0 = \pi radians$

(6) $360^0 = 2\pi radians$

Example

Convert the following to radians

(a) 135^0

(b) 210^0

Solutions

(a) if $180^0 = \pi radians$

then 135^0 = less

so $\frac{135^0}{180^0} \times \pi radians = o.75\pi radians (\frac{3}{4}\pi rad)$

(b) If $180^0 = \pi radians$

Then 210^0 = more

So $\frac{210^0}{180^0} \times \pi radians = 1.167\pi radians$

Example

Convert the following into degrees

(a) $\frac{3\pi}{2} rad$

(b) 3 rad

Solutions

(a) If π rad $= 180^0$

Then $\frac{3\pi}{2} rad$ = more

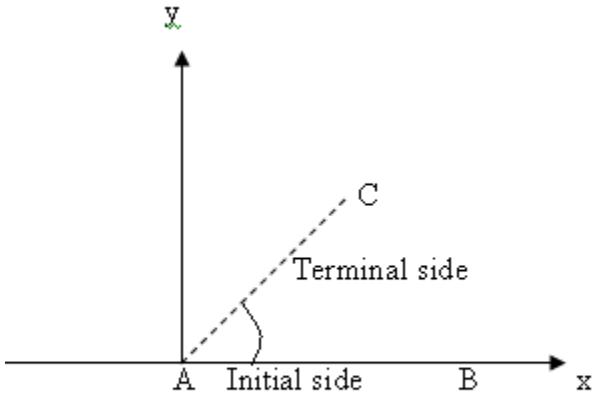
Therefore $\frac{\frac{3\pi}{2} rad \times 180^0}{\pi} = 270^0$

(b) If π rad $=180^0$

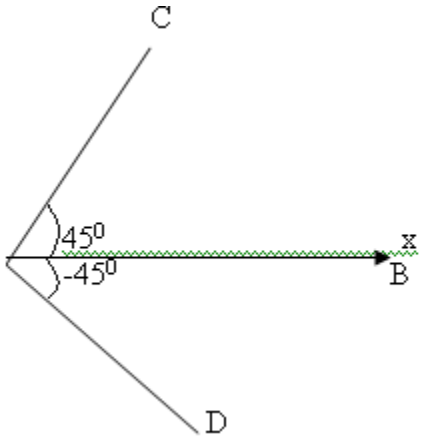
Then 3 rad = less

Therefore $\frac{3rad \times 180^0}{\pi} = 171.9^0$

STANDARD POSITION



We say that an angle is in the standard position when its vertex A is at the origin of the x-y plane and its initial side lies along the positive x-axis, where its terminal side has rotated.

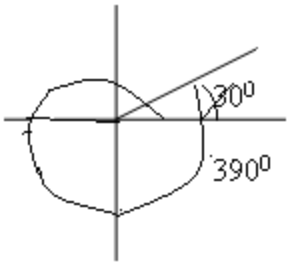


If the terminal side AC has rotated in a counter-clockwise direction, we say that the angle is positive. While if is has rotated in a clockwise direction, as AD the angle is negative.

COTERMINAL ANGLES

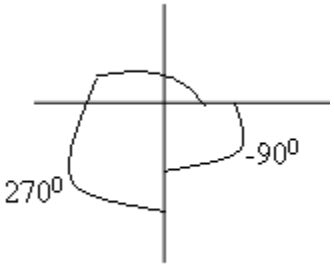
Angles are co terminal if when in the standard position, they have the same terminal side. For example, 30^0 is co terminal with $290^0(360^0 + 30^0)$

i.e



-90^0 is co terminal with 270^0

i.e



ARCLENGTH

Recall that $\frac{\text{Length of an arc AP}}{\text{Circumference}} = \frac{\theta}{360^0}$

$$\text{Length of arc AP} = \frac{\theta}{360^0} \times circumference$$

Remember also that circumference of a circle is $2\pi r$ and that $360^0 = 2\pi$

So the length of arc AP $= \frac{\theta}{2\pi} \times 2\pi r$

$= \theta r$

Thus the arc length AP= rθ

Example

Find the arc length of a sector of radius 4cm and angle 1.5 rad.

Solution

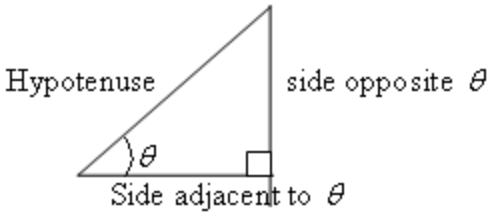
Arc length = r θ

= 4 cm × 1.5

= 6 cm

TRIGONEMETRIC FUNCTIONS

Consider the right-angled triangle below



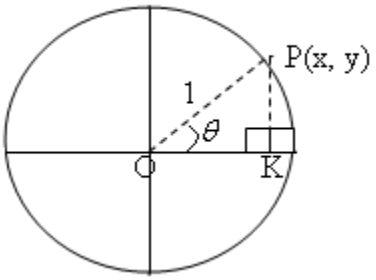
Using the right-angle triangle, we define three basic trigonometric functions

Sine θ =side opposite to θ / Hypotenuse θ

Cosine θ =side adjacent to θ / Hypotenuse θ

Tangent θ=side opposite θ / Side adjacent to θ

Consider the following unit circle



We define

1. cosine θ =cos θ= $\frac{x}{1} = x$
2. sine $\theta = \sin\theta = \frac{y}{1} = y$
3. $\tan gent\,\theta = \tan\theta = \frac{y}{x} = \frac{\sin\theta}{\cos\theta}$
4. $\sec ant\,\theta = \sec\theta = \frac{1}{\cos\theta} = \frac{1}{x}$
5. $\cos ecant\,\theta = \cos ec\,\theta = \frac{1}{\sin\theta} = \frac{1}{y}$
6. $\cot angent\,\theta = \cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{x}{y}$

These six functions of θ are called trigonometric functions

Using the above right-angle triangle OKP

$OP^2=OK^2 +OP^2$ (Pythagoras theorem)

$1^2 =x^2 + y^2$

$1 =\cos^2\theta +\sin^2\theta$ since cosθ= x and sinθ= y

Or $\cos^2\theta + \sin^2\theta = 1$ … (1)

Dividing (1) by $\sin^2\theta$ (when $\sin^2\theta \neq 0$)

i.e $\frac{\cos^2\theta}{\sin^2\theta} + \frac{\sin^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta}$

$\cot^2\theta + 1 = \operatorname{cosec}^2\theta$ . . . (2)

Dividing (1) by $\cos^2\theta$

i.e $\frac{\cos^2\theta}{\cos^2\theta} + \frac{\sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$

$1 + \tan^2\theta = \sec^2\theta$ . . . (3)

x and y-axisdivide the plane into four quadrants.

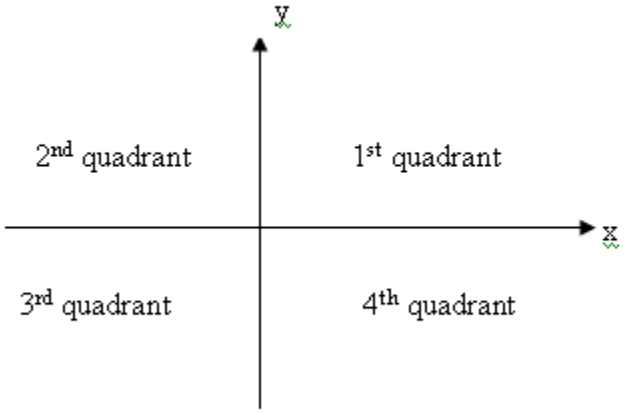
A quadrant between the positive x-axis and the positive y-axis is called first quadrant.

The second quadrant is between the positive y-axis and the negative x-axis.

The third quadrant is between the negative x-axis and the negative y-axis.

The fourth quadrant is between the positive x-axis and the negative y-axis

i.e



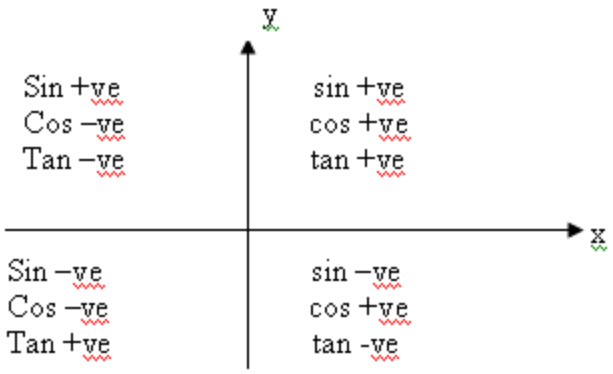
If P is in the first quadrant, $\sin\theta > 0$, $\cos\theta > 0$, $\tan\theta > 0$

If P is in the second quadrant,$\sin\theta > 0$, $\cos\theta < 0$, $\tan\theta < 0$

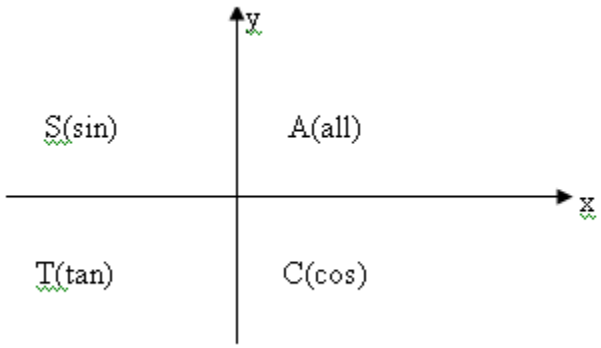
If P is in the third quadrant,$\sin\theta < 0$, $\cos\theta < 0$, $\tan\theta > 0$

If P is in the fourth quadrant, $\sin\theta < 0$, $\cos\theta > 0$, $\tan\theta < 0$

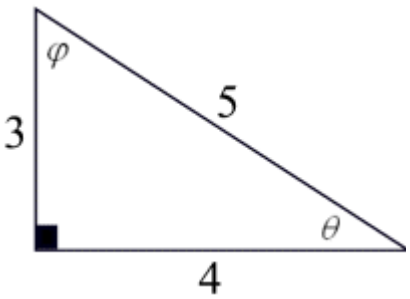
All this information is summarized in the figure below



The signs of sin, cos and tan can be remembered using the letters *CAST*, placed as in the figure below.



COFUNCTIONS



The sine of an acute angle is equal to the cosine of its complement.

In the diagram at the left, the measures of the angles designated by φ and θ add to 90º. These angles are complementary angles.

$$\theta + \varphi = 90^\circ \begin{cases} \varphi = 90^\circ - \theta \\ \theta = 90^\circ - \varphi \end{cases}$$

In this triangle, $\sin\theta = \frac{3}{5}$ and $\cos\varphi = \frac{3}{5}$.

The cosine of an acute angle is equal to the sine of its complement.

$$\sin\theta = \cos\varphi \begin{cases} \sin\theta = \cos(90^\circ - \theta) \\ \cos\varphi = \sin(90^\circ - \varphi) \end{cases}$$

Such examinations, lead us to three sets of "cofunction" identities:

Sine and **cosine** are cofunctions.	**Tangent** and **cotangent** are cofunctions.	**Secant** and **cosecant** are cofunctions.
$\sin\theta = \cos(90^\circ - \theta)$	$\tan\theta = \cot(90^\circ - \theta)$	$\sec\theta = \csc(90^\circ - \theta)$
$\cos\theta = \sin(90^\circ - \theta)$	$\cot\theta = \tan(90^\circ - \theta)$	$\csc\theta = \sec(90^\circ - \theta)$

Also written:

Sine and **cosine** are cofunctions.	**Tangent** and **cotangent** are cofunctions.	**Secant** and **cosecant** are cofunctions.
$\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$	$\tan\theta = \cot\left(\frac{\pi}{2} - \theta\right)$	$\sec\theta = \csc\left(\frac{\pi}{2} - \theta\right)$
$\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$	$\cot\theta = \tan\left(\frac{\pi}{2} - \theta\right)$	$\csc\theta = \sec\left(\frac{\pi}{2} - \theta\right)$

Notice the connection of the letters ***C***&***O***:
* sine and***co***sine ***co***functions
* tangent and ***co***tangent ***co***functions
* secant and ***co***secant ***co***functions
* ***co***mplementary

Examples:

1.	Find a value of θ for which $\sin\theta = \cos 15^\circ$ is true.	Solution: Since sine and cosine are cofunctions, we know that the value will be the complement of 15º. Answer: 75^0

2.	Write the expression tan 265º as the function of an acute angle of measure less than 45º.	Solution: angle is in the third quadrant with a reference angle of 85º. So we could write: tan 265º = tan 85º. But the question

		wants an angle less than 45º, so we use our cofunctions. tan 85º = cot 5º. Answer: cot 5º

EVEN AND ODD FUNCTIONS

A function is called even if f(x)=f(-x) and a function is odd if f(x)=-f(-x) and function is called odd if f(x) = - f(-x).

Example

The functions f, g and h are defined by $f(x)=x^3-2x$, $g(x)=1+x^2-5x^4$ and $h(x)=3x^2+2x$. Determine which of these functions are even or odd.

Solutions

(a) $f(x)=x^3-2x$

$=(-x)^3-2(-x)$

$=x^3-2x$

Therefore, f(x) is an odd function

(b) $g(x)=1+x^2-5x^4$

$g(-x)=1+(-x)^2-5(-x)^4$

$g(-x)=1+x^2-5x^4=g(x)$

since g(-x)=g(x) then , $g(x)=1+x^2-5x^4$ is an even function.

(c)$h(x)=3x^2+2x$

$h(-x)=3(-x)^2+2(-x)$

$h(-x)=3x^2-2x$

since $h(-x) \neq h(x)$ and $h(-x) \neq -h(x)$, the function h(x) is neither even nor odd

EXAMPLE

Prove the identity $\sin\theta\tan\theta \equiv \sec\theta-\cos\theta$

Proof

The way to proceed is to start with one side and show that it equals the other . Letting LHS and RHS stand for left-hand side and right hand side respectively, we have

LHS $=\sin\theta\tan\theta$

$$=\sin\theta\frac{\sin\theta}{\cos\theta}$$

$$=\frac{\sin^2\theta}{\cos\theta}$$

$$=\frac{1-\cos^2\theta}{\cos\theta}$$

$$=\frac{1}{\cos\theta}-\frac{\cos^2\theta}{\cos\theta}$$

$$=\sec\theta-\cos\theta=\text{RHS}$$

Example

Prove the identity $(\cot\theta - \operatorname{cosec}\theta)^2 = \dfrac{1-\cos\theta}{1+\cos\theta}$

Proof

LHS= $(\cot\theta - \operatorname{cosec}\theta)^2$

$$=\left(\frac{\cos\theta}{\sin\theta}-\frac{1}{\sin\theta}\right)$$

$$=\left(\frac{\cos\theta-1}{\sin\theta}\right)^2$$

$$=\frac{(\cos\theta-1)(\cos\theta+1)}{1-\cos^2}$$

$$=\frac{(\cos\theta-1)(\cos\theta+1)}{(\cos\theta+1)(\cos\theta+1)}$$ factorizing the denominator

$$=\frac{\cos\theta-1}{\cos\theta+1}=\text{RHS}$$

ADDITION AND DOUBLE ANGLE FORMULAE

ADDITION FORMULAE FOR COSINE

Cos(A+B)=cosAcosB-sinAsinB

Cos(A-B)= cosAcosB+sinAsinB

ADDITION ANGLE FORMULA FOR SINE

Sin(A+B) = sinAcosB+sinBcosA

Sin(A-B) = sinAcosB-sinBcosA

ADDITION ANGLE FORMULA FOR TANGENT

$\text{Tan(A+B)} = \dfrac{\text{tanA+tanB}}{1-\ \text{tanAtanB}}$

$\text{Tan(A-B)} = \dfrac{\text{tanA- tanB}}{1+\ \text{tanAtanB}}$

ADDITION AND DOUBLE ANGLE FORMULAE

ADDITION FORMULAE FOR COSINE

Cos(A+B)=cosAcosB-sinAsinB

Cos(A-B)= cosAcosB+sinAsinB

ADDITION ANGLE FORMULA FOR SINE

Sin(A+B) = sinAcosB+sinBcosA

Sin(A-B) = sinAcosB-sinBcosA

ADDITION ANGLE FORMULA FOR TANGENT

$\text{Tan(A+B)} = \dfrac{\text{tanA+tanB}}{2-\ \text{tanAtanB}}$

Tan(A-B) = $\frac{\text{tanA- tanB}}{\text{1+ tanAtanB}}$

Example

Prove that Tan(A+B) = $\frac{\text{tanA+tanB}}{\text{1-tanAtanB}}$

Proof

LHS=tan(A+B)

$$=\frac{\sin(A+B)}{\cos(A+B)}$$

$$=\frac{\sin A\cos B+\sin A\cos B}{\cos A\cos B-\sin A\sin B}$$

Dividing the numerator and denominator by cosAsinB, we get

$$=\frac{\frac{\sin A\cos B}{\cos A\cos B}+\frac{\sin A\cos B}{\cos A\cos B}}{\frac{\cos A\cos B}{\cos A\cos B}-\frac{\sin A\sin B}{\cos A\cos B}}$$

$$=\frac{\tan A+\tan B}{1-\tan A\tan B}=\text{RHS}$$

Example

(a) Find $\sin 15^0$ in surd form

(b) Find the exact value of $\cos 75^0$

Solutions

(a) $\sin 15^0=\sin(60^0-45^0)$

$=\sin 60^0\cos 45^0-\sin 45^0\cos 60^0$

$$=\frac{\sqrt{3}}{2}\times\frac{1}{\sqrt{2}}-\frac{1}{2}\times\frac{1}{\sqrt{2}}$$

$$=\frac{\sqrt{3}}{2\sqrt{2}}-\frac{1}{2\sqrt{2}}$$

$$=\frac{\sqrt{3}-1}{2\sqrt{2}}$$

(b) $\cos 75^0=\cos(45^0+30^0)$

$=\cos 45^0\cos 30^0-\sin 45^0\sin 30^0$

$$=\frac{1}{\sqrt{2}}\times\frac{\sqrt{3}}{2}-\frac{1}{2}\times\frac{1}{\sqrt{2}}$$

DOUBLE ANGLE FORMULA FOR COSINE

Recall that cos(A+B)= cosAcosB-sinAsinB

Then cos(A+A)= cosAcosA-sinAsinA

$\text{Cos2A}=\cos^2 A-\sin^2 A \quad (1)$

Since $\sin^2 A+\cos^2 A=1$ then it follows that $\cos^2 A=1-\sin^2 A$ and $\sin^2 A=1-\cos^2 A$

So (1) becomes $\cos 2A=1-\sin^2 A-\sin^2 A$

$=1-2\sin^2 A \quad (2)$

So (1) becomes $\cos 2A=\cos^2 A-(1-\cos^2 A)$

$=2\cos^2 A-1$

In summary, $\cos 2A=\begin{cases}\cos^2 A-\sin^2 A \\ 1-2\sin^2 A \\ 2\cos^2 A-1\end{cases}$

DOUBLE ANGLE FORMULA FOR SINE

Recall that sin(A+B)=sinAcosB+sinBcosA so letting a=b, we obtain

Sin(A+A)=sinAcosA+sinAcosA

$\text{Sin2A}=2\sin A\cos A$

DOUBLE ANGLE FORMULA FOR TANGENT

Recall that $\tan(A+B)=\dfrac{\tan A+\tan B}{1-\tan A\tan B}$. Letting A=B, we have

$\text{Tan}(A+A)=\dfrac{\tan A+\tan A}{1-\tan A\tan A}$

$\text{Tan2A}=\dfrac{2\tan A}{1-\tan^2 A}$

Example

Find $\cos 2\theta$ when $\cos\theta=-\dfrac{1}{2}$

Solution

$\text{Cos2}\theta=2\cos^2\theta-1$ (double angle formula)

$=2\text{x}\left(\dfrac{-1}{2}\right)^2-1$

$=2\text{x}\dfrac{1}{4}-1$

$=\dfrac{1}{2}-1$

$=-\dfrac{1}{2}$

Example

Prove that $2\sin^2 x=\tan x\sin 2x$

Proof

RHS=tanxsin2x

=2tanxsinxcosx since sin2x=2sinxcosx

$=2\frac{\sin x}{\cos x}\sin x\cos x$

=2sinxsinx

$= 2\sin^2 x$ =LHS

PRODUCT AND FACTOR FORMULAE

Recall that sin(A+B)=sinAcosB+sinBcosA (1)

sin(A-B)=sinAcosB-sinBcosA (2)

(1)+(2), we get

Sin(A+B)+sin(A-B)= sinAcosB+sinBcosA + sinAcosB-sinBcosA

=sinAcosB+ sinAcosB

=2sinAcosB *

(1)-(2), we get

Sin(A+B)-sin(A-B)= sinAcosB+sinBcosA – sinAcosB+sinBcosA

=sinAcosB+sinAcosB

=2sinAcosB #

Now let A+B = u and A-B = v

Adding and subtracting, we have

2A= u +v and 2B= u-v or

A= Cos(A+B)=cosAcosB-sinAsinB

Cos(A-B)= cosAcosB+sinAsinB

and B= $\frac{u-v}{2}$

Substituting $\frac{u+v}{2}$ for A, $\frac{u-v}{2}$ for B, u for A+B and v for A-B in * and #, we get

Sinu+sinv= $2\sin\frac{u+v}{2}\cos\frac{u-v}{2}$

Sinu-sinv= $2\sin\frac{u-v}{2}\cos\frac{u+v}{2}$

These are called factor sine formulae.

Recall that Cos(A+B)=cosAcosB-sinAsinB

Cos(A-B)= cosAcosB+sinAsinB

Adding and subtracting yield two more product formulae

Cos(A+B)+cos(A-B)=2cosAcosB

Cos(A+B)-cos(A-B)=- 2sinAsinB

Making the same substitution as before we obtain the cosine factor formulae

Cosu+cosv=$2\cos\frac{u+v}{2}\cos\frac{u-v}{2}$

Cosu-cosv= $-2\sin\frac{u-v}{2}\sin\frac{u+v}{2}$

Example

Find (a) $\sin 105^0 - \sin 15^0$

(b) $\cos 105^0 + \cos 15^0$. Without using a calculator

Solution

(a) $\sin 105^0 - \sin 15^0 = 2\sin\frac{105^0 - 15^0}{2}\cos\frac{105^0 + 15^0}{2}$

$= 2\sin 45^0 \cos 60^0$

$= 2x\frac{1}{2}\times\frac{\sqrt{2}}{2}$

$= \frac{\sqrt{2}}{2}$

(b) $\cos 105^0 + \cos 15^0 = 2\cos\frac{105^0 + 15^0}{2}\cos\frac{105^0 - 15^0}{2}$

$= 2\cos 60^0 \cos 45^0$

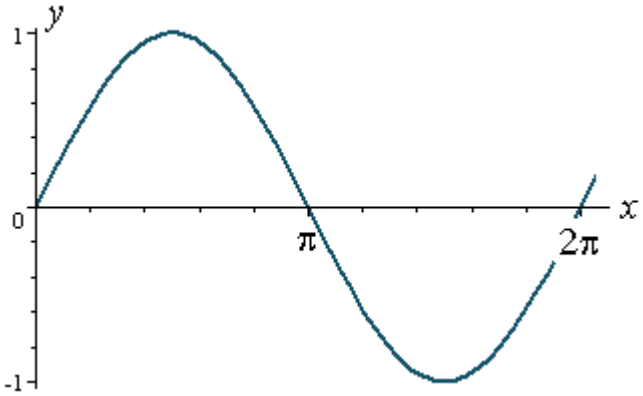
$= 2\times\frac{1}{2}\times\frac{1}{\sqrt{2}}$

$= \frac{1}{\sqrt{2}}$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

THE SINE CURVE

The sine curve is a wave which lies between the values +1 and -1; it is zero at multiplies of π (180^0)



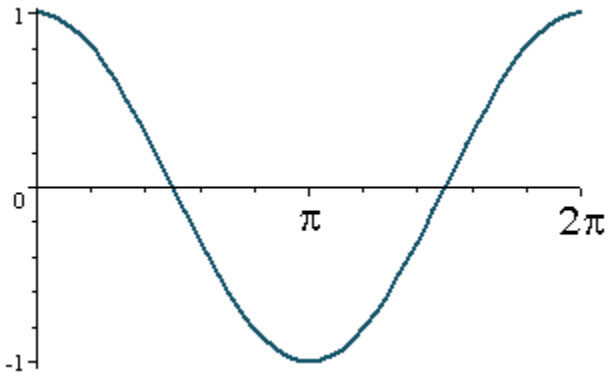
The **period** is the time it takes to go through one cycle and then start over again

Its amplitude is 1 and its period is 2π . The *a* in the expression ***y =a* sin*x*** represents the **amplitude** of the graph. It is an indication of how much **energy** the wave contains.

The amplitude is the distance from the "resting" position (otherwise known as the **mean value** or **average value**) of the curve.

THE COSINE CURVE

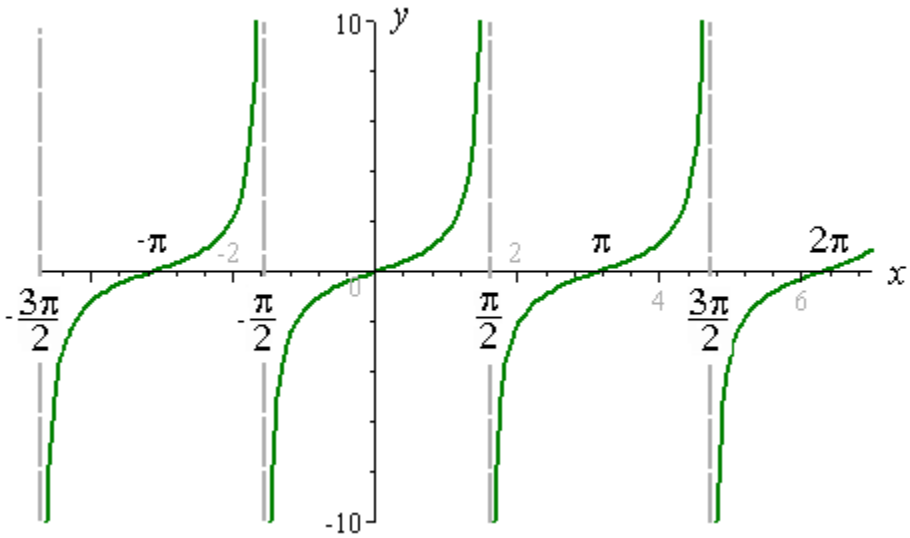
The cosine curve is a wave which lies between the values +1 and -1; it is zero at multiples of $\frac{\pi}{2}(90^0)$



Notice that the shape is the same as that of the graph of sine θ only that it has been move to the left by $\frac{\pi}{2}$ i.e $\sin\theta$ is obtained by moving the graph of $\cos\theta$ to the right by $\frac{\pi}{2}$

THE TANGENT CURVE

The tangent curve is quite different. Since $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\cos\theta = 0$ when θ is an odd multiple of $\frac{\pi}{2}$, $\tan\theta$ is infinite when θ is an odd multiple of $\frac{\pi}{2}$. The curve

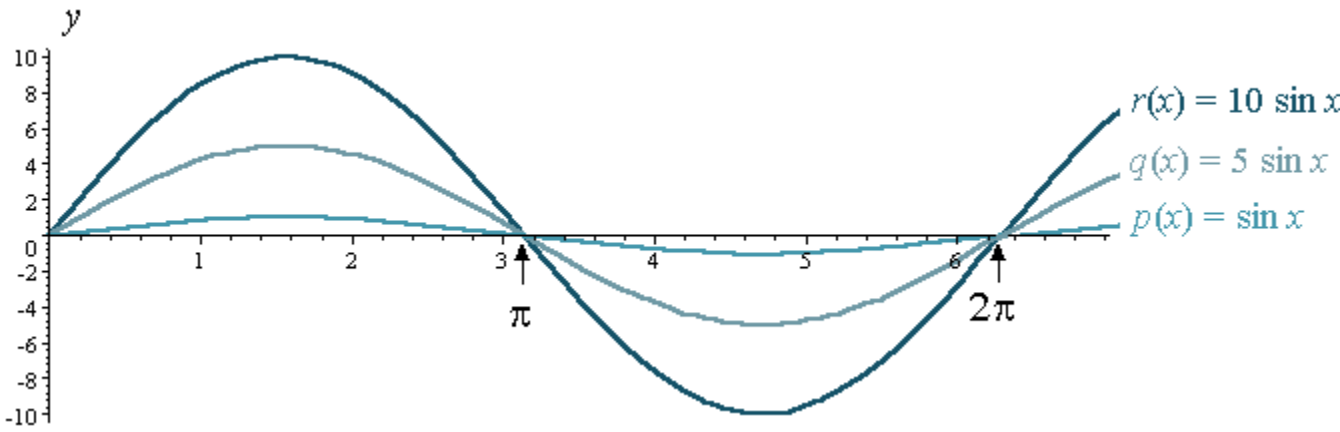


Consists an infinite number of branches, each formed by translating the branch for- $\frac{\pi}{2} < \theta < \frac{\pi}{2}$

Example

Sketch the graphs of y = sinx, y=5sinx and y = 10 sinx on the same axes.

Solutions



The graph of 5sinx is just like sinx except that the y values are five times as large, so that the graph will vary between y=-5 and y=5.

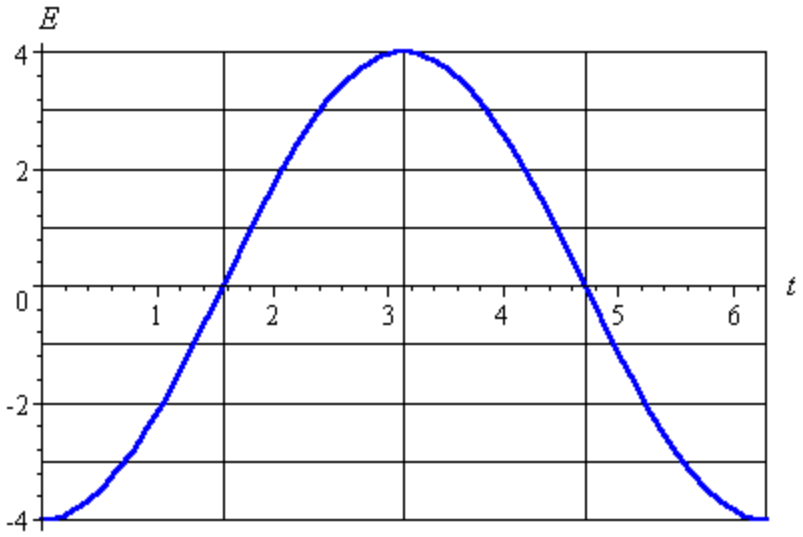
The graph of 10sinx is just like sinx except that the y values are ten times large, so that the graph will vary between y= 10 and y= -10.

Example

Sketch the graph of$E$ = -4 cos t

Solution

The variable $E$is used for "electro-motive force", another term for voltage.



Period = 2π
Amplitude = 4
Notice that:
The negative in front of the cosine has the effect of turning the cosine curve "upside down". That is, it is a mirror image in the horizontal *t* axis.
Amplitude is a positive number (it is a distance)

TRIGONOMETRIC EQUATIONS

EQUATION OF COS θ =a

Recall that its amplitude is 2π and cosine has positive sign in the 1st and 4th quadrants. All solutions is the first quadrant looks like $\theta_1+2n\pi$ where n is an integer and θ_1 is in the solution (principle value-that value that a calculator gives)
All solutions in the fourth quadrant looks like $-\theta+2n\pi$.
In general all solutions are $\pm\theta_1 + 2n\pi$.

Example

Solve $\cos\theta=\frac{1}{2}$

Solution

$\cos\theta=\frac{1}{2}$

$\theta=\cos^{-1}\frac{1}{2}$

$\theta=\frac{\pi}{3}$

Therefore all solutions are $\pm\frac{\pi}{3}+2n\pi$

EQUATION OF SIN $\theta = a$

Recall that sine is positive in the 1st and 2nd quadrant. All solutions in the first quadrant looks like $\theta_1 + 2n\pi$ where n is an integer θ_1 is the principle value. All solutions in the second quadrant are $\pi - \theta + 2n\pi$.

In general the solution of sin is $\theta_1 + 2n\pi$ or $\pi - \theta + 2n\pi$ where n is an integer

Example

Solve the equation $4\sin\theta = 3$

Solution

$4\sin\theta = 3$

$\sin\theta = \frac{3}{4}$

$\theta = \sin^{-1}(0.75)$

$\theta = 48.6^0$ (3 s.f)

The solution in the first quadrant is 48.6^0.

The solution in second quadrant is $180^0 - 48.6^0 = 131.4$

EQUATION OF TAN $\theta = a$

Recall that tan is positive in the 1st quadrant and 3rd quadrant.

The general solution for tangent is $\theta_1 + n\pi$ (recall coterminal angles)

Example

Solve the following equation $\tan\theta = -2$ between 0^0 and 360^0

Solution

$\tan\theta = -2$

$\theta = \tan^{-1}-2$

$= -63.4^0$

The principle value is 63.4^0. Since 2 has a negative sign then the solutions are found in the second and fourth quadrants

1st solution is in the second quadrant $= 180^0 - 63.4^0$

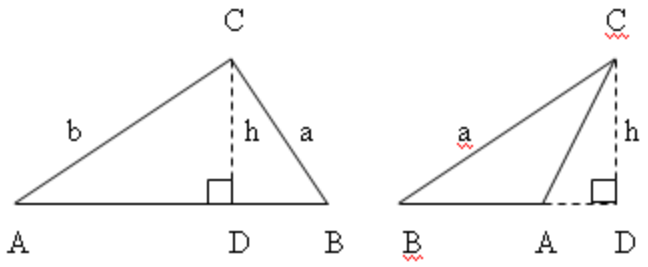
$= 116.6^0$

2nd solution is $360^0 - 63.4^0$

$= 296.6^0$

COSINE RULE

Consider the following triangle



A line drawn from C, perpendicular to AB produced if necessary divides triangle ABC into two right-angled triangles, CDA and CDB.

In triangle CDA, sinA=$\frac{h}{b} \Rightarrow h = b\sin A$ (sin180-A)=sinA

In triangle CDB, sinB=$\frac{h}{a} \Rightarrow h = a\sin B$

Therefore asinB=bsinA

i.e $\frac{a}{\sin A} = \frac{b}{\sin B}$

we can equally well divide triangle ABC into two right-angled triangles by drawing the perpendicular from A to BC (or from B toAC). This gives a similar result

i.e $\frac{a}{\sin A} = \frac{c}{\sin C}$

by combing the two results we produce the sine rule, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

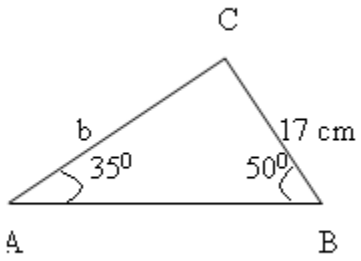
this particular version of the sine rule is useful for finding sides. To find angles the more convenient form to use is

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

Example

In triangle ABC, A=35^0, a=17cm and B=50^0. Find b.

Solution



$\frac{a}{\sin A} = \frac{b}{\sin B}$ (sine rule)

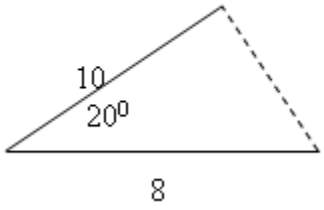
$$\frac{17}{\sin 35^0} = \frac{b}{\sin 50^0}$$

$$b = \frac{17 \times 0.7660}{0.5736}$$

B= 22.7 cm (1 dp)

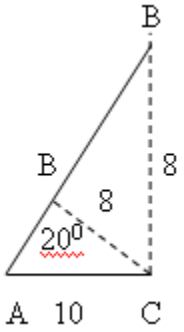
NOTE: If the angle is between the two given sides there is only one possible triangle.

i.e



If the angle is not between the two given sides it is sometimes possible to draw two triangles from the given data

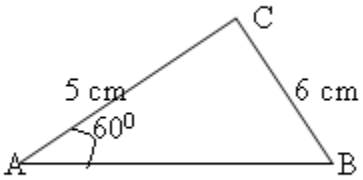
i.e



Example

In triangle ABC, A=60^0, a=6 cm and b=5 cm. Find B

Solution



$$\frac{\sin A}{a} = \frac{\sin B}{b}$$ (sine rule)

$$\frac{\sin 60^0}{6} = \frac{\sin B}{5}$$

$$\text{sinB} = \frac{\sin 60^0 \times 5}{6}$$

$$= \frac{0.8660 \times 5}{6}$$

$$= 0.7217$$

The two angles with a sin of 0.4771 are 46.2^0 and 133.8^0

Checking whether 133.8^0 is a possible value for B i.e. angle A +angle B=60^0+133.8^0

$=193.8^0$,

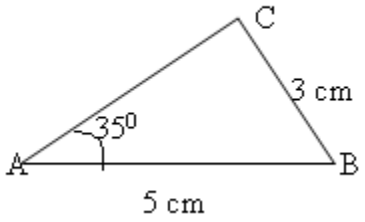
Therefore, it is not positive for the angle B to have the value 133.8^0

Therefore, the only possible value of B is 46.2^0

Example

In the triangle ABC, find C given that AB=5 cm, BC=3 cm and A=35^0

Solution



$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 35^0}{3} = \frac{\sin C}{5}$$

$$\text{SinC} = \frac{5 \sin 35^0}{3}$$

$$= 0.959$$

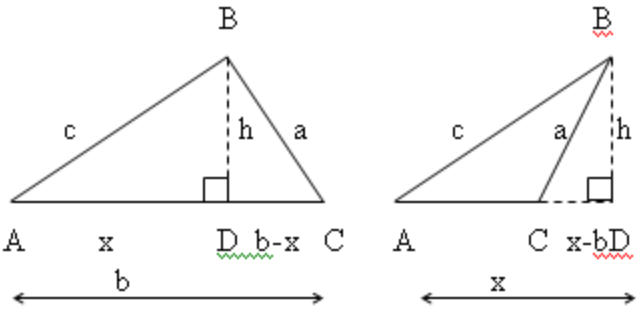
One angle whose sine is o.959 is 73^0 (to the nearest degree) but there is also an obtuse angle with the same sine which is 107^0 (to the nearest degree). If C=107^0, then A+C= 107^0+35^0

=142^0 less than 180^0 so 107^0 is another possible value of C

Therefore, angle C is either 73^0 or 107^0

COSINE RULE

Look at triangle ACD below



$h^2 = c^2 - x^2$ and $h^2 = c^2 - (x-b)^2$ (Pythagoras theorem)

Therefore $c^2-x^2=a^2-(x-b)^2$

$c^2-x^2=a^2-x^2+2bx-b^2$

$a^2=c^2+b^2-2bx$

but x=cosA, so now $a^2=b^2+c^2-2bccosA$

By using a symmetrical argument, the two other equivalent versions can be found

i.e. $b^2=a^2+c^2-2accosB$

$c^2=a^2+b^2-2abcosC$

So the cosine rule can be used to find the length of the third side, when the lengths of two sides in a triangle and the angle between them are known

The three cosine rule formulae can be rearranged to give expressions for cosA, cosB and cosC interms of a, b and c.

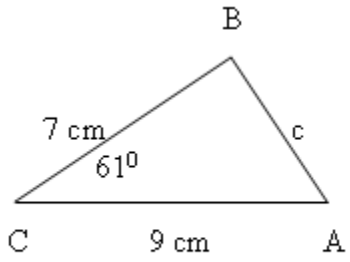
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \qquad \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \qquad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

This means that if the lengths of each of the three sides are known then each of the three angles can be found.

Example

In triangle ABC, BC=7 cm, AC=9 cm and $C=61^0$. Find AB.

Solution



$c^2=a^2+b^2-2abcosC$ (cosine rule)

$c^2=7^2+9^2-2x7x9cos61^0$

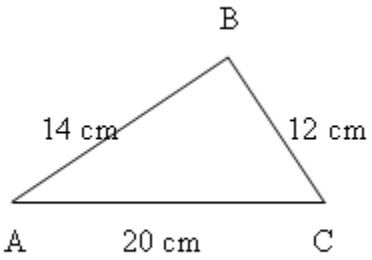
$=49+81-126x0.48480962$

$=130-61.08601215$

$C=8.301$

Therefore AB=8.30 cm correct to 3 s.f

Example
In triangle ABC if a=12 cm ,b=20 cm and c=14 cm, find A
Solution



$\cos A=\frac{b^2+c^2-a^2}{2bc}$ (cosine rule)

$=\frac{20^2+14^2-12^2}{2x20x14}$

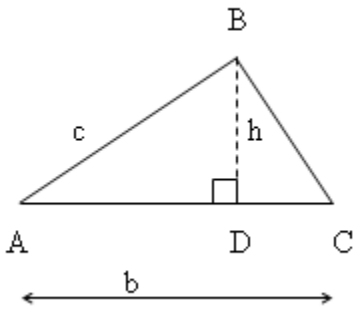
$=0.8071$

So $A=\cos^{-1}0.8071$

$A=36.2^0$ (1 d.p)

AREA OF TRIANGLE

Suppose that ABC is any triangle with angle BAC=θ. Let D be the foot of the perpendicular from B to AC.



Now $\sin\theta=\frac{BD}{BA}$ or $\sin\theta=\frac{h}{c}$, so h=c$\sin\theta$

But the area of triangle ABC= $\frac{1}{2} base \times height$

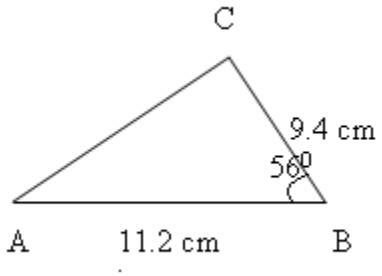
$$= \frac{1}{2}.b.h$$

$$= \frac{1}{2}.b.c \sin\theta \text{ or } \frac{1}{2}.b.c \sin A$$

Thus area of triangle= $\frac{1}{2}.b.c \sin A$

Similarly, area of triangle= $\frac{1}{2}.b.a \sin C = \frac{1}{2}.a.c \sin B$

Example
Find the area of the following triangle



Solution

Area of triangle= $\frac{1}{2}.a.c \sin B$

$$= \frac{1}{2} \times 11.2 \times 9.4 \times \sin 56^0$$

$$= 43.6405378$$

Therefore, the area of triangle ABC is 43.6 (3 s.f)

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TUTORIAL 9: TRIGONOMERTY

1. Convert the following to degrees

(i). $\frac{3\pi}{2} rad$

(ii). $\frac{\pi}{10} rad$

(iii). $\frac{9\pi}{2} rad$

2. Convert the following into radians

(i) . 45^0

(ii). 270^0

(iii). 520^0

3. Evaluate the following without using a calculator or four figure tables

(i). $\cos 75^0$

(ii). $\text{Sin}75^0+\sin 15^0$

4. Prove the following identities

(a) $2\text{secA}=\frac{\cos A}{1+\sin A}+\frac{1+\sin A}{\cos A}$

(b) $(1-\sin^2 A)(1+\tan^2)=1$

(c) $\sec\theta \cos ec\theta \equiv 2\cos ec2\theta$

5. (a). Solve the following equations

(i) $2\text{secA}=4$ between $0 \le \theta \le 360^0$

(ii) $\cos 3x=\cos x$

(iii) $2\sin^2 x-5\sin+2=0$

(b). Find all solutions x, lying in the interval $[0,2\pi]$

i. $\cos x=0$

ii. $\tan^2 x=1$

iii. $\sin 2x=1$

TOPIC 10: DIFFERENTIATION

OBJECTIVES

By the end of this topic, studentsbe should be able to;

(a) Find the limits of functions.
(b) Decide whether the given function is continuous or not.
(c) Differentiate functions using different rules.
(d) Sketching graphs using differentiation
(e) Calculating maximum or minimum values.

LIMITS

We shall say $\lim_{x\to a} f(x) = L$ if f(x) can be made arbitrarily close to L whenever we choose x close enough to a, but without actually allowing x to be equal to a.

It is very important in what follows that x should be allowed to get closer and closer to a in any fashion, but that it should never actually reach a.

THEOREM

Suppose $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$ .Then

(a) $\lim_{x\to a}\{\ f(x) \pm g(x)\ \} = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x) = L \pm \mathrm{M}$

(b) $\lim_{x\to a} f(x)\, g(x) = \lim_{x\to a} f(x) . \lim_{x\to a} g(x) = \mathrm{LM}$

(c) $\lim_{x\to a} f(x)/\, g(x) = \dfrac{lim f(x)}{\lim_{x\to a} g(x)} = \dfrac{L}{M}$ provided $\mathrm{M} \neq 0$

(d) $\lim_{x\to a} kf(x) = k \lim_{x\to a} f(x) = kL$ for all k

THEOREM

(A) $\lim_{x\to a} x = a$

(b) If f(x)=c then $\lim_{x\to a} f(x) = c$

Example

Evaluate the following

(a) $\lim_{x\to -2} x^5$

(b) $\lim_{x\to 2} (3x^2 - 5x + 4)$

(c) $\lim_{x\to a} \dfrac{x^3 - 5x + 4}{x^2 - 2}$

(d) $\lim_{x\to -2} x^5$

Solutions

(a) $\lim_{x\to -2} x^5 = (-2)^5$

$= -32$

(b) $\lim_{x\to 2}(3x^2 - 5x + 4) = \lim_{x\to 2} 3x^2 - \lim_{x\to 2} 5x + \lim_{x\to 2} 4$

$= 3\lim_{x\to 2} x^2 - 5\lim_{x\to 2} x + \lim_{x\to 2} 4$

$= 3x2^2\text{-}5x2+4$

$= 3x4\text{-}5x2+4$

$= 12\text{-}10+4$

$= 6$

(c) $\lim_{x\to 3} \dfrac{x^3 - 5x + 4}{x^2 - 2} = \dfrac{\lim_{x\to 3}(x^3 - 5x + 4)}{\lim_{x\to 3}(x^2 - 2)}$

$= \dfrac{\lim_{x\to 3}(x^3) - \lim_{x\to 3} 5x + \lim_{x\to 3} 4}{\lim_{x\to 3}(x^2) - \lim_{x\to 3} 2}$

$= \dfrac{3^3 - 5\times 3 + 4}{3^2 - 2}$

$= \dfrac{27 - 15 + 4}{9 - 2}$

$= \dfrac{16}{7}$

Remarks: The limits can be evaluated by direct substitution provided this does not lead to division by zero. If after substitution the denominator becomes zero then try to factorize both the numerator and denominator if possible and cancel the common factor.

Example

Evaluate $\lim_{x\to 3} \dfrac{x^2 - x - 6}{x - 3}$

Solution

$\lim_{x\to 3} \dfrac{x^2 - x - 6}{x - 3} = \lim_{x\to 3} \dfrac{(x-3)(x+2)}{x-3}$

$= \lim_{x\to 3}(x+2)$

$= 3+2$

$= 5$

CONTINUOUS FUNCTIONS
DEFINITION

A function f is continuous at x=a when (i) f(a) is defined (ii) $\lim_{x\to a} f(x)$ exists (iii) $\lim_{x\to a} f(x) = f(a)$. Otherwise, f is said to be discontinuous at x=a.

Example

Determine whether f(x)=$\frac{x^2+2x-3}{x-1}$ is continuous.

Solution

f(x)=$\frac{x^2+2x-3}{x-1}=\frac{(x+3)(x-1)}{x-1}$

=x+3

The denominator is zero when x=1.

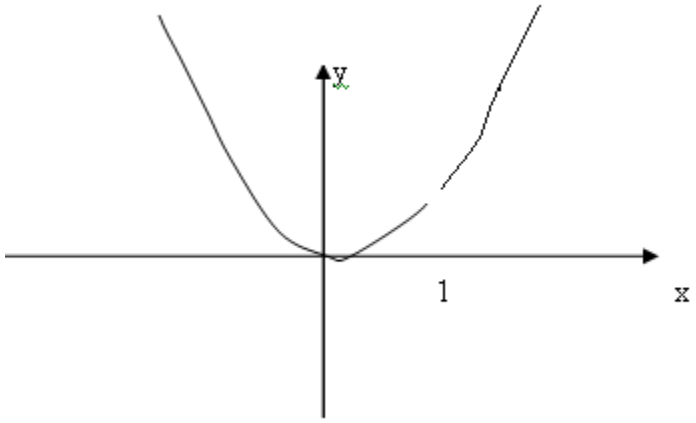
Therefore the function f(x) is discontinuous at x=1, but continuous elsewhere.

Example

By sketching the graph, decide whether the following function f(x)=$\begin{cases} x^2 when x \neq 1 \\ 0 when x = 1 \end{cases}$

Is continuous or not.

Solution



$\lim_{x\to 1} f(x) = 0$ and $\lim_{x\to 1} f(x) = \lim_{x\to 1} x^2 = 1$

Since f(1)$\neq \lim_{x\to 1} f(x)$, f is not continuous.

THE DERIVATIVE
DEFINITION

Suppose $\lim_{x\to 0}\frac{f(x+h)-f(x)}{h}$ exists for some x. then this limit is called the derivative of f at x written $f'(x)$ and f is said to be differentiable at x. i.e. $f'(x)=\lim_{x\to 0}\frac{f(x+h)-f(x)}{h}$

This process of finding the derivative using the limits is called*differentiation from firstprinciples*

Example

Differentiate the following from first principles

(a) $f(x)=x^2$

(b) $f(x)=2x^2-5x+7$

Solutions

(a) $f'(x)=\lim_{x\to 0}\frac{f(x+h)-f(x)}{h}$

$=\lim_{x\to 0}\frac{(x+h)^2-x^2}{h}$

$=\lim_{x\to 0}\frac{x^2+2xh+h^2-x^2}{h}$

$=\lim_{x\to 0}\frac{2xh+h^2}{h}$

$=\lim_{x\to 0}(2x+h)$

$=2x$

(b) $f'(x)=\lim_{x\to 0}\frac{2(x+h)^2-5(x+h)+7-(2x^2-5x+7)}{h}$

$\lim_{x\to 0}\frac{2(x^2+2xh+h^2)\ -5x+5h+7-2x^2+5x-7}{h}$

$=\lim_{x\to 0}\frac{2x^2+4xh+2h^2\ -5x+5h+7-2x^2+5x-7}{h}$

$=\lim_{x\to 0}\frac{4xh+2h^2-5h}{h}$

$=\lim_{x\to 0}(4x+2h^2-5)$

$=4x-5$

NOTATION

If f(x) denotes any function of x, that is any expression containing x, we call its derivative function $f'(x)$. The other commonly used notation of derivative function is $\frac{df}{dx}$

The following are all alternatives for denoting the derivatives

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{df(x)}{dx}$$

THE POWER RULE
THEOREM

For any real number r, $\frac{d}{dx}x^r = rx^{r-1}$

Example
Differentiate the following with respect to x

(a) $y=x^{10}$

(b) $y=5x^6$

(c) $y=\sqrt{x}$

(d) $y=\frac{3}{x^2}$

Solutions

(a) $y=x^{10}$

Then $\frac{dy}{dx} = 10x^{10-1}$

$=10x^9$

(b) $y=5x^6$

then $\frac{dy}{dx} = 5 \times 6x^{6-1}$

$=30x^5$

(c) $y=\sqrt{x}$

$y=x^{\frac{1}{2}}$

Then $\frac{dy}{dx} = \frac{1}{2}x^{\frac{1}{2}-1}$

$=\frac{1}{2}x^{\frac{-1}{2}}$

(d) $y=\frac{3}{x^2}$

$y=3x^{-2}$

then $\frac{dy}{dx} = 3 \times -2x^{-2-1}$

$=-6x^{-3}$

THE CONSTANT FUNCTION RULE
The derivative of a constant, c is zero.

Example
Differentiate the following with respect to x

(a) 7
(b) 208
(c) -347

Solutions

(a) If y =7, then $y^{/} = 0$
(b) If y =208 then $y^{/} = 0$
(c) If y =-347, then $y^{/} = 0$

THE LINEAR FUNCTION RULE

The derivative of a linear function is the coefficient of the x (unknown).

Example
Differentiate the following with respect to x

(a) y =5x
(b) y = -12x +9

Solutions

(a) If y = 5x, the $y^{/} = 5$
(b) If y = -12x+9, then $y^{/} = -12$

PRODUCT AND QUOTIENT RULES

(a) PRODUCT RULE

Suppose that f and g are differentiable at x, then

$$\frac{d}{dx}\{f(x)g(x)\ \} = f^{/}(x)g(x) + f(x)g^{/}(x)$$

Example
Use the product rule to differentiate each of the following with respect to x.

(a) y= $x^2 \times x^5$
(b) y=$(x^2+3x-5)x^{23}$

Solutions

(a) y= x^2 x x^5
$y^{/} = 2x.x^5 + 5x^4.x^2$
$= 2x^6 + 5x^6$
$= 7x^6$

(b) y = $(x^2 +3x-5)x^{23}$
$y^{/} = (2x+3)x^{23} + 23x^{22}(x^2 +x-5)$

QUOTIENT RULE

Suppose that f and g are differentiable at x and g(x) $\neq 0$. Then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\ \right) = \frac{f^{/}(x)g(x) - f(x)g^{/}(x)}{[\ g(x)\ ]^2}$$

Example

Use the quotient rule to differentiate each of the following with respect to x

(a) $y= \frac{x^5}{x^2}$

(b) $y= \frac{x^2-2}{x^2+1}$

Solutions

(a) $y = \frac{x^5}{x^2}$

$y' = \frac{5x^4.x^2 - 2x.x^5}{[\ x^2\ ]^2}$

(b) $y = \frac{x^2-2}{x^2+1}$

$y' = \frac{2x(x^2+1) - 2x(x^2-2)}{[\ x^2+1\ ]^2}$

THE CHAIN RULE

If g is differentiable at x and f is differentiable at g(x) then

$\frac{d}{dx}[f\{\ g(x)\ \}\ ] = f'g(x)g'(x)$

EXAMPLE

Differentiate the following with respect to x using the chain rule

(a) $y =(x+1)^3$

(b) $y= \frac{1}{4x-5}$

(c) $y=(x^2+2x-3)^6(2x^2+5)^7$

Solutions

(a) $y =(x+1)^3$

$y'=3(x+1)^2.1$

$=3(x+1)^2$

(b) $y= \frac{1}{4x-5}$

$y= \frac{0(4x-5) - 4.1}{[\ 4x-5\ ]^2}$

$y= \frac{-4}{[4x-5\ ]^2}$

(c) $y=6(x^2+2x-3)^5(2x+2)(2x^2+5)^7 +7(2x^2+5)^6(4x)(x^2+2x-3)^6$

IMPLICIT DIFFERENTIATION

Suppose f is any differentiable function of y where y=y(x). Then

$$\frac{d}{dx} f(y) = \frac{df}{dy}\frac{dy}{dx}$$

Example

Find $y'(x)$ for $x^2 + y^3 - 2y = 3$

Solution

$$\frac{d}{dx}(x^2 + y^3 - 2y) = \frac{d}{dx}3$$

$$\frac{dx^2}{dx} + \frac{dy^3}{dx} - \frac{d2y}{dx} = 0$$

$$2x + 3y^2\frac{dy}{dx} - 2\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(3y^2 - 2) = -2x$$

$$\frac{dy}{dx} = \frac{-2x}{3y^2 - 2}$$

The process of differentiating both sides of an equation with respect to x and then solving for $y'(x)$ 0r $\frac{dy}{dx}$ is called implicit differentiation.

Example

A curve has the equation $x^3+x^2y+y^3=11$. Show that $\frac{dy}{dx} = -\frac{2xy + 3x^2}{x^2 + 3y^2}$

Solution

$$\frac{dx^3}{dx} + \frac{dx^2y}{dx} + \frac{dy^3}{dx} = \frac{d11}{dx}$$

$$3x^2 + 2xy + x^2\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(x^2 + 3y^2) = -3x^2 - 2xy$$

$$\frac{dy}{dx} = -\frac{3x^2 + 2xy}{x^2 + 3y^2}$$

DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

The following table shows the derivatives of trigonometric functions

F(x)	$f'(x)$
sinx	Cosx
Cosx	-sinx
Tanx	Sec^2x
Cotx	$-cosec^2x$
Cosecx	-cosecx cotx
Secx	secxtanx

Example

Show that $\frac{d}{dx}\tan kx = k\sec^2 kx$

Solution

$$\frac{d}{dx}\tan kx = \frac{d}{dx}\frac{\sin kx}{\cos kx}$$

$$= \frac{k\cos kx\cos kx - \sin kx(-k\sin kx)}{(\cos kx)^2}$$ product rule and chain rule

$$= \frac{k\cos^2 kx + k\sin^2 kx}{(\cos kx)^2}$$

$$= \frac{k}{\cos^2 kx}$$ since $\cos^2x + \sin^2x = 1$

$=k\sec^2kx$ since $\frac{1}{\cos x} = \sec x$

Example

Differentiate the following with respect to x

(a) y= sin2x

(b) $y=\sin^2x$

(c) $y=\sec(x^2)$

(d) $y=\left[\sin(x^2-1)\right]^4$

Solutions

(a) y= sin2x

$y'=\cos2x.2$ chain rule

$=2\cos2x$

(b) $y=\sin^2x$

$y'=2\sin x.\cos x$ chain rule

$=2\sin x\cos x$

$=\sin2x$ since sin2A=2sinAcosA

(c) $y=\sec(x^2)$

$y'=\sec x^2\tan x^2.2x$ chain rule

$y^{/}=2x\sec x^2\tan x^2$

(d) $y=\left[\sin(x^2-1)\ \right]^4$

$y^{/}=4\left[\sin(x^2-1)\right]^3\cos(x^2-1).2x$ chain rule

$=8x\left[\sin(x^2-1)\right]^3\cos(x^2-1)$

DERIVARTIVE OF EXPONENTAIL FUNCTIONS

Theorem: If $y=e^{kx}$ then $y^{/}=ke^{kx}$

Example

Differentiate the following with respect to x

(a) $y=e^x$

(b) $y=e^{3x+1}$

(c) $y=xe^x$

(d) $e^{2x}\cos 3x$

Solutions

(e) (a) $y=e^x$

Then $y^{/}=e^x$

(f) (b)$y=e^{3x+1}$

Then $y^{/}=3e^{3x+1}$

(g) (c) $y=xe^x$

Then $y^{/}=e^x+xe^x$ product rule

(h) (d) $y=e^{2x}\cos 3x$

Then $y^{/}=2e^{2x}\cos 3x+(-\sin 3x.3)e^{2x}$

$=2e^{2x}\cos 3x-3e^{2x}\sin 3x$

DERIVATIVE OF LOGARITHMIC FUNCTIONS

The derivative of the natural logarithmic function y=*ln*f(x) is

$\frac{d}{dx}[\ \ln f(x)\ ]=\frac{f^{/}(x)}{f(x)}$ i.e. derivative of function/original function.

Example

Differentiate the following with respect to x

(a) $y=\ln(x^2+3x-2)$

(b) $y=\frac{x}{\log x}$

(c) $y=\frac{\ln x}{e^x}$

Solutions

(a)$y=\ln(x^2+3x-2)$

$y^{/}=\frac{2x+3}{x^2+3x-2}$

(c) $y=\frac{x}{\log x}$

(d) $y' = \dfrac{1.\log x - \frac{1}{x}.x}{(\ln x)^2}$ quotient rule

$= \dfrac{\log x - 1}{\ln^2 x}$

(c) $y = \dfrac{\ln x}{e^x}$

$y' = \dfrac{\frac{1}{x}.e^x - e^x \ln x}{(e^x)^2}$ quotient rule

THE DERIVATIVE AS A RATE OF CHANGE

Suppose a variable w is a function of time such that at time t, w is given by w = gt where g is differentiable function. The difference between the interval and final values of w in the time interval (t, t+h) is given by g(t+h) –g(t). We formulate the following definition;

The average rate of change of w =g(t) in the interval (t, t+h) is

$$\frac{g(t+h)-g(t)}{h}$$

The rate of change of w =g(t) with respect to t is

$$\frac{dw}{dt} = g'(t) = \lim_{h\to 0} \frac{g(t+h)-g(t)}{h}$$

Sometimes dw/dt is referred to as the *instantaneous rate of change* of w with respect to t.

Example

A scientists finds that if a certain substance is heated, the Celsius temperature t minutes, where $0 \le t \le s$, is given by the formula $g(t) = 30t + 6\sqrt{t} + 8.$

(a) Find the rate of change of g(t) during the interval (4, 4.41).

(b) Find the rate of g(t) at t= 4.

Solutions

(a) Letting t =4 and h = 0.41 in the definition, the average rate of change of g in (4, 4.41) is $\dfrac{g(4.41)-g(4)}{0.41} = \dfrac{\left[30(4.41)+6\sqrt{4.41}+8\right]-\left(120+6\sqrt{8}+8\right)}{0.41}$

$= \dfrac{12.9}{0.41}$

$\approx 31.46^0 c/\min$

(b) $g'(t) = 30 + \dfrac{3}{\sqrt{t}}$

$g'(t) = 30 + \dfrac{3}{\sqrt{4}}$

$=31.5^0$c/min

SMALL INCREMENTS

Consider two variables x and y related by the equation y = f(x). If x increases by a small δx then y increases by a corresponding small amount δy. Now $\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{dy}{dx}$

So, provided that δx is small, $\frac{\delta y}{\delta x} \approx \frac{dy}{dx} \Rightarrow \delta y \approx \frac{dy}{dx}(\delta x)$

This approximation can be used to estimate the values of a function close to a known value, i.e. y + δy can be estimated if y is known at a particular value of x.

Example

Using y = ln x and ln 1 =0, estimate the value of ln 1.1

Solution

y= ln x $\Rightarrow \frac{dy}{dx} = \frac{1}{x}$

So $\delta y \approx \frac{dy}{dx}(\delta x) = \frac{1}{x}\delta x$

But $\delta x = 0.1$ and x =1 then $\delta y \approx \frac{1}{1} \times 0.1$

Then ln 1.1= y+ δy

$\approx \ln x + \delta y$

$\approx 0 + 0.1$

≈ 0.1

Example

Using $y = \sqrt{x}$, estimate the value of $\sqrt{101}$

Solution

$y = \sqrt{x} \Rightarrow \frac{dy}{dx} = \frac{1}{2}x^{\frac{-1}{2}}$ Taking x=100, y = $\sqrt{100}$

=10

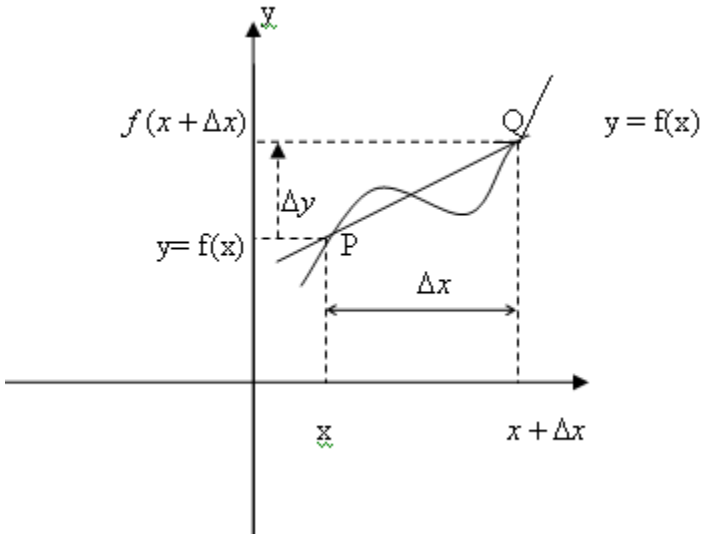
But $\delta y \approx \frac{dy}{dx}\delta x$ gives $\delta y \approx \frac{1}{2\sqrt{x}} \times 1$ and $\delta x = 1$

Then $\sqrt{101} = y + \delta y \approx \sqrt{100} + \frac{1}{20}$

=10.05

TANGENTS AND GRAGHS

Consider the following diagram



The gradient of the chord PQ is given by

$$\text{Gradient}=\frac{change in y-coordinate}{change in x-coordinate}$$

$$=\frac{\Delta y}{\Delta x}$$

$$=\frac{f(x+\Delta x)-f(x)}{\Delta x}$$

For small values of Δx, point Q approaches point P. The gradient of the chord PQ becomes closer to the gradient of the tangent to the curve at point P. So the limiting case is as $\Delta x \to 0$

Gradient of the curve at point P= $\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x}=\frac{dy}{dx}$

Remarks: $\frac{dy}{dx}$ is the gradient function on the graph of y=f(x)

Example

Find the gradient of $y=x^2+6x-3$ at (2, 13)

Solution

Gradient=f'= 2x+6

At (2, 13), gradient=$2\times 2+6$

=10

Example

Find the coordinate of the point(s) on the following curve $y=x^2+4x+1$, $\frac{dy}{dx}=8$

Solution

$\frac{dy}{dx} = 2x + 4$

If $\frac{dy}{dx} = 8$ then 2x+4= 8

2x=4

X= 2

To find the value of y, we substitute the value of x in the original equation
i.e. substituting 2 for x in $y=x^2+4x+1$, we get

$y=2^2 +4\times 2 +1$

y=4+8+1

y=13

Therefore, the coordinate is (2, 13)

Example

Find the equation to the tangent at x=2 to $y=x^2$. Find also the normal when x=2.

Solution

$y=x^2$

$\frac{dy}{dx} = 2x$

Gradient of the tangent at x=2 is 2×2 0r 4

So $f'(2) = 4$

When x=2, y=4, so the tangent passes through (2,4)

But the equation of the tangent is y-b=m(x-a)

y-4=4(x-2)

y=4x-4

(ii) The gradient of the normal that passes through (2, 4) is $\frac{-1}{4}$ (recall $m_1m_2=-1$)

So its equation is y-4= $\frac{-1}{4}$(x-2)

Y=$\frac{-1}{4}$x+$\frac{9}{2}$

INCREASING AND DECREASING FUNCTIONS

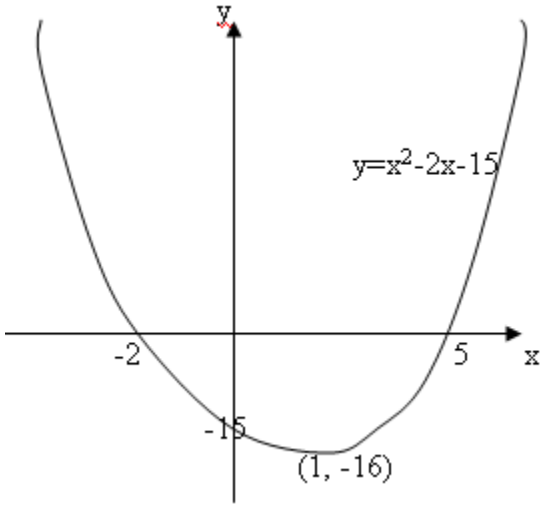
Suppose y is a differentiable function of x. At points on the graph of y= f(x) where the gradient is not zero, y must be either an increasing or decreasing function of x

1. If $\frac{dy}{dx} > 0$ (a positive gradient), then y is an increasing function of x.
2. $\frac{dy}{dx} < 0$ (a negative gradient) , then y is a decreasing function of x.

Example

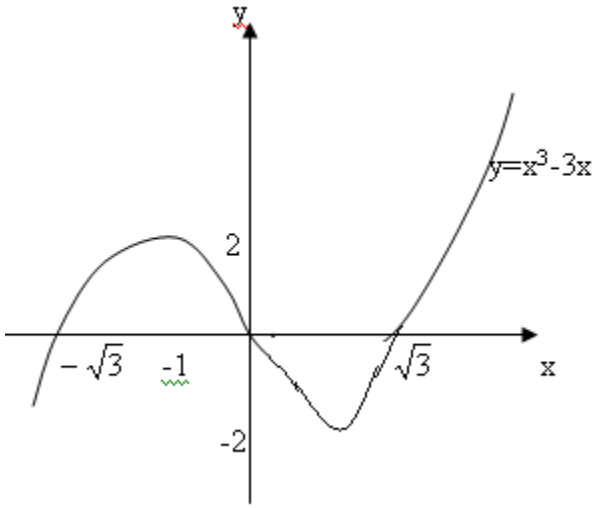
Sketch the graph of $y= x^2-2x-15$

Solution
$y= x^2-2x-15$
$y' =2x-2$
the turning points, $y' =2x-2= 0$
$x=1$
when x=1, then y=1-2-15
$y=-16$
therefore the turning point is (1, -16)
for x<1, $y'< 0$ decreasing for $x<\frac{1}{2}$ for x>1, $y'>0$ increasing for $x><\frac{1}{2}$
the y-intercept is -15 (occurs when x=0) and x-coordinate occurs when y=0
i.e. $x^2-2x-15= 0$
$(x-5)(x+2)= 0$
Either x=5 or x=-2
So its graph is



Example
Sketch the graph of $y=x^3-3x$
Solution
$y=x^3-3x$
$y' =3x^2 -3$
equating y' to zero we have, $3x^2 -3=0$
$x^2 =1$
$x=\pm 1$
when x= -1 , y=2 and when x= 1, y=-2
so the turning points are (-, 2) and (1, -2)
nature of turning points
for x<1, $y'<0$ decreasing function
for x>1, $y'>0$ increasing function
thus (1, -2) is a minimum point
for x,<-1, $y'<0$ and x>-1, $y'>0$. Thus (-1,2) is a maximum point
the y-intercept is 0 and x-intercept is $x^3-3x=0$

x(x-3)=0
either x=0 or x=3
So its graph is



HIGHER ORDER DERIVATIVES

Sometimes it is useful to know the gradient of the gradient function at a particular point on the curve y= f(x). This is called the second derivative. It is found by differentiating the first derivative, $\frac{dy}{dx}$ with respect to x to give $\frac{d}{dx}(\frac{dy}{dx})$ written $\frac{d^2y}{dx^2}$ or $f''(x)$. The expression $\frac{d^2y}{dx^2}$ is read d-two-y by d-x-squared, and shows that the differentiation process has happened twice. It is not the same as squaring $\frac{dy}{dx}$. Differentiating again with respect to x would give the third derivative, $\frac{d^3y}{dx^3}$ 0r $f'''(x)$

Example

Find $\frac{d^2y}{dx^2}$ when $y= 3x^3 +2x+1$

Solution

$y= 3x^3 +2x+1$

$\frac{dy}{dx} =9x^2 +2$

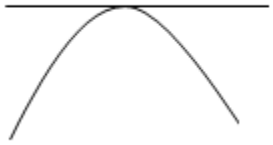
$\frac{d^2y}{dx^2} =18x$

Example

Find $f^{///}(x)$= when $y= x^4 +5x^3 -9x^2 +2x-7$
Solution
$y= x^4 +5x^3 -9x^2 +2x-7$
$f^{/}= 4x^3 +15x^2 -18x+2$
$f^{//}=12x^2 +30x-18$
$f^{///} =24x+30$

CONCAVITY AND THE SECOD DERIVATIVE TEST
The concept of concavity is useful for describing the graph of a differentiable function. If $f^{/}(x)$ exists, then f has a tangent line with slope $f^{/}(c)$ at the point P(c, f(c)).
Let f be a function that differentiable at a number c.

(i) The graph is concave upward at the point P(c, f(c)). If there exists an open interval (a, b) containing c such that on (a, b) the graph of f is above the tangent line through P.



(iii) The graph is concave downward at the point P(c, f(c)). If there exists an open interval (a, b) containing c such that on (a, b) the graph of f is below the tangent line through P.



TEST OF CONCAVITY
If the function f is differentiable on an open interval containing c, then the point P(c, f(c)), the graph is;
(j) concave upward if $f^{//}(c) >0$
(ii) concave downward if $f^{//}(c) < 0$

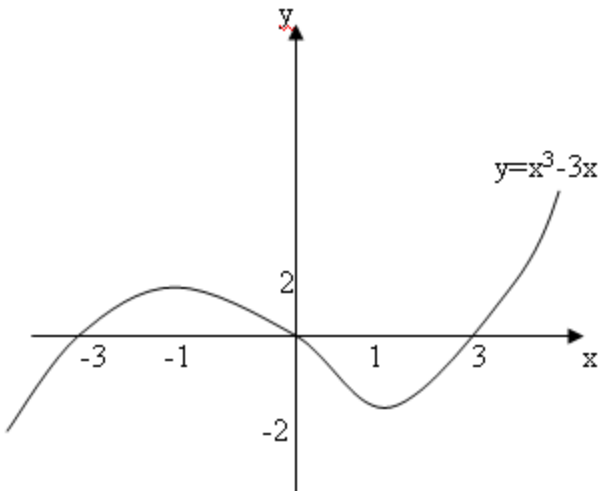
POINT OF INFLECTION
Points on the graph of *f* where the concavity changes from up-to-down or down-to-up are called*inflection points* of the graph.

A point P(c, f(c)) on the graph of a function f is a point of inflection if there exists an open interval (a, b) containing c such that one of the following statements holds;
(i) $f^{//}(x) >0$ if $a< x< c$ and $f^{//}(x)$ if $c <x<b$ or
(ii) $f^{//}(x) <0$ if $a<x<c$ and $f^{//}(x) >0$ if $c<x<$
SECOND DERIVATIVE TEST
If f is twice differentiable and
(a) $f^{/}(x_0) =0$ and $f^{//}(x_0)>0$ then f has a minimum point at x_0
(b) $f^{/}(x_0) =0$ and $f^{//}(x_0)<0$ then f has a maximum point at x_0

(c) A point x_0 is a point of inflexion 0f y= f(x) if $f''(x_0)=0$ and for all small positive h
either i. $f'''(x-h)>0$ or
ii. $f'''(x-h)<0$

Example
Use the second derivative test to find the local extrema of $f(x) =x^4 -8x^2 +10$
Solution
$y=x^4 -8x^2 +10$
$y' =4x^3 -16x$
$=4x(x^2 -4)$
$=4x(x-2)(x+2)$
To get the turning point, equate y' to zero, we have
$4x(x-2)(x+2)= 0$
Either $4x=0$, $x-2 =0$ or $x+2= 0$
$\Rightarrow$ x= 0, x=2 or x=-2
When x=0, y=10 i.e the critical point is (0, 10)
When x=2, $y=2^4-8\times 2^2 +10$
y=-6 thus the critical point is (2, -6)
when x= -2, $y= (-2)^4-8(-2)^2+10$
y= -6 thus the critical point is (-2, -6)
nature of the critical points
$y'' =12x^2 -16$
$f''(0) =-16$; (0,10) is a minimum point
$f''(2) = 32$; so (2,-6) is a minimum point
$f''(-2)= 32$ so (-2, -6) is a minimum point

Example
(a) Find the coordinates of the turning points on the curve $y=x^3-3x$
(b) Use the second derivative test to determine their nature
(c) Sketch the curve
Solutions
y-intercept is 0 and x-intercept can be found by solving the following equation
$x^3-3x=0$
$x(x^2-3)=0$
either x=0 $x=-\sqrt{3}$ or $x=\sqrt{3}$
turning points; $3x^2-3=0$
$x^2 -1=0$
$x=\pm 1$
when x=1, y=-2 thus one of the turning point is (1, -2)
when x= -1, y= 2 thus the other turning point is (-1, 2)
nature of critical points
$y'' =6x$
$y''(1)=6>0$ so (1,-2) is a local minimum point
$y''(-1)=-6<0$ so (-1, 2) is a local maximum point
so its graph is



APPLICATION OF MAXIMUM AND MINIMUM

Example

A farmer has 100m of fencing. He wishes to fence in a rectangular field one side of which needs no fencing as it is formed by a straight river. What is the greatest area that he can enclose?

Solution



Perimeter=x+x+y=100

=2x+y=100 …(1)

Area=xy … (2)

Making y the subject of (1), we get y=100-2x

Substituting 1-2x for y in (2), we obtain A=x(100-2x)

$A=100x-2x^2$

$\frac{dA}{dx} = 100 - 4x$

$\frac{d^2A}{dx} = -4 < 0$ so it is indeed a maximum

To find the maximum value of A i.e. $\frac{dA}{dx} = 0 \Rightarrow$ 100-4x=0

100=4x

So x= 25

Substituting 25 for x in (1) we get $2 \times 25 + y = 100$

y=50

Therefore, the greatest area is 25×50 (1250m^2)

Example

The volume of a right circular cylinder open at one end is to be 8π cm^3. Find its minimum surface area.

Solution

Volume of cylinder = $\pi r^2 h = 8\pi$

Surface area $= 2\pi rh + \pi r^2$

Making r the subject of (1), we obtain, $r = \sqrt{\frac{8}{h}}$

Substituting $\sqrt{\frac{8}{h}}$ for r in (2), we obtain

$$A = 2\pi \frac{\sqrt{8}}{\sqrt{h}} + \pi \frac{8}{h}$$

$$A = 2\pi\sqrt{8}h^{\frac{-1}{2}} + 8\pi h^{-1}$$

$$\frac{dA}{dx} = 2\pi \times \frac{1}{2} h^{\frac{-1}{2}} \sqrt{8} - 8\pi h^{-2} = 0$$

$$\pi\sqrt{8}h^{\frac{-1}{2}} - 8\pi h^{-2} = 0$$

$$\frac{\pi\sqrt{8}}{h^{\frac{1}{2}}} = \frac{8\pi}{h^2}$$

Squaring both sides, we get $\frac{8\pi^2}{h} = \frac{64\pi^2}{h^4}$

$8h^3 = 64$

$h^3 = 8$

$h = 2$

Substituting2 for h in (1) we get $\pi r^2 \times 2 = 8\pi$

$r^2 = 4$

$r = 2$

thus the minimum value $= 2\pi rh + \pi r^2$

$$= 2\pi \times 2 \times 2 + \pi \times 2^2$$
$$= 8\pi + 4\pi$$
$$= 12\pi$$

TUTORIAL 10:DIFFERENTIATION

1. evaluate the following limits where possible
   (a) $\lim_{x\to 2}(x+5)$
   (b) $\lim_{x\to 1}\frac{x-1}{x^2-1}$
   (c) $\lim_{x\to 3}\frac{x^2-2x-3}{x-3}$

2. Differentiate from the first principles
(a) $f(x) = 6x^2 + 4$
(b) $f(x) = 2x^2 - 3x + 3$

3. Differentiate the following with respect to x
   (a) $y = 6x^2 + 4x - 7$
   (b) $y = 2^7$
   (c) $y = (x^2 + 4)(x^7 + 8) + x^5$
   (d) $y = \sqrt{x - x^2}$
   (e) $y = \frac{x+2}{x^2+3x}$
   (f) $x^2 - xy + 3y^2 = 3$
   (g) y= ta7x
   (h) $y = \frac{\ln x}{\sin x}$
   (i) $y = e^x \tan 2x$

4. If $y = 3x^3 + x - 4$, solve the equation $y' = 0$

5. A curve has the equation $x^2 + x^3y + y^3 = 0$. Use implicit differentiation to show that

$$\frac{dy}{dx} = -\frac{2x + 3x^2y}{x^3 + 3y^2}$$

6. For the following curves find the critical points and points of inflexion and sketch the graphs.
   (a) $y = x^3 - 3x^2 - 18x$
   (b) $y = -x^3 + 3x - 2$
   (c) $y = x^3 + 6x$

TOPIC 12: INTEGRATION

OBJECTIVES

By the end of this topic, students should be able to;

(c) Define integration.
(d) Integrate functions using any method/technique.
(e) Differentiate between indefinite and definite integrals.
(f) Calculate areas and volumes using integration.
(g) Approximate areas using rectangle, trapezoidal and Simpson's rules.

THE INVERSE OF DIFFERENTIATION

We know that the derived function of x is 1. Therefore, we wish to find an expression whose derived function is 1. One answer is x. but it is not the only answer because x+2, x+3 and x+$18\frac{1}{2}$ are all expressions whose derived functions equal to 1. In fact, (x+c) is the general solution where c is called a constant.

DEFINITION

Let F be any ant derivative of f. The indefinite integral of f(x) with respect to x is defined by. $\int f(x)dx = F(x) + c$ Where c is a constant.

NOTATION

$\int$ is called the symbol for integration. The term to be integrated f(x) is called the integral. dx means integrate with respect to x. If y is a function of x, $\int ydx$ stands for the integral of y with respect to x. The integral sign $\int$ cannot be divorced from dx if the integral is with respect to x.

INTEGRATION FORMULAE AND TECHNIQUES

CONSTANT RULE

The integral of a constant k is

$\int kdx = kx + c$

Example

Evaluate the following integrals;

(a) $\int 5dx$

(b) $\int -12dx$

(c) $\int dx$

Solutions

(a) $\int 5dx$ = 5x + c

(b) $\int -12dx$ = -12x +c

(c) $\int dx$ =x +c

THEOREM (POWER RULE)

For any rational power r ± -1

$$\int x^r dx = \frac{x^{r+1}}{r+1} + c$$

Example

Evaluate the following

(a) $\int x^{17} dx$

(b) $\int \sqrt{x} dx$

Solutions

(a) $\int x^{17} dx = \frac{x^{17+1}}{17+1} + c$

$$= \frac{x^{18}}{18} + c$$

(b) $\int \sqrt{x} dx = \int x^{\frac{1}{2}} dx = \frac{1}{2} x^{\frac{1}{2}-1}$

$$= \frac{1}{2} x^{\frac{-1}{2}}$$

THEOREM:Suppose that f(x) and g(x) have ant derivatives. Then

(a) $\int \{f(x) \pm g(x)\} dx = \int f(x) dx \pm \int g(x) dx$

(b) $\int af(x) dx = a \int f(x) dx$

Example

Evaluate the following

(a) $\int (x^3 + 2x^2) dx$

(b) $\int (x^4 - 3x^2 + x + 1) dx$

SOLUTIONS

(a) $\int (x^3 + 2x^2) dx = \int x^3 dx + \int 2x^2 dx$

$$= \frac{x^4}{4} + 2\frac{x^3}{3} + c$$

(b) $\int (x^4 - 3x^2 + x + 1) dx = \int x^4 dx - \int 3x^2 dx + \int x dx + \int 1 dx$

$$= \frac{x^5}{5} - 3\frac{x^3}{3} + \frac{x^2}{2} + x$$

$$= \frac{x^5}{5} - x^3 + \frac{x^2}{2} + x$$

INTEGRATION BY SIMPLIFICATION OF THE INTEGRAND

Here you simplify the integrand.

Example

Evaluate the following

(a) $\int (x+2)^2 dx$

(b) $\int \frac{x^2+1}{\sqrt{x}} dx$

Solutions

(a) $\int (x+2)^2 dx = \int (x+2)(x+2)dx = \int (x^2+4x+4)dx$

$$= \frac{x^3}{3} 4\frac{x^2}{2} + 4x$$

$$= \frac{x^3}{3} - 2x^2 + 4x$$

(b) $\int \frac{x^2+1}{\sqrt{x}} dx = \int \left(\frac{x^2+1}{x^{\frac{1}{2}}} \right) dx = \int \left(x^{2-\frac{1}{2}} + x^{\frac{-1}{2}} \right) dx$

$$= \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c$$

$$= \frac{2x^{\frac{5}{2}}}{5} + 2x^{\frac{1}{2}} + c$$

INTEGRATION BY SUBSTITUTION

The technique of integration by substitution consists of the following steps

1. Choose an expression for u: a common choice is the innermost expression or inside term of a composition of function
2. compute $\frac{du}{dx}$
3. Replace all terms in original integrand with expressions involving u and du
4. evaluate the resulting (u) integral

NOTE: If part of the integrand can be obtained by differentiating part, then use substitution method.

Example

Evaluate the following

(a) $\int (x^3+5)^{100}(3x^2)dx$

(b) $\int x^2\sqrt{x^3+4}dx$

Solution

(a) Note $3x^2$ can be obtained by differentiating x^3+4. So let $u = x^3 + 5$

Then $\frac{du}{dx} = 3x^2$

Or $du = 3x^2dx$

We can replace $3x^2dx$ by du and $x^3 + 5$ by u

It follows that $\int (x^3+5)^{100}(3x^2)dx = \int u^{100}du$

$$= \frac{u^{101}}{101} + c$$

$$= \frac{(x^3+5)^{101}}{101} + c$$

(b) $\int x^2\sqrt{x^3+4}dx$

A quick look at the integrand shows that x^2 can be obtained by differentiating $x^3 + 4$

$\therefore$ let $u = x^3 + 4$

$du = 3x^2\,dx$

So $\int x^2\sqrt{x^3+4}dx = \int \sqrt{u}\,\frac{du}{3}$

$$= \frac{1}{3}\int u^{\frac{1}{2}}du$$

$$= \frac{1}{3}\frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c$$

$$= \frac{2u^{\frac{3}{2}}}{9} + c$$

$$= \frac{2(x^3+4)^{\frac{3}{2}}}{9} + c$$

INTEGRATION BY PARTIAL FRACTIONS

Two or more fractions can be combined to give a single fraction.

For example: $\frac{3}{x+2} - \frac{2}{x-5} = \frac{3(x-5)-2(x+2)}{(x+2)(x-5)}$

$$= \frac{3x-15-2x-4}{(x+2)(x-5)}$$

$$= \frac{x-19}{x^2-3x-10}$$

Equally, a fraction can be expressed as a fraction or difference of two or more proper fraction, known as partial fractions

Example

Suppose $\frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}, x \neq 1,-2$. What are the values of A and B?

Solution

$$\frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

$$= \frac{A(x+2) + B(x-1)}{(x-1)(x+2)}$$ simplifying the right hand side

Since the denominators are the same, we can equate the numerators

1 = A (x+2) + B(x-1)

1 = Ax +2A +Bx – B

1= Ax+ Bx +2A – B

Equating the coefficients, we have

A + B = 0 and

2A – B = 1

Solving these two simultaneous equations, we have A = $\frac{1}{3}$ and B = $\frac{-1}{3}$

$$\therefore \frac{1}{(x-1)(x+2)} = \frac{\frac{1}{3}}{x-1} + \frac{\frac{-1}{3}}{x+2}$$

$$= \frac{1}{3}\left(\frac{1}{x-1} - \frac{1}{x+2}\right)$$

This process is called decomposing into partial fractions.

Example

Decompose into partial fractions

(a) $\frac{1}{x^2+3x-10}$

(b) $\frac{5x+7}{x^2+2x-3}$

(c) $\frac{x^3+5x^2+7x+6}{x^2+3x+2}$

Solutions

(a) $\frac{1}{x^2+3x-10} = \frac{1}{(x+5)(x-2)} = \frac{A}{x+5} + \frac{B}{x-2}$

$$= \frac{A(x-2) + B(x+5)}{(x+5)(x-2)}$$

Equating the numerators, we have

1= A(x-2) +B(x+5)

1 =Ax +Bx -2A +5B

Equating the coefficients, we get

A + B = 0 and

-2A +5B = 1

Solving the simultaneous equations, we obtain A = $\frac{-1}{7}$ and B = $\frac{1}{7}$

Thus $\frac{1}{x^2+3x-10} = \frac{-\frac{1}{7}}{x+5} + \frac{\frac{1}{7}}{x-2}$

$= \frac{1}{7}\left(\frac{1}{x-2} - \frac{1}{x+5}\right)$

(b) $\frac{5x+7}{x^2+2x-3} = \frac{5x+7}{(x+3)(x-1)} = \frac{A}{x+3} + \frac{B}{x-1}$

$= \frac{A(x-1)+B(x+3)}{(x+3)(x-1)}$

Equating the numerators, we have

5x +7 = A(x-1) + B(x+3)

5x +7 =Ax + bx –A +3B

Equating the coefficients, we have

A + B = 5 and

-A +3B = 7

Solving these simultaneous equation, we get B =3 and A = 2

Thus $\frac{5x+7}{x^2+2x-3} = \frac{2}{x+3} + \frac{3}{x-1}$

(c) $\frac{x^3+5x^2+7x+6}{x^2+3x+2}$

Note: If the numerator of a rational expression has the same or higher degree than the denominator, you must first perform a long division and following this with a partial fractions decomposition of the remaining proper fractions.

Using long division

$$\begin{array}{r} x^2+2 \\ x^2+3x+2 \overline{)\, x^3+5x^2+7x+6} \\ \underline{x^3+3x^2} \\ 2x^2+7x \\ \underline{2x^2+6x} \\ x+6 \end{array}$$

$\frac{x^3+5x^2+7x+6}{x^2+3x+2}$ =x +2 + $\frac{x+6}{x^2+3x+2}$

= x+ 2 + $\frac{x+6}{(x+2)(x+1)}$

i.e. $\dfrac{x+6}{(x+2)(x+1)} = \dfrac{A}{x+1}+\dfrac{B}{x+2}$

$$\frac{x+6}{(x+2)(x+1)} = \frac{A(x+2)+B(x+1)}{(x+1)(x+2)}$$

Equating coefficients, we have

A + B =1 and

2A + B=6

-A = -5 $\Rightarrow A = 5$ and B = -4

Thus $\dfrac{x^3+5x^2+7x+6}{x^2+3x+2} = x+2+\dfrac{5}{x+1}-\dfrac{4}{x+2}$

To integrate an improper fraction, first writes it as the sum or difference of proper algebraic fractions.

Example

Evaluate $\int \dfrac{dx}{x^2-x-2}$

Solution

First, we decompose the integrand into partial fractions

i.e. $\dfrac{1}{x^2-x-2} = \dfrac{1}{(x-2)(x+1)} = \dfrac{A}{x-2}+\dfrac{B}{x+1}$

Then 1= A(x-2) + B(x+1)

$A = \dfrac{-1}{3}$ and $B = \dfrac{1}{3}$

So $\int \dfrac{dx}{x^2-x-2} = \int \dfrac{\frac{-1}{3}}{x+1}+\dfrac{\frac{1}{3}}{x-2}dx$

$$= \frac{1}{3}\int\left(\frac{1}{x-2}-\frac{1}{x+1}\right)dx$$

$$= \frac{1}{3}\left(\ln|x-2|-\ln|x+1|\right)+c$$

$$= \frac{1}{3}\ln\left|\frac{\ln x-2}{x+1}\right|+c$$

Example

Evaluate $\int \dfrac{6x+6}{x^3+4x^2+x-6}dx$

Solution

Using the factor theorem; $x^3 + 4x^2 + x - 6 = (x-1)(x+2)(x+3)$

By decomposing into partial fractions

$$\int \frac{6x+6}{x^3+4x^2+x-6}dx = \int\left(\frac{1}{x-1}+\frac{2}{x+2}-\frac{3}{x+3}\right)dx$$

$$= \ln|x-1|+2\ln|x+2|-3\ln|x+3|+c$$

Example

Evaluate $\int \frac{x^2+x+1}{2x^2-5x+3}dx$

Solution

Using long division, we get $\frac{x^2+x+1}{2x^2-5x-3} = \frac{1}{2}+\frac{1}{2}\frac{7x+5}{2x^2-5x-3}$

Then $\frac{7x+5}{2x^2-5x-3} = \frac{A}{2x+1}+\frac{B}{x-3}$

Solving A and B we find A= $\frac{-3}{7}$ and $B = \frac{26}{7}$

So $\int_0^2 \frac{x^2+x+1}{2x^2-5x+3}dx = \int_0^2 \left[\frac{1}{2}+\frac{1}{2}\left(\frac{\frac{-3}{7}}{2x+1}+\frac{\frac{26}{7}}{x-3}\right)\right]dx$

$$= \int_0^2 \frac{1}{2}dx + \frac{1}{14}\int_0^2 \left(\frac{26}{x-3}-\frac{3}{2x+1}\right)dx$$

$$= \frac{1}{2}x\Big|_0^2 + \frac{1}{14}\left(26\ln|x-3| - \frac{3}{2}\ln\left|2x+1\right|_0^2\right|$$

$$= 1 + \left(26\ln|-1| - \frac{3}{2}\ln|5| - 26\ln|-3| - \frac{3}{2}\ln|1|\right)$$

$$= 1 + \frac{1}{14}\left(-26\ln 3 - \frac{3}{2}\ln 5\right)$$

DETERMINATION OF EQUATIONS FROM DERIVATIVES

We can find the equation or function from the derivative by integrating the derivative.

Example

The gradient of a curve y = f(x) is given by $\frac{dy}{dx} = (2-x)(2-3x)$. Find the equation of the curve, given that it passes through the point (3, 2)

Solution

To find y, we need to integrate

$y = \int (2-x)(2-3x)dx$

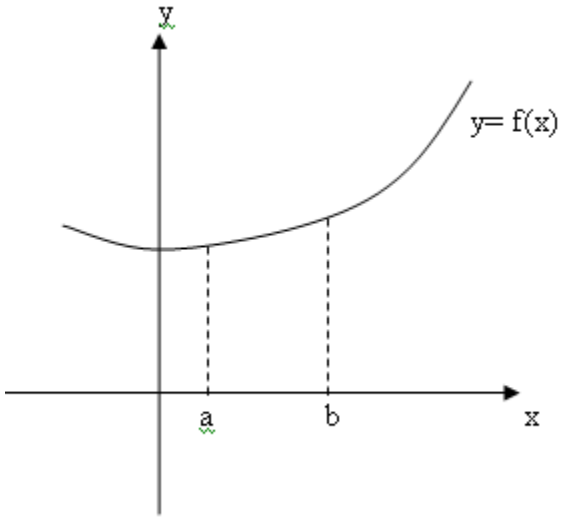
$= \int (4-8x+3x^2)dx$

$= 4x - \frac{8x^2}{2} + \frac{3x^3}{3} + c$

$=4x-4x^2+x^3$

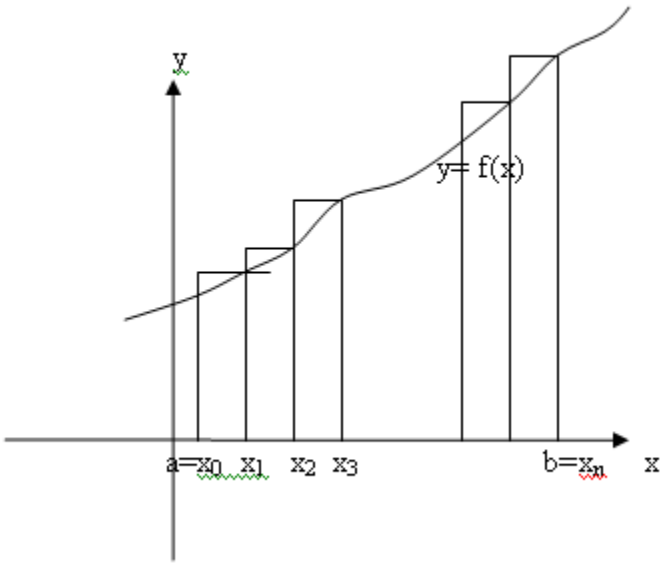
THE DEFINITE INTEGRAL

Let us consider the following graph



We would like to find the area under the curve between a and b.

We start by dividing the interval $[a,b]$ into n equal pieces. The width of each subinterval in the partition is then $\frac{b-a}{n}$, which we denote by Δx (meaning a small change in x). The points in the partion are denoted by x_0= a, $x_1 = x_0 + \Delta x$, $x_2 = x_1 + \Delta x$, $x_3 = x_2 + \Delta x$, and so on and x_n= b



The height of each rectangle is $f(x_i)$ where i= 1, 2, 3, …, n. It should be clear from the above figure that the area under the curve between a and b is approximately the same as the sum of the areas of the rectangles.

i.e $A \cong f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x + \ldots f(x_n)\Delta x$

$$= \sum_{i=1}^{n} f(x_i)\ \Delta x$$

DEFINITION

The definite integral of f from a to b is $\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\ \Delta x$

NOTE: If f is continuous and $f(x) \geq 0$ on $[a,b]$ then the $\int_a^b f(x)dx =$ the area under the curve on $[a,b]$

We call 'b' the upper limit of the integral and 'a' the lower limit of the integral. An integral with limitsis called a definite integral.

THEOREM

Suppose f is continuous on $[a,b]$ and F is an ant derivative of f. Then

$$\int_a^b f(t)dt = F(b) - F(a)$$

Example

Evaluate the following

(a) $\int_1^2 x^2 dx$

(b) $\int_0^2 (x^2 - 2x)dx$

(c) $\int_1^3 x(3x^2 - x)dx$

Solutions

(a) $\int_1^2 x^2 dx = \frac{x^3}{3}\Big|_1^2$

$$= \frac{1}{3}(2^2 - 1^3$$

$$= \frac{1}{3}(4-1)$$

$$= 1$$

(b) $\int_0^2 (x^2 - 2x)dx = \frac{x^3}{3} + \frac{x^2}{2}\Big|_0^2$

$$= \frac{2^3}{3} + 4 - (\frac{0^3}{3} - 0^2)$$

$$= \frac{8}{3} + 4$$
$$= \frac{20}{3}$$

(c) $\int_1^3 x(3x^2 - x)dx = \int_1^3 (3x^3 - x^2 + x)dx$

$$= \frac{3x^4}{4} - \frac{x^3}{3} + \frac{x^2}{2}\Big|_1^3$$
$$= \frac{243}{4} - \frac{108}{3} + \frac{9}{2} - (\frac{3}{4} - \frac{1}{3} + \frac{1}{2})$$

$$= \frac{729 - 108 + 54}{12} - \frac{9 - 4 + 6}{12}$$
$$= \frac{675}{12} - \frac{11}{12}$$
$$= \frac{675 - 11}{12}$$
$$= \frac{664}{12}$$

Example
Find the area under the curve $y=x+x^2$ between x=1 and x=3.
Solution

$$\int_1^2 (x + x^2)dx = \frac{x^2}{2} + \frac{x^3}{3}\Big|_1^3$$
$$= \frac{3^2}{2} + \frac{3^3}{3} - (\frac{1^2}{2} + \frac{1^3}{3})$$
$$= \frac{27 + 54}{6} - \frac{3 + 2}{6}$$
$$= \frac{81}{6} - \frac{5}{6}$$
$$= \frac{76}{6}$$

THE INTEGRAL OF EXPONENTIAL FUNCTIONS

Recall that $\frac{d}{dx}e^{kx} = ke^{kx}$. Since integration is the opposite of differentiation, then

$$\int e^{kx}dx = \frac{e^{kx}}{k} + c$$

NOTE: We divide the function by the derivative of the power.

Example

Evaluate the following

(a) $\int e^{2x}dx$

(b) $\int e^{-x}dx$

(c) $\int_0^1 e^{2-5x}dx$

Solutions

(a) $\int e^{2x}dx = \frac{e^{2x}}{2} + c$

(b) $\int e^{-x}dx = \frac{e^{-x}}{-1} + c$

$= -e^{-x} + c$

(c) $\int_0^1 e^{2-5x}dx = \frac{e^{2-5x}}{-5}\Big|_0^1$

$= \frac{-1}{5}\left(e^{2-5\times1} - e^{2-5\times0}\right)$

$= \frac{-1}{5}\left(e^{-3} - e^{2}\right)$

THE INTEGRALS OF LOGARITMIC FUNCTIONS

A further set of important integrals can be found by reversing the differentiation of logarithmic functions.

Recall that $\frac{d}{dx}\ln x = \frac{1}{x}$

Reversing it, we have $\int \frac{1}{x}dx = \ln x + c$

NOTE: The derivative of the denominator is the numerator.

In general, if f(x) $\neq 0$ and f is differentiable, then $\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + c$

Example

Evaluate the following

(a) $\int \frac{3}{x}dx$

(b) $\int \frac{6}{1+3x}dx$

(c) $\int \frac{2x+3}{x^2+3x+5}dx$

Solutions

(a) $\int \frac{3}{x}dx = 3\int \frac{1}{x}dx$

$=3\ln|x|+c$

(b) $\int \frac{6}{1+3x}dx = 6\int \frac{1}{1+3x}dx$

Let u = 1 +3x then du =3dx

So (b) $\int \frac{6}{1+3x}dx = 6\int \frac{1}{u}\frac{du}{2}$

$=\frac{6}{2}\int \frac{1}{u}du$

$=3\ln|u|+c$

$= 3\ln|1+3x|+c$

(c) $\int \frac{2x+3}{x^2+3x+5}dx = \ln|x^2+3x+5|+c$

i.e let u = x^2+3x+5 then du = (2x+3) dx

so (c) $\int \frac{2x+3}{x^2+3x+5}dx = \int \frac{1}{u}du$

$=\ln|u|+c$

$=\ln|x^2+3x+5|+c$

THE INTEGRALS OF TRIGONOMETRIC FUNCTIONS

Recall that $\frac{d}{dx}\frac{\sin kx}{k} = \cos kx$ and $\frac{d}{dx}\frac{\cos kx}{k} = -\sin kx$. It follows that

$\int \cos kxdx = \frac{\sin kx}{k}+c$ and $\int \sin kx = -\frac{\cos kx}{k}+c$

Recall that $\frac{d}{dx}\tan kx = k\sec^2 kx \Rightarrow \int \sec^2 kxdx = \frac{\tan kx}{k}+c$

$\frac{d}{dx}\cot kx = -k\cos ec^2 kx \Rightarrow \int \cos ec^2 kxdx = -\frac{\cot kx}{k}+c$

Recalling that tanx= $\frac{\sin x}{\cos x}$ and $\int \frac{1}{x}dx = \ln x + c$, then

$\int \tan xdx = \int \frac{\sin x}{\cos x}dx$

Let u =cosx, so du = -sinx dx

So $\int \tan xdx = \int \frac{\sin x}{\cos x}dx = -\int \frac{1}{u}du$

$=-\ln|u|+c$

$= -\ln|\cos x|+c$

$=\ln|\cos x|^{-1}+c$

$$= \ln\left|\frac{1}{\cos x} + c\right|$$

$$= \ln|\sec x| + c \text{ since secx} = \frac{1}{\cos x}$$

Recall that cotx = $\frac{cox}{\sin x}$, then $\int \cot dx = \int \frac{\cos x}{\sin x} dx$

Let u= sinx, so du = cosx dx

So $\int \cot dx = \int \frac{\cos x}{\sin x} dx = \int \frac{1}{u} du$

$$= \ln|\sin x| + c$$

Thus $\int \cot x dx = \ln|\sin x| + c$

Example

Evaluate $\int \sec x dx$

Solution

$$\int \sec x dx = \int \sec x \left[\frac{\sec x + \tan x}{\sec x + \tan x}\right] dx$$

$$= \int \left[\frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\right] dx$$

Let u = secx +tanx, then du = (secxtanx +sec^{2}x) dx

So $\int \left[\frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\right] dx = \int \frac{1}{u} du$

$$= \ln|u| + c$$

$$= \ln|\sec x + \tan x| + c$$

Similarly $\int \cos ecx dx = -\ln|\cos ecx + \cot x| + c$

Example

Find

(a) $\int \sin 3x dx$

(b) $\int \cos(1 + 2x) dx$

Solutions

(a) $\int \sin 3x dx = -\frac{\cos 3x}{3} + c$

(b) $\int \cos(1 + 2x) dx = \frac{\sin(1 + 2x)}{2} + c$

Example

Evaluate the following

(a) $\int_0^{\frac{\pi}{4}} \sin 2\theta d\theta$

(b) $\int_0^1 \tan 3x dx$

Solution

(a) $\int_0^{\frac{\pi}{4}} \sin 2\theta d\theta = -\frac{\cos 2\theta}{2}\Big|_0^{\frac{\pi}{4}}$

$$= -\frac{1}{2}(\cos 2\times\frac{\pi}{4} - \cos 2\times 0)$$

$$= \frac{-1}{2}(\cos\frac{\pi}{2} - \cos 0)$$

$$= \frac{-1}{2}(0-1)$$

$$= \frac{1}{2}$$

(b) $\int_0^1 \tan 3x dx = \int_0^1 \frac{\sin 3x}{\cos 3x} dx$

Let u = cos3x, then du = -3sin3x dx

So $\int_0^1 \tan 3x dx = \int_0^1 \frac{\sin 3x}{\cos 3x} dx = \int -\frac{1}{3} du$

$$= \frac{-1}{3}\ln|u| + c$$

$$= \frac{-1}{3}\ln|\cos 3x| + c$$

GROWTH AND DECAY LAWS

Let P(t) denote the size of the population at time t. The derivatives $P^{/}$ then records the rate of change of the population. It is reasonable to except the rate of growth P(t) to be proportion to the size of the population P(t).

We assume that there is a fixed number k independent of time such that

$P^{/}(t) = k\,P(t)$

Or $\frac{P^{/}(t)}{P(t)} = k$ *

Integrating both sides of * with respect to t, we obtain.

$\int \frac{P^{/}(t)}{P(t)} dt = \int k dt$ #

You should notice that we can substitute u=P(t) in the integral on the left-hand side. For this choice of u, we have $du = P'(t)dt$ and so # becomes

$$\int \frac{1}{u} du = \int k dt$$

Evaluating these integrals, we obtain

$$\ln|u| + c_1 = kt + c_2$$

Where c_1 and c_2 are constants of integration. Subtracting c_1 from both sides yields,

$$\ln|u| = kt + (c_2 - c_1) = kt + c$$

Where we have combined the two constants of integration into one on the right-hand side.

Since u= P(t) > 0, we have

$$\ln P(t) = kt + c$$

Taking exponentials of both sides, we get,

$$P(t) = e^{\ln P(t)} = e^{kt+c} = e^{kt} e^{c}$$

Since c is an arbitrary constant, we write

$A = e^c$ and get

$$P(t) = Ae^{kt}$$

(1) If k>0 then the above formula is called an exponential growth law and
(2) If k<0 then it is an exponential decay law.

Example

The size of the world population at the beginning of 1998 was approximately 5.14 billion. At the beginning of 1999, it was 5.23 billion. Assume that the growth rate remains constant

(a) What is the growth constant k?
(b) What will the population be in 2010?

Solutions

Let P(t) be the population in billion at time t. For convenience measures time starting in the year 198; that is, t – 0 corresponds to 1998 and t=1 to 1999. Thus P(0) = 5.14 and P(1) = 5.23. The equation describing the population in billions at time t is

$P(t) = 5.14e^{kt}$

(a) $P(1) = 5.14\, e^{k.1}$

So, $5.23 = 5.14e^{k}$

$$e^{k} = \frac{5.23}{5.14}$$

$$k = \ln \frac{5.23}{5.14}$$

$$\approx 0.0174$$

(b) Hence at t, $P(t) = 5.14e^{0.0174t}$

The year 2010 corresponds to t = 12, so in the year 2010 the population in billion will be

$P(12) = 5.14e^{0.0174 \text{ X } 12}$

$= 5.14e^{0.21}$

$= 6.33.$

Example

Radium decays exponentially and has half-life of approximately 1600 years; that is, given any quantity, 0ne-half of it will disintegrate in 1600 years. Find a formula for the amount q(t) remaining from 50 milligrams of pure radium after t years. When will there be 20 mg left?

Solution

By hypothesis, q(0)= 50 and q(1600)= 25

But $q(t) = Ae^{kt}$, so q(0) = A = 50 and at t= 1600, we get

$25= 50e^{1600k}$, or $e^{1600k} =1/2$

Hence $1600k= \ln(1/2)= \ln 1 - \ln 2 = -\ln 2$

$$k = \frac{-\ln 2}{1600}$$

At t, we have $q(t) = Ae^{\frac{-\ln 2}{1600}t}$

(b) when q(t) = 20 mg,

$$20 = 50e^{\frac{-\ln 2}{1600}t}$$

$$\frac{2}{5} = e^{\frac{-\ln 2}{1600}t}$$

Taking the natural logarithm of both sides, we obtain,

$$\ln\frac{2}{5} = -\ln 2 \times \frac{t}{1600}$$

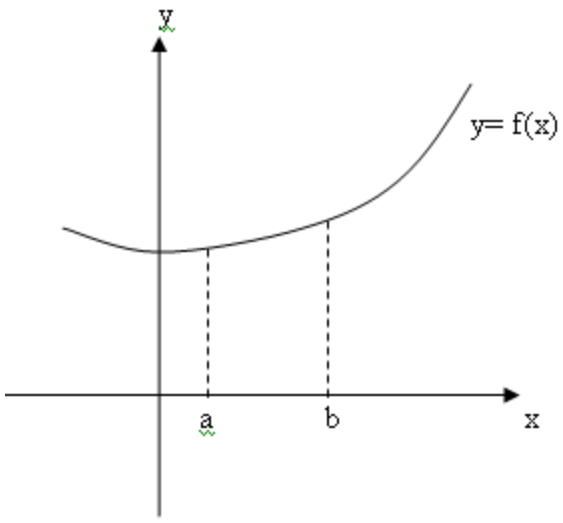
$$t = \frac{\ln\left(\frac{2}{5}\right) \times 1600}{-\ln 2}$$

$$t = \frac{-1466.065171}{-0.69314718}$$

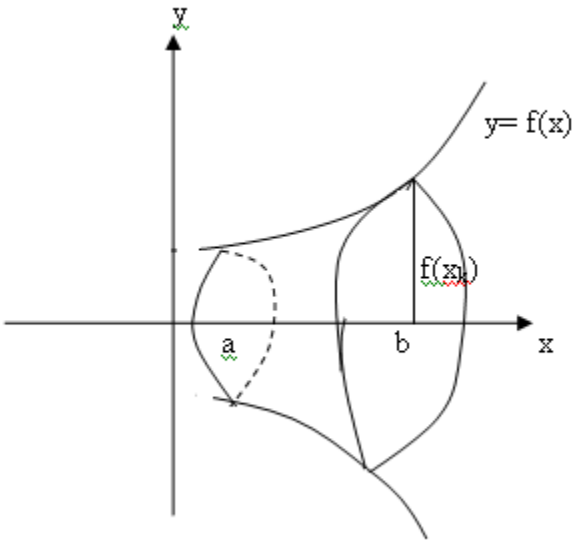
$$t = 2115 years$$

VOLUMES OF REVOLUTIONS

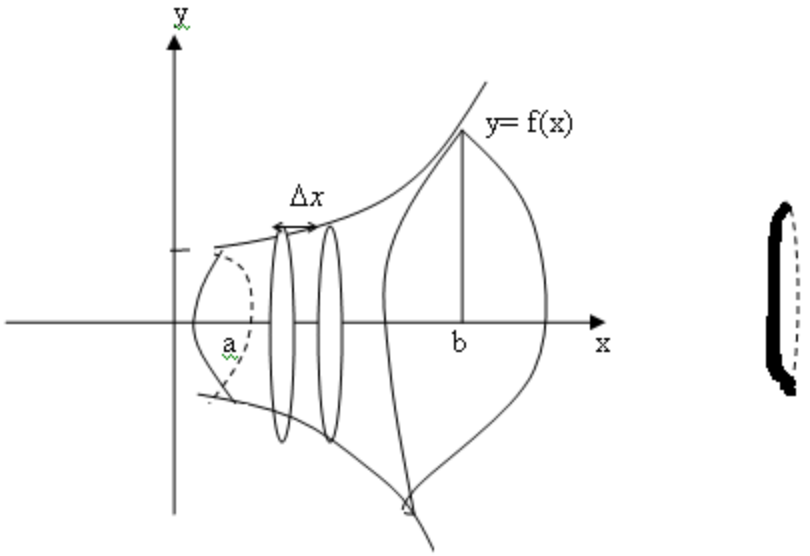
Consider the following diagram



If we rotate this curve once around the x-axis, a solid figure is described



The basic idea is to divide the solid into a number of thin slices and then add the volumes of the slices together.



Volume of slice= $\pi r^2 h$

$= \pi y^2 \Delta x$ where r=y and h= Δx

The volumes can be evaluated accurately by adding together the volumes of all the discs/slices

Volume= $\sum_a^b \pi y^2 \Delta x$ where Δx is infinitely small.

This sum can be evaluated accurately using integration

V= $\int_a^b \pi y^2 dx$ 0r $\int_a^b \pi f(x) dx$ or $\pi \int_a^b f(x) dx$

Of course, if the curve is rotated instead around the y-axis, the formula becomes V= $\pi \int_a^b x^2 dy$

Example

Find the volume generated by rotating the area bounded by the curve x= -1 and x= 1 about the x=axis.

Solution

$V = \pi \int_a^b y^2 dx$

$= \pi \int_{-1}^{1} (x^2 + 1)^2 \ dx$

$= \pi \int_{-1}^{1} (x^4 + 2x^2 + 1) \ dx$

$$= \pi\left(\frac{x^5}{5} + 2\frac{x^3}{3} + x \right|_{-1}^{1}$$

$$= \pi\left(\frac{1}{5} + \frac{2}{3} + 1 \right) - \left(\frac{-1}{5} - \frac{2}{3} - 1 \right)$$

$$= \pi\left(\frac{28}{15} + \frac{28}{15} \right)$$

$$= \frac{56\pi}{15}$$

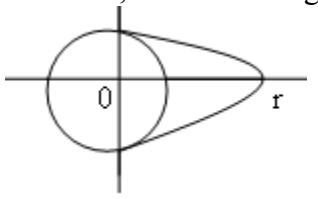
Example
The curve $y= x^2$ between x=0 and x=2 is rotated about the y-axis. Find the volume of the parabolic dish formed.
Solution

V= $\pi \int_a^b x^2 dy$ since we are rotating about the y-axis, the limits must be in terms of y.
When x =2, y= 4 and when x= 0, y= 0
Thus V= $\pi \int_0^4 y dy$

$$= \pi \frac{y^2}{2} \Big|_0^4$$

$$= 8\pi$$

Example
Find the volume of a sphere.
Solution
A sphere is produced by rotating a circle around either axis. Recall that the equation of the circle, centre the origin and radius r is $x^2 + y^2 = r^2$ or $y^2 = r^2 - x^2$



Thus the volume of a hemisphere is

$$V = \pi \int_0^r (r^2 - x^2) dx$$

$$= \pi\left(r^2 x - \frac{x^3}{3} \right|_0^r$$

$$= \pi\left(r^2 \times r - \frac{r^3}{3} - 0 \right)$$

$$= \frac{2\pi r^3}{3}$$

So the volume of a whole sphere = 2 x volume of a hemisphere

$$= 2 \text{ x } \frac{2\pi r^3}{3}$$

$$= \frac{4\pi r^3}{3}$$

WORK AND ENERGY

The work done by a constant force on a body in a straight line is the product of *F*, the component of the force in the direction of movement, and *x*the total distance moved. Thus

$W = Fx$

The work done represents the increase of kinetic energy of the body. For example, for a falling body of mass m, the force is constant mg and in falling a distance h, the work done is mgh. This represents the increase in kinetic energy, which is $\frac{1}{2} mv^2$, so, we have

$mgh = \frac{1}{2} mv^2$

and hence $v = \sqrt{2gh}$

If the force F varies, such as in a spring where $F(x) = kx$, it can be shown that

$W = \int_a^b F(x)dx$, where $x = a$ and $x = b$ are the end points of the motion.

Example

A spring is stretched o.5 meter longer that its rest length. The force required to keep it at that length is 3 newtons. Find

(a) total work done accomplished in stretched the spring o.5 meters from its rest position.

(b) total work accomplished in stretched the spring 1 meter from its rest position.

Solution

Recall that $F = kx$

$3 = k \times 0.5$

$k = 6$

But $W = \int_a^b F(x)dx$

$= \int_0^{0.5} 6xdx$

$= 3x^2 \Big|_0^{0.5}$

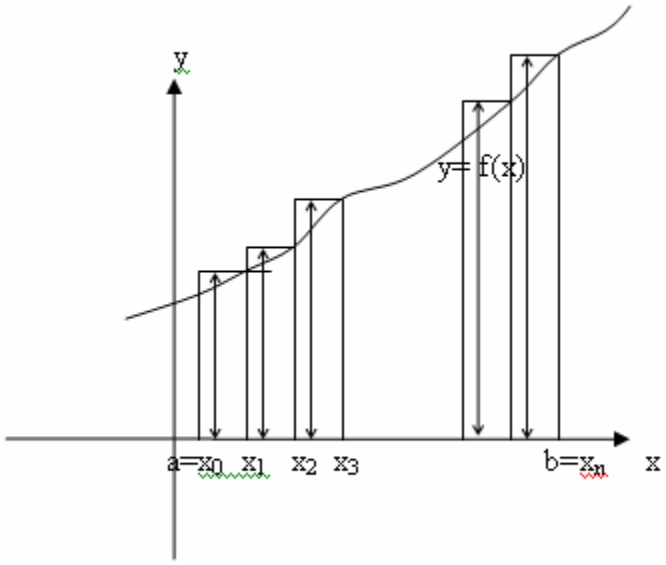
$= 0.75 Joules$

NUMERICAL INTEGRATION

We have going to discuss how to approximate $\int_a^b f(x)dx$ by the area under the curve y = f(x) between x=a and x=b

RECTANGLE RULE

Suppose A is the area bounded by the graph of y= f(x), the x-axis and the lines x=a and x=b. An approximate value for A can be found by dividing this area up into n rectangles. Each strip has width h = $\frac{b-a}{n}$ and the height of each of these rectangles is determined by the value of the function f(x) at the mid point of each strip.



Area under the curve between a and b $\cong hf_1 + hf_2 + hf_3 ... + hf_{n-1} + hf_n$

$\cong h(f_1 + f_2 + f_3 + ... + f_{n-1} + f_n)$

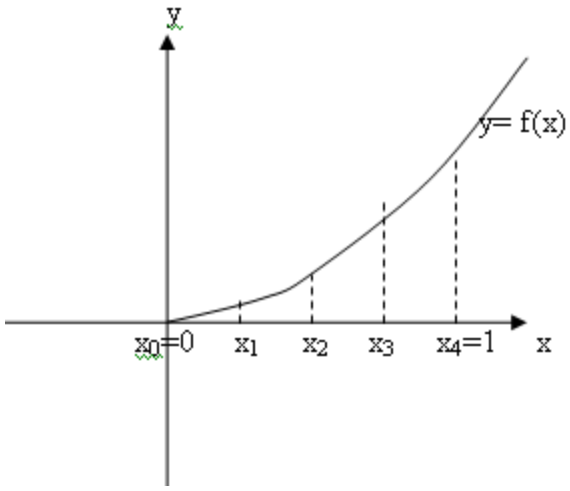
This result can be written as:

$$\int_a^b f(x)dx \cong h\left(f_1 + f_2 + f_3 + ... + f_{n-1} + f_n\right)$$

Example

Evaluate $\int_0^1 x^2 dx$ using rectangle rule with four strips

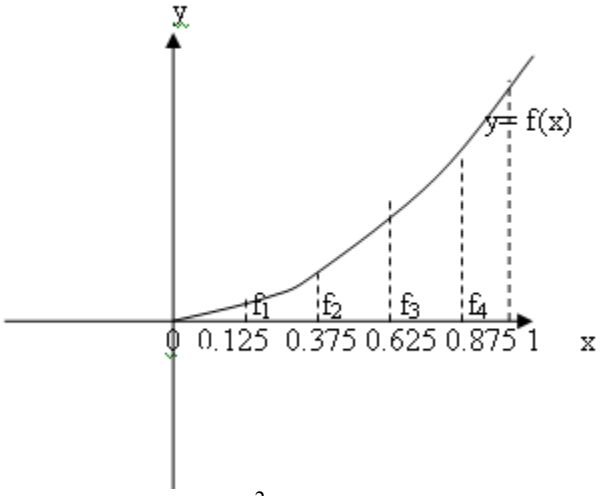
Solution



But h = $\frac{b-a}{n}$

$=\frac{1-0}{4}$

$=\frac{1}{4}$ or 0.25



f_1=f(0.125)= $(0.125)^2 = 0.015625$

$f_2 = f(0.375) = (0.375)^2 = 0.140625$

$f_3 = f(0.625) = (0.625)^2 = 0.455625$

$f_3 = f(0.75) = (0.75)^2 = 0.5625$

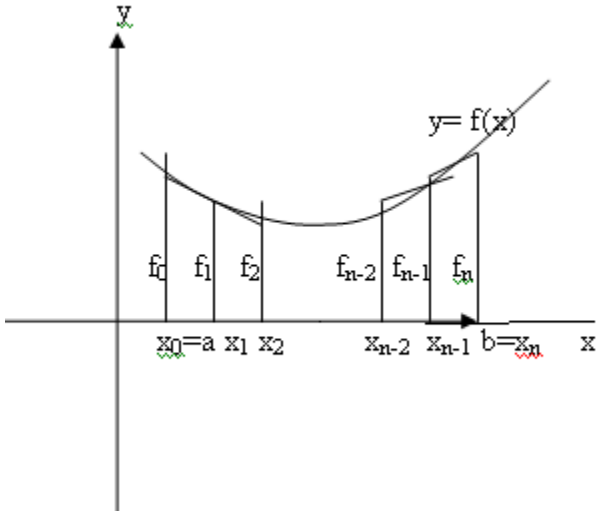
Using the rectangle rule

$$\int_0^1 x^2 dx \cong h(f_1 + f_2 + f_3 + f_4)$$

$\cong 0.25(0.015625 + 0.140625 + 0.455626 + 0.5625)$
$\cong 0.25 \times 1.174375$
$\cong 0.29359375$
$\cong 0.294$ (to 3 d.p)

TRAPEZIUM RULE

You divide the area under the curve y=f(x) into n trapezia with equal height $h = \frac{b-a}{n}$



Area under the curve between a and b $\approx h\left(\frac{f_0 + f_1}{2}\right) + h\left(\frac{f_1 + f_2}{2}\right) + \ldots + h\left(\frac{f_{n-1} + f_n}{2}\right)$

$\approx \frac{h}{2}(f_0 + f_1 + f_3 + \ldots + f_{n-1} + f_n)$

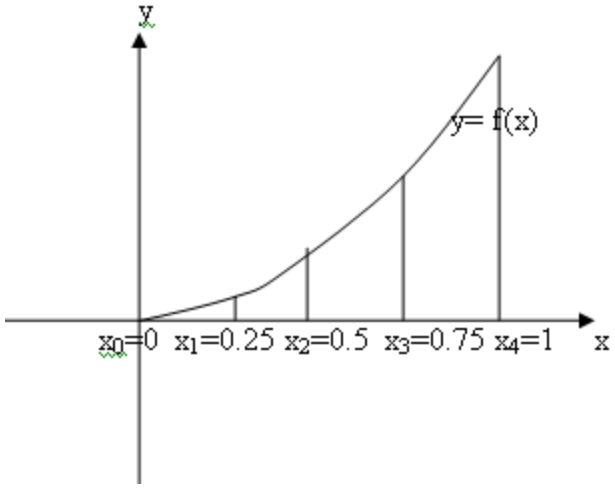
$\approx \frac{h}{2}(f_0 + f_n + 2f_1 + 2f_2 + 2f_3 + \ldots + 2f_{n-1})$

$\approx \frac{h}{2}[f_0 + f_n + 2(f_1 + f_2 + f_3 + \ldots + f_{n-1})]$

Example

Evaluate $\int_0^1 x^2 dx$ using trapezium rule with four strips

Solution



But $h = \frac{b-a}{n} = \frac{1+0}{4}$

$= \frac{1}{4} or 0.25$

$f_0 = f(0) = 0^2 = 0$

$f_1 = f(0.25) = (0.25)^2 = 0.0625$

$f_2 = f(0.5) = (0.5)^2 = 0.25$

$f_3 = f(0.75) = (0.75)^2 = 0.5625$

$f_4 = f(1) = 1^2 = 1$

But $\int_0^1 x^2 dx \approx \frac{h}{2}\left[f_0 + f_4 + 2(f_1 + f_2 + f_3)\right]$

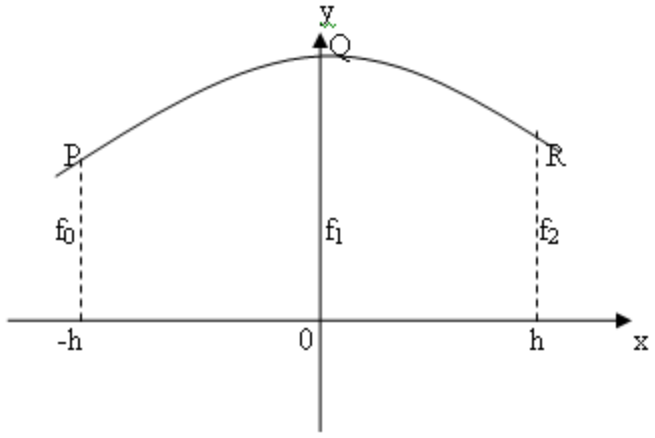
$\approx \frac{0.25}{2}\left(0 + 2\left[0.0625 + 0.25 + 0.5625\right]\right)$

$\approx 0.125 \times 2.75$

≈ 0.344 (t0 3 d.p)

SIMPSON RULE

We approximate the curve by a number of parabolic arcs. Now any parabola is totally determined by three points. Let us then find the area under the curve through three given points. Let the three points be $(-h, f_0)$, $(0, f_1)$ and (h, f_2) and $f(x) = ax^2 + bx + c$



If the parabola passes through the points (-h, f_0), (0, f_1) and (h, f_2) then

$f_0 = ah^2 - bh + c$ (1)

$f_1 = c$ (2)

$f_2 = ah^2 + bh + c$ (3)

$f_0 + f_2 = 2ah^2 + 2c$

$f_0 + f_2 + 4f_1 = ah^2 + 2c + 4c$

$=ah^2 + 6c$

Area under the parabola is

$$\int_{-h}^{h} (ax^2 + bx + c)dx = \frac{ax^3}{3} + \frac{bx^2}{2} + cx \Big|_{-h}^{h}$$

$$= \frac{ah^3}{3} + \frac{bh^2}{2} + ch - \left(\frac{-ah^3}{3} + \frac{bh^2}{2} - ch \right)$$

$$= \frac{ah^3}{3} + \frac{bh^2}{2} + ch + \frac{ah^3}{3} - \frac{bh^2}{2} + ch$$

$$= \frac{2}{3} ah^3 + 2ch$$

$$= \frac{1}{3} h(ah^2 + 6c) \quad *$$

But $ah^2+6c = f_0+f_2+4f_1$

So returning to *, we have

$$\int_{-h}^{h} (ax^2 + bx + c)dx = \frac{h}{3}(f_0 + f_2 + 4f_1)$$

Dividing up the interval $[a,b]$ into an even number, 2n of strips of width h. We approximate the area under the curve

$$\int_a^b f(x)dx \approx \frac{h}{3}(f_0 + 4f_1 + f_2) + \frac{h}{3}(f_2 + 4f_3 + f_4) + \frac{h}{3}(f_4 + 4f_5 + f_6) + \ldots + \frac{h}{3}(f_{2n-2} + 4f_{2n-1} + f_{2n})$$

$$\approx \frac{h}{3}[(f_0 + f_n) + 4f_1 + 4f_3 + 4f_5 + 4f_7 + \ldots 4f_{2n-1} + 2f_2 + 2f_4 + 2f_6 + \ldots + 2f_{2n}]$$

$$\approx \frac{h}{3}[(f_0 + f_n) + 2(f_2 + f_4 + f_6 + \ldots + f_{2n-2}) + 4(f_1 + f_3 + f_5 + \ldots f_{2n-1})]$$

Example

Evaluate $\int_0^1 x^2 dx$ using Simpson rule with four strips.

Solution

$$h = \frac{b-a}{n} = \frac{1-0}{4}$$

$$=0.25$$

So $x_0 = 0$, $x_1 = 0.25$, $x_2 = 0.5$, $x_3 = 0.75$ and $x_4 = 1$

$$f_o = 0^2 = 0$$

$$f_1 = (0.25)^2 = 0.0625$$

$$f_2 = (0.5)^2 = 0.0625$$

$$f_3 = (0.75)^2 = 0.25$$

$$f_4 = (1)^2 = 1$$

But $\int_0^1 x^2 dx \approx \frac{h}{3}[(f_o + f_4) + 2f_2 + 4(f_1 + f_3)]$

$$\approx \frac{0.25}{3}[(0+1) + 2 \times 0.25 + 4(0.0625 + 0.5625)]$$

$$\approx 0.83333(1 + 0.5 + 2.5)$$

$$\approx 0.83333 \times 4$$

TUTORIAL 12: INTEGRATION

1.Evaluate the following:

(a) $\int x^9 dx$ (e) $\int (x+1)^2 dx$

(b) $\int \left(x^4 + 2x^3 - 5x^2 + x - 10\right) dx$ (f) $\int (\sin 3x + \cos 2x) dx$

(c) $\int \frac{x^2 - 1}{x+1} dx$ (g) $\int 2 \sin x \cos x dx$

(d) $\int \left(x^3 + 2\right)^5 x^2 dx$ (h) $\int \frac{x}{x^2 - 4x - 5} dx$

2.Evaluate:

(a) $\int_0^1 \left(x^2 - 1\right) x dz$

(b) $\int_0^1 x \cos x^2 dx$

(c) $\int_0^2 e^{-7x} dx$

3. Show hat $1 + \frac{1}{x-1} = \frac{x}{x-1}$. Hence find $\int \frac{x}{x-1} dx$

TOPIC 13: SERIES

OBJECTIVES

By the end of this topic, you should be able to;

(a) Define the following terms; sequences, series, arithmetic progression and geometric progression.
(b) Find the sum of series.
(c) Simplify binomial series using Pascal's triangle.
(d) Calculate the limits and sum of infinite series.

SEQUENCES

A sequence is a set of numbers occurring in order. Some examples of sequences are 2, 4, 6, 8, 10 … and 1, 4, 9. 16, 25,..

The terms of a sequence are often represented by ordered lower case letters. Thus a_1 denotes the first term and a_2 denotes the second term and the sequence becomes a_1, a_2, a_3, …

A sequence can also be described as an nth term which gives a general formula for computing all the terms of the sequence.

Example

If $a_n = \frac{n+1}{n}$ then find a_1, a_2 and a_3.

Solutions

(i) $a_1 = \frac{1+1}{1} = 2$

(ii) $a_2 = \frac{2+1}{2} = \frac{3}{2}$

(iii) $a_3 = \frac{3+1}{3} = \frac{4}{3}$

Example

If $a_n = \frac{(-1)^n}{2^n}$ then find out a_1, a_2 and a_4.

Solution

$a_1 = \frac{(-1)^1}{2^1} = \frac{-1}{2}$

$a_2 = \frac{(-1)^2}{2^2} = \frac{1}{4}$

$a_{10} = \frac{(-1)^{10}}{2^{10}} = \frac{1}{2^{10}}$

SERIES

It is a sequence in which the terms have been added.

Some examples of series are

(a) 1+2+4+8+16+32+64

(b) 1+4+9+16+…

There are two types of series namely finite and infinite. Example (a) above is an example of finite series (we know how many terms are there) and example (b) is an infinite series because we can not know the total number of terms.

Consider again the series 1+2+4+8+16+64. As each tem is a power of 2, we can write this series in the form of: $2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6$

All terms of this series are of the form 2^n, so 2^n is a general term. Instead of a series appearing as a string of terms added together, the series can is written using sigma notation. i.e. $1+2+4+8+16+32+64 = \sum_{0}^{6} 2^n$

$\sum$ is the Greek letter, sigma. It is used in mathematics to represent summation. Placing the lowest and highest values that takes below and above the sigma symbol respectively, indicates that n takes all integral values between these extreme values.

Example

Write the following in the sigma notation

(1) 1+8+27+64+125 given that n^{th} term = n^3

(2) 2+4+6+8+… given that n^{th} term = 2n

Solutions

(i) $1+8+64+125 = \sum_{n=1}^{5} n^3$

(ii) $2+4+6+8+\ldots = \sum_{n=1}^{\infty} 2n$

Example

Find the first three terms if $a_n = \sum_{r=0}^{5} r(r+1)$

Solutions

N^{th} term = r(r+1)

1st term = 0(0+1)

=1

2nd term =1(1+1)

=2

3rd term =2(2+1)

=6

Example

Evaluate $\sum_{3}^{5} (r+1)(r+2)$

Solution

$$\sum_{3}^{5} (r+1)(r+2) = (3+1)(3+2) + (4+1)(4+2) + (5+1)(5+2)$$

$$= 4\times 5 + 5\times 6 + 6\times 7$$

=20 +30 +42
=92

ARITHMETIC PROGRESSION (SERIES)
An Arithmetic Progression (A.P) is a sequence where any term differs from the preceding term by a constant, called the common difference. The common difference may be positive or negative and is usually denoted by d.
In general, if an AP has a first term a, and a common difference d,
the nth term AP = a +(n-1)d

Example
Find an expression for the nth term of the AP 10, 7, 4, 1, -2, . .
Solution
$a = 10$ and d= 7-10(-3)
But nth term= a+ (n-1)d
=10 + (n-1)-3
=10-3n+3
=13-3n

Example
The nth term of an AP is 12-4n. Find the first term and the common difference
Solutions
$n^{th} term = 12 - 4n$
1st term = 12-4
=8
2nd term = 12-8
=4
So d= 4-8
=-4

SUM OF A. P
In general, the sum of an AP is given by
$S_n = \frac{n}{2}\{2a + (n-1)d\}$ or $S_n = \frac{n}{2}(a + l)$ where n= number of terms
A= first term
d = common difference
l= last term

Example
Find the sum of 2+5+8+11+…+47
Solution
Given a= 2 and l = 47
But n^{th} term = a +(n-1)d
2+(n-1)3 =47
2+3n-3 =47
3n= 47+1

$3n = 48$

$n = 16$

but $S_n = \frac{n}{2}(a + l)$

$S_{16} = \frac{16}{2}(2 + 47)$

$S_{16} = 8 \times 49$

$= 392$

Example

Show that the terms of $\sum_{r=1}^{n} \ln 2^r$ are in the arithmetic progression

Solution

$\sum_{r=1}^{n} \ln 2^r = ln2 + ln2^2 + ln2^3 + ln2^4 + \ldots + ln2^{n-1} + ln2^n$

$= ln2 + 2ln2 + 3ln2 + 4ln2 + \ldots + (n-1)ln2 + nln2$

We see that there is a common difference of *ln2* between successive terms, so the terms of this series are in AP.

GEOMETRIC PROGRESSION (G.P)

A geometric progression is a sequence in which each term is produced from the preceding term by multiplying by a fixed number called a common ratio. The common ratio may be positive or negative.

In general if a GP has a first term *a* and a common ratio *r*, then

n^{th} *term* $= ar^{n-1}$ and

$S_n = \frac{a(r^n - 1)}{r - 1}$ if r>1 or

$S_n = \frac{a(1 - r^n)}{1 - r}$ if r< 1 where *a*= first term

n = number of terms

r = common ratio

S = sum

Example

Find an expression for the nth term of the GP 3, 6, 12 24, . . .

Solution

Given a =3 and r =2

But n^{th} *term* $= ar^{n-1}$

$= 3 \times 2^{n-1}$

Example

A general series has first term 8 and common ratio 1.4 Find the sum of the first 10 terms.

Solution

Given a =8, r =1.4>1 and n= 10

But $S_n = \dfrac{a\left(r^n - 1\right)}{r-1}$

$S_{10} = \dfrac{8\left(1.4^{10} - 1\right)}{1.4-1}$

=558.51 (2 d.p)

INFINITE SERIES

Suppose we start adding the terms together. We define the individual sums by

$S_1 = a_1$

$S_2 = a_1 + a_2 = S_1 + a_2$

$S_3 = a_1 + a_2 + a_3 = S_2 + a_3$

$S_4 = a_1 + a_2 + a_3 + a_4 = S_3 + a_4$

.

.

.

$S_n = a_1 + a_2 + a_3 + a_4 + ... + a_{n-1} + a_n = S_{n-1} + a_n$

We refer to S_n as the *n^{th} partial sum*

DEFINITION

If the sequence of partial sums $S_n = \sum_{k=1}^{n} a_k$ converges (to some number *S*), then we say that the series $\sum_{k=1}^{\infty} a_k$ converges (to *S*). We write $\sum_{k=1}^{\infty} a_k = \lim \sum_{k=1}^{n} a_k = \lim_{n\to\infty} S_n = S$

In this case, we call *S* the sum of the series.

For any A.P. $S_n = \dfrac{n}{2}\{2a + (n-1)d\}$, which always approaches infinity as $n \to \infty$.

Therefore any AP is divergent.

GPs with common ratio r such that -1<r<1 generates convergent infinite series where

$S_{n\to} S_\infty$ as $n \to \infty$ and $S_\infty = \dfrac{a}{1-r}$

Example

Determine whether each series converges. If it does, give its sum to infinity

(a) 3 +5 +7 +…

(b) $1 - \dfrac{1}{4} + \dfrac{1}{16} - \dfrac{1}{64} + ...$

(c) $3 + \dfrac{9}{2} + \dfrac{27}{4} + ...$

Solutions

(a) 3 +5 +7 +…

Here a=2 and d=2. This is an example of an AP. Therefore, it does not converge.

(b) $1-\frac{1}{4}+\frac{1}{16}-\frac{1}{64}+...$

Here a=1,and $r=\frac{-1}{4}\div\frac{1}{1}$

$=\frac{-1}{4}$ i.e -1<$\frac{-1}{4}$<1 then it converges

So $S_\infty=\frac{1}{1--\frac{1}{4}}$

$=\frac{1}{1+\frac{1}{4}}$

$=\frac{1}{\frac{4+1}{4}}$

$=\frac{4}{5}$

(c) $3+\frac{9}{2}+\frac{27}{4}+...=3+3\times\frac{3}{2}+3\times\frac{9}{2}+...$

$=3+3\times\frac{3^1}{2}+3\times\frac{3^2}{2}+...$

This series is a G.P where $r=\frac{3}{2}$>1

Therefore, the series does not converge.

Example

Find $\sum_{r=1}^{\infty}\frac{3}{(r+1)(r+2)}$

Solution

nth term $=\frac{3}{(n+1)(n+2)}$

We need to decompose this into partial fractions i.e $\frac{3}{(n+1)(n+2)}=\frac{A}{n+1}+\frac{B}{n+2}$

We see that A = 3 and B =-3 (verify this)

So $\sum_{r=1}^{\infty}\frac{3}{(r+1)(r+2)}=\sum_{r=1}^{\infty}\left[\frac{3}{r+1}-\frac{3}{r+2}\right]$

But

$\sum_{r=1}^{\infty}\left[\frac{3}{r+1}-\frac{3}{r+2}\right]=\left(\frac{3}{2}-\frac{3}{3}\right)+\left(\frac{3}{3}-\frac{3}{4}\right)+\left(\frac{3}{4}-\frac{3}{5}\right)+...+\left(\frac{3}{n}-\frac{3}{n+1}\right)+\left(\frac{3}{n+1}-\frac{3}{n+2}\right)$

All terms except the first and last cancel each other out. So

$\sum_{r=1}^{\infty}\left[\frac{3}{r+1}-\frac{3}{r+2}\right]=\frac{3}{2}-\frac{3}{n+2}$ as $n\to\infty$, $\frac{3}{n+2}\to 0$, so the series converges to the value $\frac{3}{2}$

i.e. $\sum_{r=1}^{\infty}\frac{3}{(r+1)(r+2)}=\frac{3}{2}$

POWER SERIES

A series such as $x+x^2+x^3+x^4+...$ is called a power series because the terms involve powers of a variable quantity.

BINOMIAL SERIES

The word binomial means an expression with two terms. Some examples of binomials are $(a+b)^2, (3+x)^4$ and *(5-y)*^{7}

Recall that $(a+b)^2=a^2+2ab+b^2$

$(a+b)^3=(a+b)(a+b)^2$

$=(a+b)(a^2+2ab+b^2)$

$=a^3+2a^2b+ab^2+a^2b+2ab^2+b^3$

$=a^3+3a^2b+3ab^2+b^3$

We could pf course continue in this way indefinitely. Fortunately thee is a simple rule which tells us what the expression of $(a+b)^n$ is for any n$\in N$

```
          1   1
        1   2   1
1  3    3   1
1  4  6    4   1
1  5  10  10  5   1
```

Pascal's triangle

HOW IS PASCALS TRIANGLE CONSTRUCTED

1. Each row starts and ends with 1
2. Each other number is the sum of two numbers, in the row above it as shown by the arrows.

Points to remember when expanding binomial terms

1. The powers of the first term decreases from the original power to zero (0) and the powers of the second term increases form zero (0) to the original power on the binomial.
2. The sum of powers of the first and second terms equals the power on the original binomial.
3. The coefficients of the term come from the Pascal's triangle.

Example

Expand $(x+4)^5$

Solution

Using the fifth row of Pascal's triangle

$$(x+4)^5 = 1x^5 \times 4^0 + 5x^4 \times 4 + 10x^3 \times 4^2 + 10x^2 \times 4^3 + 5x^1 \times 4^4 + 1x^0 \times 4^5$$
$$= x^5 \times 1 + 5x^4 \times 4 + 10x^3 \times 16 + 10x^2 \times 64 + 5x \times 256 + 1 \times 1024$$
$$= x^5 + 20x^4 + 160x^3 + 640x^2 + 1024$$

Example

Expand $(2x-3)^3$

Solution

$$(2x-3)^3 = (2x+-3)^3 = 1(2x)^3(-3)^0 + 3(2x)^2(-3)^1 + 3(2x)^1(-3)^2 + 1(2x)^0(-3)^3$$
$$= 1 \times 8x^3 \times 1 + 3 \times 4x^2 \times (-3) + 3 \times 2x \times 9 + 1 \times 1 \times (-27)$$
$$= 8x^3 - 36x^2 + 54x - 27$$

Example

Find an approximation correct to four decimal places of

(a) $(1.01)^5$

(b) $(0.99)^4$

Solutions

(a) $(1.01)^5 = (1+0.1)^5$

Using the 5th row of the Pascal's triangle, we have

$$(1.01)^5 = (1+0.1)^5 = 1 \times 1^5 \times (0.1)^0 + 5 \times 1^4 \times (0.1)^1 + 10 \times 1^3 \times (0.1)^2 + 10 \times 1^2 \times (0.1)^3 + 5 \times 1^1 \times (0.1)^4 + 1 \times 1^0 \times (0.$$
$$= 1 \times 1 \times 1 + 5 \times 1 \times 0.1 + 10 * 1 \times 0.01 + 10 \times 1 \times 0.001 + 5 \times 1 \times 0.0001 + 1 \times 1 \times 0.00001$$
$$= 1 + 0.5 + 0.1 + 0.01 + 0.0005 + 0.00001$$
$$= 1.61051$$

Therefore $(1.01)^5 = 1.6105$ (to 4d.p)

(b) $(0.99)^4 = (1\text{-}0.01)^4$

$=(1+\text{-}0.01)^4$

$$= 1 \times 1^4 \times (-0.01)^0 + 4 \times 1^3 \times (-0.01)^1 + 6 \times 1^2 \times (-0.01)^2 + 4 \times 1^1 \times (-0.01)^3 + 1 \times 1^0 \times (-0.01)^4$$
$$= 1 \times 1 \times 1 + 4 \times 1 \times (-0.01) + 6 \times 1 \times 0.0001 + 4 \times 1 \times (-0.000001) + 1 \times 1 \times 0.00000001$$
$$= 1 - 0.04 + 0.0006 - 0.000004 + 0.00000001$$
$$= 0.96059601$$

Therefore $(0.99)^4 = 0.9606$ (to 4 d.p)

TUTORIAL 13: SERIES

3. Evaluate the following;

   (a) $\sum_{r=1}^{3} r^3$ (b) $\sum_{r=2}^{4} (r^2+1)$ (c) $\sum_{r=3}^{r=5} (r+1)(r+2)$

4. Find the first five terms of the sequence whose n^{th} term is given by

$$n^{th} = \frac{n(n-1)(n-2)}{6}$$

5. Find the sums of the following series;
   (a) 1+5+9+13+…+41
   (b) 2+4+8+…+256
   (c) 27+24+21+…+9
   (d) 2-6+18-54+…+13122
6. Find the expansion of the following binomials using Pascal's triangle:
   (a) $(1+x)^6$
   (b) $(3-x)^4$
   (c) $(2+3x)^5$
7. Find the value of $(1.01)^7$ to four decimal places using Pascal's Triangle.

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